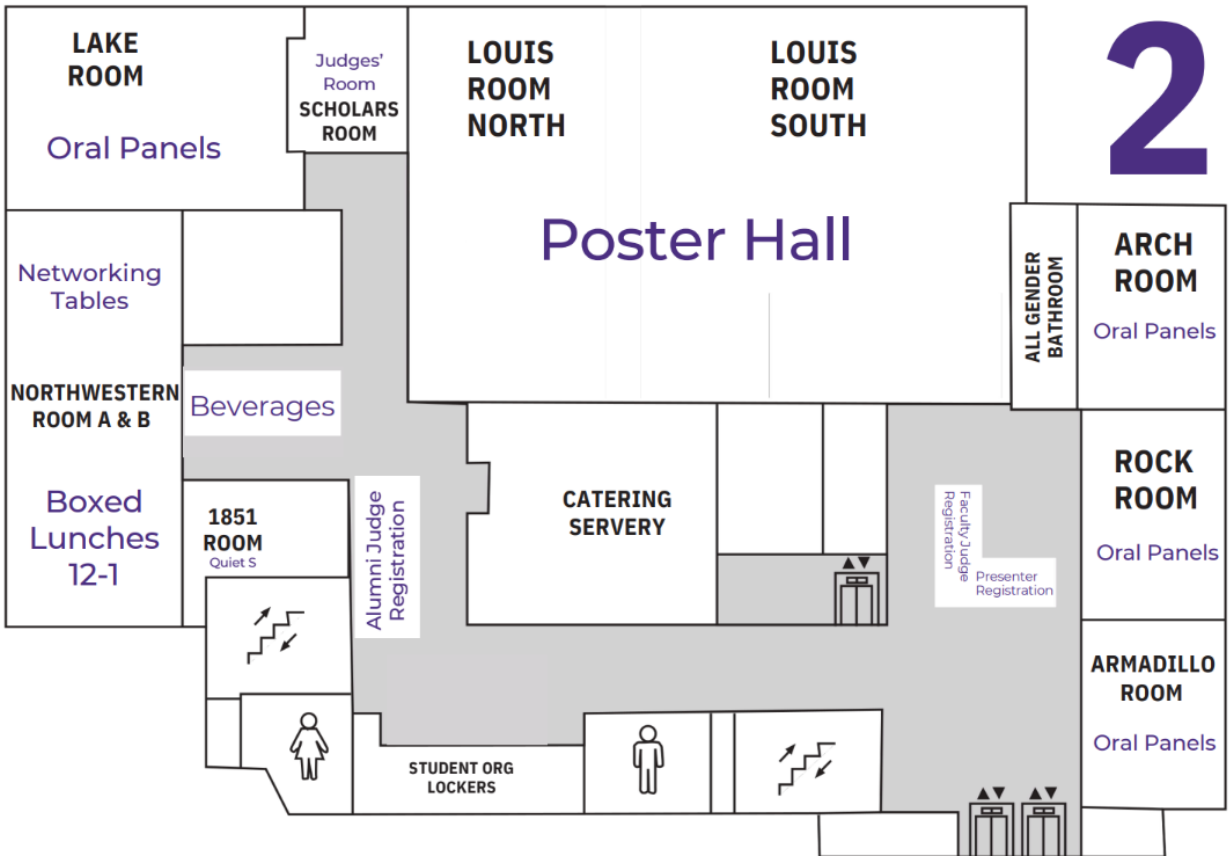
Abstract Guide

Abstracts are organized alphabetically by last name

To search by name:

- Press Download on the bottom of the page
- Use your device's Find fuction
    - iPhone: Tap the Share icon, scroll to Find on Page, enter student name
    - Android: Tap More, then tap Find in Page, enter student name

Norris Second Floor Map



Afacan, Can Faculty Advisor: Hai Zhou

Deductive Proof vs. Model Checking for Safety Verification of Autonomous Physical Systems

Autonomous systems are widely used in practice, from drones to robotic arms in factories, because they improve productivity and efficiency across many settings. Yet even when their controllers come with theoretical guarantees, real-world environments remain noisy and uncertain.

Formal verification uses mathematical logic to prove that a formal model satisfies a specification, or to exhibit a counterexample when it does not. This project studies how formal verification can be used to analyze the safety of a 3D drone model controlled by a hardware-oriented controller. The goal is to understand which verification methods are most effective for systems that combine continuous dynamics, discrete control, and safety constraints.

We model the drone mathematically, design a simple controller that keeps it within a safe region, and analyze the resulting closed-loop system using three approaches: explicit-state model checking in TLA+/TLC, symbolic model checking in nuSMV, and deductive proof in Lean 4. We also study the model across three tiers of complexity, with each tier introducing richer physics, such as wind, or more involved controller behavior. This allows us to compare the methods in terms of guarantee strength, specification, proof effort, and scalability as the system becomes more complex.

Although the project is still ongoing, it already provides a common framework for comparing major verification techniques on the same control problem. The broader contribution is a clearer view of the tradeoffs between these approaches for safety-critical autonomous systems.

Presentation Format: Oral
Time: 12:45-2:15PM
Room: Lake Room

Alshubbak, Sanad

Scrolling Into the Future: Social Media and Gen Z Men's Education and Career Pathways

For Gen Z men coming of age today, the familiar script that "college is the surest path to success" feels less certain, while social media has become a major place where ideas about work, money, and status circulate. Despite growing attention to social media, its role in shaping pathway perceptions and career decision-making for young men remains understudied. This project asks: How does social media inform Gen-Z males' career aspirations and education choices, and how does it shape perceptions of college, trades/alternative credentials, and entrepreneurship?

Data collection uses a qualitatively driven mixed-methods design. I recruit U.S. men ages 18-25 from two contrasting contexts (Northwestern University and Chicago's southwest suburbs), administer an online screener and pre-interview survey, and conduct ~21 semi-structured interviews. Interviews include a brief projective vignette exercise, and transcripts are analyzed using grounded theory-inspired thematic coding (iterative coding, constant comparison, and refinement of themes that explain how participants link social media to pathway beliefs). Preliminary conversations suggest participants organize post-high school decisions around three pathways: four-year degree-to-employment, trades/alternative credentials, and entrepreneurship/self-employment, and describe social media as legitimizing non-college paths, increasing the appeal of entrepreneurship, and reshaping definitions of success toward autonomy and status. The study examines how algorithmically curated content reshapes these pathways alongside influences from families, schools, and local opportunity structures, including masculinity/status "scripts." Findings contribute to research on social media, education-to-work transitions, and gendered expectations of "success," with implications for educators and career advisors.

Presentation Format: Oral
Time: 10:45AM-12:15PM
Room: Arch Room

Alvarez, Julia Faculty Advisor: Andrew Papachristos

Building or Breaking Trust: The Effectiveness of Chicago Community Policing Programs on Building LGBTQ+ Trust

The consent decree was issued by the Department of Justice to the city of Chicago in 2019, mandating community policing programs to help build a relationship with groups struggling to interact with the police, including the LGBTQ+ community. In alignment with these mandates, the Chicago Police Department implemented the LGBTQ+ Liaison Program, but no evaluation has yet measured its effects on the community. Utilizing interview data of Chicago LGBTQ+ community stakeholders and ethnographic observations of beat meetings in the 19th police district, I analyze whether this program has been working to increase trust in police for the LGBTQ+ community. Leveraging existing theory on what shapes trust in the police, the analysis found that the program's limited visibility results in a neutral effect on trust. However, respondents viewed the program's goals positively and want to see more visible implementation in the future. The findings suggest that the program may positively influence trust. This exploratory study indicates that the LGBTQ+ liaison program could be successful in Chicago if properly implemented.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 61P

Ametovski, Jasmine Faculty Advisor: Eyal Kimchi

The Impact of Systemic Inflammation and Cholinergic Antagonism on Slow Wave Activity in Cortical Areas

Delirium is a neuropsychiatric condition characterized by acute changes in attention and awareness as well as electroencephalography (EEG) slowing. Delirium and attentional deficits can be precipitated by inflammation and cholinergic antagonism, but the brain regions vulnerable to slowing by these precipitants remain unknown. While EEG has excellent temporal resolution, it lacks the spatial resolution to identify regional vulnerability. Other methodologies with greater spatial resolution do not have the temporal resolution to monitor oscillations. Widefield imaging is a technique which allows monitoring of regional cortical activity with high temporal resolution. I used widefield imaging to assess how systemic inflammation and anticholinergic scopolamine affect slow wave activity (SWA) in cortical regions. The dorsal cortices of mice expressing genetically encoded calcium sensors in excitatory neurons were recorded following injections of saline, lipopolysaccharide (LPS), and scopolamine to monitor fluorescence changes in multiple brain regions. Using fluorescence, power spectral density and the ratios of delta (1-3 Hz) and theta (3-5) power were computed for each injection and brain region. LPS and scopolamine decreased locomotor activity in mice. LPS increased relative delta power throughout the entire dorsal cortex which was localized to sensory cortices, while scopolamine only increased relative delta power in somatosensory areas. These findings clarify regional vulnerability of cortical regions to SWA in response to systemic inflammation and cholinergic antagonism, suggesting that these substances utilize different modalities to alter behavior, cognition, and SWA. These findings suggest that delirium may be precipitated through heterogeneous pathways that contribute to altered attention and awareness.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 32P

Anderson, Laurel Faculty Advisor: Mark Hauser

Making Cool Water: Evaporative Cooling in Caribbean Coarse Earthenware

Porous earthenware jars, commonly known as goglets, have historically been used throughout the Caribbean to store water. These vessels are not only defined by their specific shape but also by their ability to cool the water they hold through evaporation. This study investigates how the choice of material used for these goglets impacts the water stored inside them, and how this compares to historical and archaeological understandings of their use.

The vessels used in this experiment were purchased from contemporary ceramicists in the Caribbean. Their form and composition are similar to goglets used on sugar plantations in the 18th and 19th centuries. The test vessels were filled with water then placed inside a temperature regulated chamber, where water temperature, air temperature, and relative humidity were recorded. Water temperature decreased several degrees when stored in the more porous of the test vessels. The most significant drop in temperature occurred within the first three hours, which is consistent with accounts of goglets being refilled around three times during the day.

The observation of the evaporative cooling effect of these vessels helps contextualize how they were and are used, and to understand the use of coarse earthenware as a deliberate choice informed by its properties rather than emerging solely from a lack of access to other materials. This type of experimental archaeology also helps to reconstruct the role of water and its modification in the social and material lives of the past.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 41P

Argus, Aria Faculty Advisor: Karl Scheidt

Synthesis of MAP2K4/7 Inhibitors: Overcoming Drug Resistance in Pancreatic Ductal Adenocarcinoma

Pancreatic ductal adenocarcinoma (PDAC) is the third leading cause of cancer-related mortality, characterized by low survivability and the absence of early-stage symptoms. Due to its low post-surgery survivability and resistance to conventional radiation and chemotherapies, designing a drug to inhibit the Kirsten Rat Sarcoma virus protein (KRAS), whose mutations are linked to approximately 90% of PDAC cases, is a promising therapeutic approach. However, undesired reactivation of KRAS through intricate intracellular signaling networks, particularly the Mitogen-Activated Protein Kinase (MAPK) pathway, diminishes the efficacy of current KRAS-inhibition therapies. One possibility for resensitizing KRAS to such inhibition-based therapies is deactivating MAP2K4/7, two proteins involved in this MAPK pathway. To this end, we designed and synthesized MAP2K4/7-inhibitors with drug-like properties. By switching out the portion of these MAP2K4/7 inhibitor molecules that bind to their target, thereby imparting inhibition, we fine-tuned these small-molecule drugs to be more potent and more precise in targeting MAP2K4/7. The most optimized MAP2K4/7 inhibitors demonstrated that stronger deactivation of a particular branch of the MAPK pathway drives more complete suppression of the KRAS-reactivating feedback loop, resensitizing KRAS to its existing inhibition therapies. Most significantly, this MAP2K4/7 inhibition strategy holds promising clinical significance for overcoming drug resistance in PDAC and other KRAS-driven cancers.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 05P

Avellar, Andre Faculty Advisor: Joe Mazzulli

Investigating Drug-Based Strategies to Improve Lysosomal Function in CLN7 Neuropediatric Batten Disease

Batten disease is a group of neurodegenerative lysosomal storage disorders that primarily affect children and are characterized by progressive loss of neuronal function due to impaired cellular waste processing. Mutations in MFSD8, encoding the lysosomal membrane protein CLN7, cause CLN7 Batten disease and lead to selective disruption of lysosomal function. Despite advances in disease genetics, effective treatments remain limited, in part due to an incomplete understanding of how lysosomal dysfunction leads to selective protein instability. Previous studies have shown that CLN7 deficiency reduces CLN5 levels via post-translational mechanisms, suggesting dysregulated proteolysis. However, it remains unclear whether CLN5 loss reflects increased degradation or altered lysosomal regulation. As CLN5 functions as the lysosomal bis(monoacylglycero)phosphate synthase, its reduction is expected to disrupt lysosomal homeostasis. To address this, I investigated CLN5 stability using cycloheximide (CHX) chase assays and pharmacological inhibition of lysosomal proteases in MFSD8 knockout HEK293T cells. CHX experiments did not support increased CLN5 degradation, as levels were reduced at baseline and showed no consistent time-dependent decline. In contrast, broad inhibition of cysteine proteases robustly restored CLN5 levels (~9–10-fold), whereas selective inhibition produced only partial rescue, and inhibition of other protease classes had no effect. Importantly, immunocytochemistry revealed that the abundance of CLN5 puncta and their lysosomal association are reduced in MFSD8-KO cells and partially restored upon protease inhibition. Together, these findings indicate that CLN5 loss arises from dysregulated, network-level proteolysis. This work suggests that targeting cysteine protease activity may restore lysosomal function in CLN7 Batten disease.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 50A

Ayele, Soliana Faculty Advisor: Adriana Weisleder

Symbolic Play Engagement in Early Childhood: The Extent to Which Toy and Non-Toy Object Availability Explains Differences in Pretend Play

Symbolic play is an important developmental milestone in which young children begin to use objects and actions to represent things as they would occur in the real world. A key characteristic of symbolic play is that conventional actions (e.g., eating from a spoon) become progressively separated from their functional contexts (e.g., pretending to eat from an empty spoon during play), a form of decontextualization.

Symbolic play research has played an important role in shaping theories of cognitive development, however, in recent years, few have examined young children's symbolic play in naturalistic settings, like in the home environment, which is where learning and development mostly occur.

The goal of this project is to examine how children's engagement in symbolic play is shaped by differences in opportunities as measured by the availability of different objects in their immediate environment. A diverse sample of mother-child dyads (n = 90) from a variety of linguistic and cultural backgrounds participated in play interactions at home. These interactions were then coded in 10-second intervals by Spanish-English speaking coders. In a second coding pass, videos were coded for the objects the child was interacting with to determine if there were certain objects that evoked symbolic play.

Symbolic play offers a window to understanding children's emotional, cognitive, and social development. Future research includes exploring diverse samples to determine if the findings map onto the other linguistically and culturally diverse U.S. populations.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 53A

Bahena, Leisly Faculty Advisor: Marco Gallio

Substrate Temperature Preference During Egg-Laying in Drosophila

Ectotherms like fruit flies cannot regulate their internal body temperature and instead rely on behavioral strategies to avoid temperatures outside of their optimal range. Unlike adult flies, *Drosophila* larvae are limited in their ability to escape unfavorable conditions, and thus, female flies have to select a suitable substrate on which to lay eggs to increase the survival of the progeny; however, we know little about the specific choices a female fly makes when choosing the egg-laying substrate. The Gallio Lab studies the molecules and neurons underlying temperature-guided behaviors in *Drosophila*. In this project, we explored how substrate temperature influences where female fruit flies lay their eggs. We developed a new 2-choice temperature assay, where gravid female flies are given a choice between substrates kept at different temperatures. Using this assay, we found a strong preference for egg laying at 25C over lower or higher temperatures. Surprisingly, the main thermosensory organ of the fly, the antennae, is not required for this behavior. To identify the relevant thermosensory neurons, we have tested the role of candidate cell types using genetic tools to silence their activity. Lastly, we found that *Drosophila* species adapted to different thermal environments prefer to lay eggs at temperatures matching their native environments. This work is expected to reveal new aspects of decision-making during egg-laying behavior in *Drosophila*.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 90P

Bai Yang, Aileen Faculty Advisor: Jeffrey Lewis

Foreign Domestic Workers in Hong Kong: A Study of the Impacts of Low-Skill Immigrants on Native Female Labour Decisions

Behind Hong Kong's gleaming economy is an invisible workforce that keeps it running: roughly 340,000 Foreign Domestic Workers (FDWs) who cook, clean, and raise children. Over the past 40 years, the number of FDWs in Hong Kong has grown elevenfold, yet their economic effects on native female labour outcomes remain understudied. Using district-level Hong Kong census data from 2006–2021 and US American Community Survey microdata from 2006–2024, this paper conducts the first district-level analysis of FDW effects in Hong Kong and updates the US analysis of Cortés and Tessada (2011) to a more recent period. Hong Kong results show positive and significant associations between district FDW concentration and native female employment across all three model specifications. US results find no robust effects on whether women work at all, but consistent positive effects on full-time employment, primarily among unmarried women without young children. Results showed smaller effects for more highly educated women, suggesting that the mechanisms linking low-skill immigration to native women's labour supply vary by education level. Together, the results suggest that immigrant domestic workers enable native women to work more, not just more often, with important implications for immigration and gender equity policy.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 69A

Barnickol, Karl Faculty Advisor: Milan Mrksich

High-Throughput Discovery of Trispecific Antibodies for Cancer Therapies Using MegaMolecules

Immunotherapy is transforming oncology by leveraging antibodies to target specific cancer cell surface receptors, thereby reducing the off-target side effects associated with nonspecific treatments like chemotherapy. Once bound, these antibodies eliminate tumors by blocking growth signals, recruiting immune cells, or delivering toxic payloads. Multispecific antibodies—which bind to multiple receptors simultaneously—offer even greater potential by disrupting several signaling pathways at once, increasing binding strength through enhanced avidity, and increasing selectivity for cancer cells. Modern drug discovery is built on high-throughput screening of large and structurally diverse libraries of candidates to identify those with activity and narrow down millions of possibilities to a few promising leads. However, synthesis of large libraries of trispecific antibodies is a significant challenge. MegaMolecules provide a modular platform that enables rapid, precise assembly of multispecific antibodies. Rather than designing whole, rigid IgG scaffolds, megamolecules utilize high-affinity, covalent enzyme-inhibitor reactions to prepare multivalent antibodies with unprecedented scale and stability. In this project, we utilized the megamolecule architecture to construct a diverse library of 300 trispecific antibodies targeting numerous commonly expressed breast cancer epitopes from a base library of 23 fusion proteins. By successfully synthesizing and characterizing this library, this project demonstrates the unique efficacy of the megamolecule platform to address the prevailing limitations of immunotherapeutic design, establishing a scalable foundation for the discovery of the next generation of multispecific cancer therapies.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 04P

Baumgardner, Emma

Characterizing Peptoids as an Alternative Nanoparticle Stealth Coating

Breast cancer remains the most common cancer diagnosis for women worldwide, motivating the need for targeted drug therapies. Folate receptor alpha (FOLRa), a protein overexpressed on the surface of breast cancer cells, represents an attractive therapeutic target for this approach. Previously, gold nanoparticles functionalized with folic acid (FA) have been shown to selectively bind to FOLRa-overexpressing cells, however the molecular architecture linking FA to the nanoparticle surface represents a critical design tradeoff. Traditional linear polyethylene glycol (PEG) linkers, the current standard for FA conjugation, may shroud targeting capability of FA through steric interactions at longer lengths. However, shorter linker lengths cause rapid deterioration of nanoparticle stability. Peptoids, a synthetic, biomimetic polymer with tunable side chain chemistry, offer a promising alternative linker architecture which may help retain targeting efficiency without compromising nanoparticle stability through their brush-like structure. To investigate how linker architecture influences targeting, gold nanoparticles were functionalized with FA using both linear PEG and peptoid linkers and evaluated for their interactions with FOLRa-overexpressing cancer cells. Initial results suggest that linker architecture meaningfully affects cellular interactions, pointing to peptoids as a possible alternative to retaining targeting efficacy and without compromising stability of targeted nanoparticle-based cancer therapy. Ongoing work focuses on detailed material characterization to elucidate the underlying chemical mechanisms responsible for these differences.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 79P

Bell, Chase Faculty Advisor: Shirin Vossoughi

"Why don't fidget toys work?": Embodied artifacts in a MS summer camp.

In the Hubspace Summer Program, Evanston middle schoolers of color spend 6 weeks learning about STEAM (Science, Technology, Engineering, Art, and Mathematics) and telling/listening to stories, within an environment that blends joy and learning. Educators, including Dr. Shirin Vossoughi and myself, are also researchers, documenting then analyzing the program to redesign it for the next summer and share knowledge with the broader field. This style of educational research, when done with care and rooted in relationships, can unearth understandings central for educators' and young people's growth. As an Emerging Scholar, I took ethnographic field notes, recorded the video and audio of pedagogical interactions, and began my educator journey. After the summer, I conducted interviews with educators, coded field notes to see the connections between various forms of movement and focus among campers, and co-designed a week of the summer 2026 program. Watching my fellow Hubspace educators, I noticed how they saw the body as integral to learning, giving campers the freedom to move while thinking (e.g. crocheting while listening to an educator or friend). Over the school year, I read through fieldnotes and interviewed fellow educators to understand how and why they approached learning this way. They described their unique ways of understanding focus and learning that help them honor our young people's bodies and minds. While redesigning the curriculum for next summer, we incorporated new strategies such as drawing as an intentional form of storylistening, and aim to understand how these are taken up by the campers.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 01A

Bernick, Gabriel Faculty Advisor: David Rapp

Evaluating Reliance on Misinformation for Video Stimuli

The continued influence effect (CIE) is the cognitive psychological phenomenon where people continue to rely upon or mention information that has been retracted. Previous research indicates that providing participants with a plausible alternative explanation for retracted information reduces CIEs, in contrast to when an alternative explanation is not provided. This alternative explanation may fill any related mental model gap when a cause is refuted. However, this issue has predominantly been studied using text stimuli. It remains unclear whether and how other kinds of media might lead to CIEs. This study uses a fictional news broadcast about a chemical spell to test if CIEs can be applied to video stimuli. Participants were first prompted to watch one of four fictional broadcasts: a broadcast with no relevant information, a broadcast where a causal explanation is not discounted, a broadcast with a causal explanation that is discounted, and a broadcast with a causal explanation that is discounted and corrected with an alternative cause. They were then prompted to answer open-ended questions about the events of the story. Inference scores and endorsements are calculated by using the numbering of references that participants make to the original cause in both the open-response memory, inference, and free-recall questions. Reliance on CIEs will then be assessed based on inference scores and endorsement of the original cause. This survey is in the process of being administered, and responses have not yet been recorded.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 65P

Bers, Nico Faculty Advisor: Mark Gorski

Ingredients of Our Black Hole's Diet: Correlating Different Gas Phases in the Center of the Milky Way

At the center of our Milky Way sits Sagittarius A* (Sgr A*), a supermassive black hole four million times the mass of our Sun, and our closest cosmic laboratory for understanding how black holes grow and the fate of our galaxy. Despite the Galactic Center being the best resolved region around any supermassive black hole, we still don't fully understand how gas ends up feeding Sgr A* versus being pushed away from it-a gap that limits our understanding of black hole growth across the universe. Using radio telescope observations from the Atacama Large Millimeter/submillimeter Array (ALMA) and machine learning techniques, we map two distinct phases of gas near Sgr A*, cold molecular gas and hot ionized hydrogen. These techniques isolate dominant signals in the complex, three-dimensional data and statistically group gas structures by their properties. We identify several distinct gas structures near Sgr A*, including bow shocks and expanding shells, each shedding light on a different stage of black hole activity. This work is ongoing, and our continued analysis aims at understanding how these structures connect to the cycle of black hole feeding and feedback in our galaxy.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 08P

Bhakta, Diya Faculty Advisor: Amy Paller

shRNA-Mediated Knockdown of SPTLC3 Impairs Epidermal Differentiation and Filaggrin Expression in 3D Skin Organoid Models

Atopic dermatitis (AD), commonly known as eczema, is a chronic condition that affects more than 200 million people worldwide. It often causes dry, itchy, and inflamed skin because the skin barrier does not work properly. A healthy skin barrier depends on ceramides, natural fats that help the skin hold moisture and block irritants. SPTLC3 is part of an enzyme that helps make ceramides, but its exact role in skin health is not fully understood. This study examined whether lower levels of SPTLC3 may contribute to the skin barrier problems seen in eczema.

To test this, we reduced SPTLC3 levels in human skin cells called keratinocytes using shRNA, a tool that lowers gene activity. These cells were grown in three-dimensional (3D) skin organoids, which are lab-grown models that closely resemble real human skin. We then measured changes in important skin barrier genes and proteins using molecular imaging and gene analysis techniques.

When SPTLC3 was reduced, levels of filaggrin also decreased. Filaggrin is a key protein that helps skin stay strong, hydrated, and protected. We also observed changes in other markers involved in normal skin development, suggesting the skin tissue was not maturing properly.

Overall, our findings suggest that SPTLC3 plays an important role in keeping the skin barrier healthy. Lower SPTLC3 levels may weaken the skin and contribute to eczema. Understanding this pathway could help researchers develop new treatments that repair the skin barrier and improve symptoms for people with atopic dermatitis.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 34A

Bhasin, Bhavika Faculty Advisor: Thomas Bozza

Optimization of an inducible genetic strategy to study olfactory monoallelic expression

Olfactory sensory neurons (OSNs), responsible for detecting smells, each express only one odorant receptor (OR) gene chosen from thousands of possible options. However, the biological mechanisms that control this “one neuron–one receptor” rule are poorly understood. These neurons are continuously generated from stem cells in the olfactory epithelium, and receptor choice occurs during a short developmental window in immature OSNs. Studying this process requires the ability to control gene activity before this critical window. To do this, we used a genetic system in which the drug tamoxifen turns genes on/off in developing neurons. This system relies on an enzyme (Cre recombinase) expressed from the Ascl1 locus in stem cells. A key challenge is ensuring that this system activates efficiently in the right cells. To optimize this, we used a fluorescent reporter (Ai14; ROSA-tdTomato) that marks recombined cells, and tested different conditions by varying (1) the age at which tamoxifen was administered and (2) the number of consecutive treatment days. We measured recombination using imaging and fluorescence-activated cell sorting (FACS). We found that earlier treatment, starting at postnatal day 2 (P2), produced the highest proportion of labeled neurons, while later treatment resulted in more total labeled neurons due to overall tissue growth. The optimal balance was achieved at P5. Changing the number of treatment days (1, 3, or 5) did not significantly affect labeling. These results establish an optimized method for controlling gene activity in developing olfactory neurons, providing a foundation for studying how neurons select a single receptor gene.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 49P

Brady-Peters, Isabella Faculty Advisor: Michelle Driscoll

Rupture Instabilities of Thin Colloidal Films

Thin liquid films, which have a thickness ranging from a fraction of a nanometer to micrometers, have applications across various fields including disease transmission, industrial coating, and biomedical devices. Liquid films are a convenient system for studying fluid dynamics in two dimensions. This project focuses on films made with colloidal suspensions, which are formed when micron-sized particles, called colloids, are suspended in the film. When the concentration of colloids in the liquid is high enough, colloidal suspensions exhibit non-Newtonian behavior, which means that the liquid flow direction changes when force is applied. Examples of Newtonian fluids are water and oil, whereas some non-Newtonian fluids are mucus, paint, and oobleck. These colloidal films have been shown to exhibit various instabilities during rupture that qualitatively deviate from the expectation for Newtonian film rupture. To study this behavior, colloidal films across a range of thicknesses and colloidal concentrations were recorded during spontaneous and induced rupture on a high speed camera, and then analyzed using ImageJ and Python. The films are formed with a flat horizontal film stretcher and experiments are performed in a sealed box so that humidity can be stabilized and environmental factors can be mitigated. Analysis of the films is on-going, but it has been found that different colloid geometry (rods or spheres) corresponds to different instabilities, and there appears to be a critical transition thickness between types of instabilities. Further exploration is being done into the dependence of the speed of the rupture on thickness, colloidal concentration, and colloidal geometry.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 59A

Caldwell, Benjamyn Faculty Advisor: SonBinh Nguyen

Liposomal Spherical Nucleic Acid (LSNA) for the Treatment of Inflammation

Chronic inflammatory diseases arise when immune activation and inflammation are dysregulated. Common drug therapies, such as steroids, effectively reduce acute inflammation, but bring severe side effects with long-term or high-dose usage. To this end, recent studies have suggested that inhibitory oligonucleotides (INH-ODNs) can serve as alternative therapies by inhibiting the initial activation of inflammatory responses and preventing symptoms from spreading. These molecules bind to and inhibit the activation of a target immune receptor with high specificity, making them an ideal candidate for dysfunctional immune system signaling. However, nucleic acids are susceptible to rapid nuclease degradation and have poor cellular uptake, limiting their translation into clinical treatments. To address these challenges, we employ liposomal spherical nucleic acid (LSNA) nanoparticles for the delivery of INH18, an INH-ODN, to investigate the hypothesis that a liposomal structure can act as an effective scaffold to improve nucleic acid uptake and delivery. We first construct an amphiphilic conjugate where 3'-disulfide-INH18 was conjugated to a phospholipid via a stable maleimide linker. This amphiphilic conjugate was then anchored into a DOPC liposomal core to form the desired LSNA. This LSNA treatment was tested against stimulated inflammation in an inflammation-reporter cell line, measuring the production of the reporter protein. Inflammatory cytokine production was also evaluated using enzyme-linked immunosorbent assays (ELISAs).Our results indicate that the INH18 LSNA can effectively suppress both CpG-induced activation and pro-inflammatory cytokine secretion, laying the groundwork for synergistic dual-therapy strategies that combine upstream receptor inhibition by INH-ODNs with the downstream alleviation of inflammation by steroids.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 56A

Carlson, Rose Faculty Advisor: Bradley Sageman

Earth System History Repeats Itself: Contextualizing our Current Situation Through Records of Ancient Volcanism and Acidification

Ocean acidification is a dangerous consequence of heightened carbon emissions. When the concentration of carbon dioxide in the atmosphere increases, more carbon dioxide is absorbed by the ocean, where it reacts with water molecules to ultimately increase marine acidity and negatively impact marine ecosystems. One way to understand the longer-term consequences and feedback mechanisms associated with ocean acidification is to study similar periods in Earth's past. This project focuses on Ocean Anoxic Event 2 (OAE2), a 94-million-year-old event that involved global depletion of marine oxygen (anoxia) that was caused by high levels of volcanism. Volcanic carbon dioxide emissions at this time may have also led to widespread ocean acidification, in a manner similar to modern ocean acidification. This project seeks to evaluate the spatial and temporal relationship between volcanism and acidification at two sites during OAE2 using geochemical analysis of isotope ratios preserved in the corresponding rocks. Rock samples were crushed, dissolved, and analyzed for isotopic ratios via thermal ionization and isotope ratio mass spectrometric techniques. Ongoing calcium, carbon, and osmium isotopic analysis of samples from these two sites may suggest a complex interplay between volcanism, acidification, and post-depositional chemical alteration before, during, and after OAE2. These results could enhance scientific understanding of the spatial variability of acidification during OAE2, and assist in projections of future, human-driven acidification impacts.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 46A

Carmona, Avryl Faculty Advisor: Dan McAdams

Early Adversity, Meaning-Making, and the Construction of Narrative Identity among Exceptionally Generative Latina Leaders

How do individuals sustain moral commitment amid enduring hardship? This study examines how highly generative Latina activists interpret adversity and construct narratives that sustain their work in advocacy and community organizing. Focusing on lives shaped by chronic insecurity, gendered violence, and political instability, it asks what narrative forms enable sustained commitment to others. Using an idiographic qualitative design, the study analyzes autobiographical interviews with ten Latina activists engaged in advocacy related to forced disappearance, child abuse, Indigenous political participation, and gender-based violence. Interviews were transcribed, translated, and analyzed inductively for narrative identity processes. Across cases, participants structured their narratives by foregrounding early disadvantage alongside moments of opportunity, followed by a turning point: the recognition that their suffering was shared. From this, commitment to others took shape as a responsibility they could not refuse. Suffering is not left in the past; it remains an ongoing point of connection to others' pain. Participants stay closely connected to their sources of grief and interpret both victims and perpetrators through relational and contextual understandings. Action and suffering unfold together: the same conditions that threaten participants also sustain their work by giving it meaning. What participants describe as healing is not resolution, but the ability to continue while remaining connected to pain, which enables them to support others. This study introduces collective redemption to describe this narrative form. In contrast to dominant U.S. models of redemption, these findings suggest that generativity emerges through sustained engagement with hardship rather than from its resolution.

Presentation Format: Oral
Time: 12:45-2:15PM
Room: Armadillo Room

Cesena, Gabby Faculty Advisor: Peter Locke

Language Deprivation from Structural Violence: Trauma, LDS, and Implications

While existing literature documents both the harms of language deprivation and high rates of trauma among deaf individuals, few studies explicitly connect these phenomena. This paper questions whether psychological harm and trauma in deaf populations can be primarily attributable to language deprivation rather than the condition of being deaf. It also examines the proposal of Language Deprivation Syndrome (LDS) and its implications for how these outcomes are classified in mental health contexts. This project uses a qualitative synthesis of historical accounts, policy analysis, and peer-reviewed research to examine how educational systems, healthcare practices, and language access shape early development in deaf children. The scope includes scholarship on deaf education, mental health outcomes, and emerging theoretical frameworks. Across studies, language deprivation is consistently associated with cognitive delays, reduced educational attainment, and higher rates of depression, anxiety, and trauma exposure. Findings indicate that language deprivation is primarily a structural issue shaped by educational policy, limited access to sign language, and societal attitudes toward deafness. Evidence further suggests that psychological harm in deaf populations is better explained by this structural deprivation than by deafness itself. While emerging research on LDS identifies a pattern of developmental outcomes linked to deprivation, it also raises concerns about the pathologization of socially produced harm. Recognizing language deprivation as a preventable trauma reframes it as a public health and human rights issue. These findings underscore the need for early bilingual language access and systemic reforms in education, healthcare, and family support to improve outcomes and reduce harm.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 76P

Chang, Claudia Faculty Advisor: Jennifer Tackett

Gender-Based Personality Differences in Early-Stage Entrepreneurs

This study provides a comparison of early-stage entrepreneurs and successful entrepreneurs, while investigating gender-based personality differences within early-stage entrepreneurs. Utilizing a battery of assessments, this research identifies that while early-stage entrepreneurs share a foundational profile with successful founders, distinct differences exist in key areas such as ambition, learning approach, and prudence. Successful entrepreneurs demonstrated higher achievement orientation and lower prudence, suggesting that flexibility and risk-tolerance are critical drivers of success. Early-stage entrepreneurs most deviated from successful entrepreneurs in their derailers. Higher relative scores in cautiousness, diligence, and dutifulness likely impede the early-stage entrepreneur's ability to make rapid decisions. Conversely, successful entrepreneurs scored higher on the mischievous and imaginative scales, reflecting a greater propensity for limit-testing and unconventional methodology. Gender analysis revealed significant differences in negative emotionality (neuroticism) as well as in cognitive and affective empathy, with women in both, which is also consistent with broader population trends. However, women's higher scores in empathy did not diminish their self-perceived leadership capacity or social potency, which remained equal across genders. Going against trends in the broader population, there was no significant difference in scores related to achievement motivation or competitiveness. In the general population, men tend to score higher, but the competitiveness that characterizes entrepreneurs could explain why we fail to see that trend between genders in the early-stage entrepreneurs. Overall, these findings suggest that while "successful" leadership traits are relatively gender-neutral, internal affective processing styles and factors such as developmental stage (age) play significant roles in the entrepreneurial trajectory.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 63P

Chen, Jubilee Faculty Advisor: Jennifer Krizman

Neural Correlates of Spectral and Temporal Perceptual Strategies in Music-in-Noise Performance

Listening-in-noise requires listeners to extract precise auditory cues from a mixture of target sound streams and background noise. While music-in-noise perception has garnered recent interest as an assessment of auditory processing beyond speech, the neural mechanisms underlying individual differences in performance remain unclear. In particular, it is unknown whether variability in music-in-noise perception reflects distinct perceptual strategies that map onto neural encoding differences. This study aims to identify neural correlates of performance on the Music-in-Noise Test (MINT) by examining how music-in-noise perception is supported by different perceptual cues, and how these strategies relate to measures of auditory processing. Young adults aged 18-25 without extensive music experience (N = 20) completed the Music in Noise Test (MINT), which isolates performance based on pitch, rhythm, spatial, visual, and predictive cues. Principal component analysis of MINT subtests identified a single dominant component, separating pitch-based from more rhythm and visual performance strategies. Neural responses to music presented in quiet and noise were recorded using the Frequency-Following Response (FFR) and encoding strength at the fundamental frequency and harmonics was examined as a function of the behavioral strategy. Listeners with pitch-driven MINT performance exhibited stronger neural encoding of the fundamental frequency and harmonics in noise compared to those relying on non-spectral cues. These results demonstrate that behavioral strategies for music perception in noise are directly reflected in neural encoding of acoustic cues.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 23A

Chen, Jessica Faculty Advisor: Amy Paller

The Role of Serine Palmitolyltransferase Single Small Unit B in the Skin Barrier

When skin barrier function is compromised, skin diseases, such as eczema (atopic dermatitis, AD), can develop. AD, characterized by dry, itchy, and inflamed skin, reduces the quality of life for many individuals. The cause of AD is still not fully understood, and more effective, long-term treatments are needed. Ceramides are natural fats important to the skin barrier, assisting in moisture retention and protection. People with AD often have lower ceramide levels. After analyzing skin samples from AD patients and healthy individuals, we found that those with severe eczema were associated with having shorter ceramide lengths. Longer ceramides are thought to help form stronger skin barriers, while shorter ones may weaken them. A protein complex called SPT helps determine ceramide length early in the production process. One component of the complex, SPTSSB, is important for producing longer ceramides in the skin. Previous data suggest that SPTSSB levels are lower in human skin with AD. Investigating the effects of SPTSSB expression could reveal SPT's role in AD development, providing insight into potential directions for AD treatment. This study employed three-dimensional lab grown skin to model human skin development. The amount of SPTSSB was lowered and the changes were observed. To study these changes, we used laboratory techniques that measure gene activity and visualize the skin in culture. Reducing SPTSSB resulted in lower levels of key skin development markers, indicating impaired skin development. The study findings suggest that SPTSSB is important for skin barrier function, making it a potential treatment target for eczema.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 35A

Chen, Rebecca Faculty Advisor: Atique Ahmed

Beyond the Cytoskeleton: Vimentin as a Regulator of Chemotherapy Resistance in Glioblastoma

Glioblastoma multiforme (GBM) is a highly aggressive brain tumor with a median survival of 15 months, in part due to the tumor's adaptive resistance to temozolomide (TMZ), the first-line chemotherapeutic agent in GBM treatment. However, strategies to overcome resistance and improve patient outcomes remain elusive. Recent research suggests that VIM, a cytoskeletal protein overexpressed in GBM and a validated marker of invasion and poor prognosis, can also function as a DNA-binding protein in the nucleus, indicating potential roles in gene regulation. This study investigates whether VIM contributes to TMZ resistance through its nuclear-associated functions. To evaluate the role of VIM in TMZ resistance, we utilized multiple patient-derived xenograft (PDX) GBM cell lines treated with TMZ. Changes in VIM expression and subcellular localization were assessed using Western blotting and immunocytochemistry. To map VIM-DNA interactions, we performed CUT&RUN (Cleavage Under Targets & Release Using Nuclease) followed by downstream bioinformatics analyses to identify VIM-bound gene regions and associated pathways. We found that both cytoplasmic and nuclear VIM expression is increased by TMZ treatment. CUT&RUN analysis further revealed that VIM binds to regulatory regions of genes associated with TMZ resistance programs at 72 hours post-TMZ exposure. These findings suggest that VIM may act as a regulator of gene expression contributing to chemoresistance in GBM through means beyond its conventional cytoskeletal roles. Collectively, this work highlights a previously underexplored role of VIM in GBM biology and supports its potential as a therapeutic target for overcoming TMZ resistance.

Presentation Format: Oral
Time: 10:45AM-12:15PM
Room: Lake Room

Chen, Kathleen Faculty Advisor: Jason Hsieh

Interrupted! Using a Virtual Museum Video Game to Study Task-Switching

Empirical task switching studies investigating the ability to flexibly redirect attention, often rely on strict parameters to maintain a controlled environment. However, this does not capture naturalistic conditions involving exploration and everyday distractors. *Gallery Glimpse* was developed to address this, a video game evaluating naturalistic working- and episodic-memory suitable for stationary research participants, i.e. intracranial electroencephalography (iEEG). Participants alternate between viewing art pieces in a museum (episodic memory encoding) and interpolating memorization of phone numbers (working memory distractor). This study establishes normative behavioral data on cognitive flexibility in a control cohort aged 6–57 (n=89). We hypothesized controls would exhibit a transient "switch cost,"—decreased episodic memory accuracy immediately following the working memory distractor—while the iEEG cohort would demonstrate impaired cognitive flexibility via a prolonged switch cost. A mixed-effects model confirmed a switch cost in controls. The distractor task disrupted memory performance on the first art piece with improved memory accuracy on subsequent art pieces ($\beta = 0.07$, $z = 6.54$, $p < .001$). The model also identified a significant practice effect, with improved accuracy in the second to last ($\beta = 0.05$, $z = 4.43$, $p < .001$) and last ($\beta = 0.07$, $z = 5.82$, $p < .001$) rooms. iEEG cohort analyses will identify behavioral differences during the same period. This work will establish the use of Gallery Glimpse for bridging the gap between controlled experimental settings and real-world cognitive flexibility, while also providing a framework for understanding epilepsy-related memory impairments.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 55P

Cheng, Crystal Faculty Advisor: Sarah Bartolome

Group Singing as Treatment for Children's Wellness

Out of the Covid-19 pandemic emerged a mental health crisis, resulting from the prolonged periods of isolation and online interaction. This crisis was especially prevalent amongst children and adolescents, as younger people reported higher levels of depression and anxiety. In response to this crisis, Dr. Sarah Bartolome created the Music for Childhood Well-being Initiative (MCWI), an initiative exploring the impact of singing in a group on children's mental health and wellbeing. In this initiative, 8-week long sessions were conducted where a group of children aged 8-12 were led by an instructor to sing and engage in breathing and stretching exercises. As an Emerging Scholar research assistant, my involvement in this initiative included setting up surveys for the participants to complete, helping participants put on the heart-rate monitoring device we used to collect their heart rate and heart rate variability data, accompanying the singing by playing the piano during sessions, and administering closing interviews for each session. As the sessions progressed, I observed how these sessions led the participants to bond over a sense of community. This became something that all the participants looked forward to every week, and even if participants admitted to entering the room feeling tired, they usually left feeling happy after the session's activities. Further implications of this research could involve the exploration of utilizing group singing as a potential mental health treatment for demographics beyond children, as it has proved to be effective amongst children in MCWI sessions.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 06A

Cheng, Sophia Faculty Advisor: Robin Bates

Living, Working, and Playing: Asian American Environmental Justice Before the Movement

In 1991, over one thousand environmental activists of color gathered in Washington, D.C. for the First National People of Color Environmental Leadership Summit, marking the beginning of the environmental justice (EJ) movement. To formally define EJ, they ratified the seventeen Principles of Environmental Justice, highlighting the unequal environmental burdens that low-income communities of color face and redefining the environment itself to include not only the natural wilderness, but also spaces where humans live, work, and play. Despite the movement's increased inclusivity and reach since 1991, scholars and the general public have overlooked Asian American EJ activism because of the pervasive model minority myth, which casts Asians as deferential, apolitical machines alienated from nature. To counter this misconception, my paper examines three cases of Asian American organizing that contain EJ characteristics but lack scholarly analysis of this dimension because they occurred prior to the movement. Using the seventeen Principles, I analyze San Francisco Chinatown's bubonic plague outbreak in 1900, Hawai'i's sugar strike in 1946, and San Francisco Manilatown's International Hotel movement in the 1970s. I argue that, throughout their long history of organizing around issues like civil rights, labor rights, and housing, Asian Americans were fighting for environmental justice long before the movement began. My paper demonstrates it is possible to apply the modern EJ framework to pre-movement history, and I encourage historians to reexamine their areas of study accordingly. Ultimately, I aim to dispel stereotypes about Asian American apoliticalness and uncover a hidden thread of Asian American history.

Presentation Format: Oral
Time: 12:45-2:15PM
Room: Armadillo Room

Chi, Joshua Faculty Advisor: Daniel Dombeck

Isolating the Utility of Planning Against a Dynamic Threat in Naturalistic Decision-Making

Traditionally, research on choice-based behavior has relied on simple, static experimental setups that lack ecological validity. To address this limitation, the Dombeck Lab and MacIver Lab have developed a large arena with reconfigurable obstacles and an autonomous robotic predator, allowing for the systematic manipulation of environmental structure and threat dynamics. Using high-resolution tracking of a mouse's position and velocity, this system enables detailed characterization of navigational strategies, escape behaviors, and decision-making processes during predator evasion and reward-seeking.

Previous work using this arena demonstrated that increasing environmental complexity through the addition of obstacles enhances the utility of planning; while in open, obstacle-free environments, mice tend to rely on habitual, stereotyped routes that minimize deliberation. In the present study, we ask whether planning-like behavior can emerge even in a low-complexity, open arena when a dynamic and behaviorally relevant element is introduced. Specifically, we examine whether the presence of a behaviorally-complex, mobile, interactive threat is sufficient to disrupt habitual navigation and elicit behavioral signatures consistent with planning. By isolating dynamic threat from static spatial complexity, this work aims to clarify the conditions under which planning becomes computationally advantageous in naturalistic decision-making.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 48P

Chiu, Timothy Faculty Advisor: JJ Parker

Exploring fathers' health behavior motivations to leverage healthcare engagement in the perinatal period: A qualitative study

Commonly, fathers engage in their partner's healthcare in the perinatal (e.g. obstetrics appointments) but underutilize their own healthcare and preventive care services. Thus, researching ways to better engage fathers with healthcare services by leveraging perinatal contact and fathers' perinatal-specific motivations may provide opportunities for health promotion of fathers.

Thus, semi-structured interviews with new fathers (n=30) and paired mothers (n=10) in Prentice Women's Hospital were conducted to explore perspectives on and experiences of fathers' health behaviors in the perinatal period. Interview transcripts were cleaned and thematically analyzed based on the Health Belief Model.

It was found that fathers were generally highly motivated to maintain or improve their own health in the perinatal period. Being an "active father," longevity for family's sake, and "groundedness" from new fatherhood were important motivations for engaging in positive health behaviors (i.e. exercise, healthy diet, limiting substances use). Low perceived susceptibility to chronic disease and social norms against healthcare engagement were cited as deterrents for preventive healthcare utilization. Time constraints, insurance, and difficulty navigating the healthcare system were cited as barriers to actionable healthcare engagement despite increased motivation. Both fathers and mothers mentioned that leveraging family-centered rather than individual motivations while providing ease to healthcare access (e.g. health screenings in OB settings; efficient primary care referrals) could be effective.

This study highlights the potential efficacy of the perinatal period as a leveraging period for engaging fathers with healthcare and offers insight into more broadly expanding health programming for fathers.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 57P

Choudhary, Anand Faculty Advisor: Almaz Mesghina

Effects of Independence and Interdependence Priming on Big Five Personality Trait Reporting

Personality assessments such as the Big Five are often assumed to reflect stable individual traits, yet growing evidence suggests that responses may be shaped by contextual and cultural factors, including how individuals construe themselves in relation to others. This study examined whether priming independence versus interdependence influences personality trait reporting, and whether individual differences in self-construal predict variation in responses. Participants (N = 117) first completed baseline Big Five (BFI-2) and self-construal measures. In a second session, participants were randomly assigned to an independence or interdependence priming task before completing anchoring vignettes designed to reduce reference group bias in personality judgments. The priming manipulation did not produce significant group-level differences in vignette-adjusted personality ratings. As a result, multiple regression analyses were conducted to examine whether individual differences in self-construal predicted trait reporting while controlling for baseline personality. Across all five traits, baseline Big Five scores were strong and consistent predictors of vignette-adjusted ratings (all p & .001). However, neither interdependence nor independence scores significantly predicted trait ratings for any domain (all p > .13). These findings suggest that personality reporting, even when adjusted for reference group bias, is largely stable and not easily influenced by brief contextual primes or individual differences in self-construal. This has implications for theories of cultural variation in personality, highlighting potential limits on the malleability of self-reported traits in experimental contexts.

Presentation Format: Oral
Time: 10:45AM-12:15PM
Room: Rock Room

Chuang, Una Faculty Advisor: John Rogers

Fingertip-Sized Wireless Button Sensors for Health Monitoring in Neonatal and Pediatric Patients

The button sensors project aims to develop fingertip-sized devices for wirelessly monitoring vital signs of preterm patients in the Neonatal Intensive Care Unit (NICU). Three discrete units-ECG/EEG (chest/head), PPG (head) , and mechano-acoustic (peripheral)-work in tandem to provide comprehensive measurements of key vitals: heart rate, respiration, blood oxygen level, EEG, heart rate variability, sleep, motion, and temperature. Each unit includes electronic components encapsulated by medical-grade silicone or epoxy, which protects sensitive neonatal skin from abrasion and ensures longevity of the device. One significant technical achievement of the project was successfully miniaturizing photoplethysmography (PPG) for the head unit, enabling measurement of peripheral oxygen saturation with increased space optimization and power-efficiency. The sensors are supported by two power supply options-wireless power transfer technology or a miniature magnet-connected battery-which can be easily interchanged to reduce data dropouts. Data streams continuously and wirelessly to a smartphone app and uploads to the cloud, which allows clinicians to monitor changes in real time.

By implementing wireless communication, this design overcomes the limitations of conventional wired systems with rigid interfaces, protecting neonates from further skin injuries and vascular complications. Ultimately, by offering a fully wireless, ultra-miniaturized device and an intuitive real-time monitoring platform, this integrated system represents a significant advancement in enabling parent kangaroo care, streamlining clinical workflows, allowing unobstructed movement, and delivering accurate, minimally intrusive monitoring in even the most sensitive patients.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 15P

Constante, Amber Faculty Advisor: William Leonard

Altered Foodways: The Impact of Colonialism on PCOS Prevalence in the Andes

Polycystic ovarian syndrome (PCOS) is a reproductive and metabolic disorder caused by both genetic and dietary factors. The prevalence of PCOS within a population is particularly high among the indigenous populations of the global south, compared to rates in European populations. This study examines the prevalence of PCOS among indigenous and mestizo Andean populations and considers the impact of colonialism and changing foodways on the emergence of this condition. Following PRISMA guidelines, a systematic literature review compiled academic journal articles on PCOS rates, foodways, and Spanish colonial policy in the indigenous Andes from the 16th century to 20th century. The implementation of a tribute economy and debt peonage system disrupted indigenous agricultural practices and access to domestic foods, resulting in changes in dietary patterns. Additionally, increase in trade from lowlands to highlands resulted in increased availability of non-traditional foodstuffs, creating social pressure to adopt a non-traditional diet for social mobility. Both of these developments resulted in increased reliance on imported and processed foods over traditional, domestic foods that continued into the 20th century. Comparative evidence suggests that shifting from a subsistence diet to one high in simple carbohydrates and fats resulted in body composition and metabolic health that promoted the emergence of PCOS. Understanding the effect of colonialism on contemporary agrarian practices and foodways unveils underlying systems that raise PCOS prevalence within this population, allowing them to be targeted for treatment.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 42A

Cowan, Abigail Faculty Advisor: Peter Slevin

Internal Fact-Checking and the Crisis of Trust in News Mediaa

This thesis examines the role of fact-checking in addressing the crisis of trust in American news media, asking whether current practices meaningfully improve public understanding or credibility. While fact-checking has become a prominent journalistic tool, this project argues that its dominant form-external, post-publication verification of political claims-faces structural limitations that prevent it from achieving its intended audience-facing goals. Drawing on interviews with leading practitioners, including Glenn Kessler of *The Washington Post*, Kimberly Ammiano of the *CBS Evening News* research team, and Zach Helfand of *The New Yorker*, as well as historical and case analysis and research in media psychology, the thesis compares external fact-checking with internal, pre-publication verification processes. It finds a disconnect between the "imagined" effects of external fact-checking, as described by journalists, and its "real" effects on audiences, which are shaped by factors such as selective exposure, motivated reasoning, and declining institutional trust. While fact-checking has at times imposed reputational costs on political actors, evidence suggests it rarely produces lasting changes in public beliefs. The project argues that strengthening internal fact-checking within original reporting-rather than relying primarily on reactive, standalone corrections-offers a more effective way to ensure accuracy reaches audiences at the point of consumption. By reframing fact-checking as a proactive rather than corrective practice, this research offers a potential solution in the consolidation of fact-checking efforts that prioritizes internal fact-checking, and contributes to ongoing debates about journalistic responsibility, media credibility, and the future of truth-seeking in a fragmented information environment.

Presentation Format: Oral
Time: 12:45-2:15PM
Room: Arch Room

Crew, Hazel Faculty Advisor: Louise Egerton-Warburton

Using arbuscular mycorrhizal fungi (AMF) as indicators of belowground responses to urban river corridor restoration in Chicago

Urban prairie restorations are increasingly used to recover ecosystem health and services in river corridors. However, belowground responses to restoration, particularly by soil arbuscular mycorrhizal fungi (AMF), are not well characterized in these systems. AMF form symbiotic relationships with plant roots, enhancing nutrient and water uptake, particularly nitrogen (N) and phosphorus (P), in exchange for carbon (C). Their external hyphal networks contribute to nutrient cycling, soil aggregation, and improved soil structure, supporting plant establishment in restored ecosystems. Because of these functions, AMF abundance can serve as a useful benchmark for evaluating restoration progress. In this study, we investigated the relationship between AMF hyphal abundance, soil properties and restoration status in three sites that comprised a restoration gradient along the Chicago River corridor (unrestored control; early and late-stage restoration). While this gradient allows for initial comparisons, the limited number of sites constrains broader generalization. Results to date suggest that differences in soil properties across sites, rather than distance from the river, are the primary drivers of variation along the restoration gradient. Notably, the late-stage restoration site showed lower C and higher P levels relative to unrestored and early-restoration sites. Consistent with our expectations, AMF abundance was highest in the restored site and lowest in the unrestored site. This pattern aligns with lower nutrient availability in restored soils, where plants increasingly rely on AMF to support nutrient acquisition. These results highlight the importance of belowground recovery during restoration and reinforce the value of incorporating AMF abundance into monitoring frameworks to better assess ecosystem recovery and guide future restoration practices.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 28P

Crowell, Alexander Faculty Advisor: Alfred George

Altered Kinetics and Voltage Dependence of Activation in a Pathogenic KCNH1 Potassium Channel Variant

Genetic variants in neuronal potassium channel genes are associated with epilepsy and neurodevelopmental disorders. Variants in *KCNH1*, which encodes the voltage-gated potassium channel KV10.1, cause two closely related neurodevelopmental disorders: Temple-Baraitser syndrome (TBS) and Zimmermann-Laband Syndrome (ZLS). These disorders affect young children with epilepsy, developmental delay, intellectual disability, and other non-neuronal features. Despite this clinical association, the mechanisms by which disease-causing variants alter channel function remain poorly understood. We investigated the functional effects of R357Q, the most recurrent pathogenic *KCNH1* variant. Human embryonic kidney (HEK293) cells were transiently transfected with R357Q or wild-type *KCNH1* plasmid, and potassium currents were measured using whole-cell voltage clamp electrophysiology. This technique allows precise control of membrane potential while recording ion channel activity. Cells expressing the R357Q variant exhibited markedly faster channel opening (activation kinetics) and altered dependence on voltage. We also investigated the hypothesis that strong pre-pulse hyperpolarization preceding channel activation slows kinetics in R357Q. We examined this effect using two alternatively spliced KCNH1 isoforms (isoform 1, isoform 2). Hyperpolarized pre-pulses significantly slowed activation of the R357Q variant in both channel isoforms and produced different kinetics across both wildtype isoforms. Notably, isoform 2 contains a short deletion in the extracellular S3-S4 linker region, suggesting that this region may influence gating properties. These findings suggest that altered voltage sensitivity and gating behavior contribute to channel dysfunction in TBS and ZLS. Understanding how specific mutations disrupt potassium channel function provides mechanistic insight into epilepsy pathogenesis and may inform future therapeutic strategies targeting channel gating dynamics.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 88P

Dale, Aleksandar Faculty Advisor: Heather Pinkett

Structural and Functional Insights into ProU Regulation: Investigating CBS-Mediated Inhibition in Yersinia pestis

Pathogens often encounter sudden environmental changes as they move between hosts. Changes in salt concentrations can cause osmotic shock, leading to a rapid influx or efflux of water from the pathogen to balance the difference in salt, killing it. Bacteria can protect themselves against osmotic shock by importing osmoprotectant molecules, often by utilizing ATP-binding cassette (ABC) transporters. These transporters hydrolyze ATP to move substrates against a gradient into or out of the cell. My project focuses on the osmoprotectant ABC importer ProU found in the pathogen *Yersinia pestis*, the causative agent of the Black Plague. The ProU importer imports the osmoprotectant molecule glycine betaine to stabilize intracellular salt and charge concentrations, thereby preventing water loss from the bacterium. ProU is believed to be regulated by the cystathionine-v-synthase (CBS) regulatory domain, based on research on the related transporter OpuA. The CBS domain in OpuA inhibits transporter function upon binding cyclic-di-AMP, a known stress response molecule and secondary messenger. To uncover the regulatory mechanism of the ProU transporter, the ProU CBS domain was separated from ProU and crystallized. Two apo structures of the ProU CBS domain were determined by X-ray crystallography, revealing a dramatically different orientation compared to the CBS domain of OpuA when binding cyclic-di-AMP. The CBS structure can help provide vital information on how ProU is regulated and offer a more targeted approach to treating pathogens amid increased pathogen antibiotic resistance, as many pathogens rely on osmoprotectant transporters for their infectivity.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 12P

Danielyan, Meri Faculty Advisor: Ben Gorvine

Psychological States and Responses to Constructive Friendship Criticism: Comparing Self-Affirmation, Felt Security, and Ego-Threat

Constructive criticism (CC) can promote behavioral improvement and strengthen relationships, yet recipients often respond defensively to evaluative feedback. Prior research suggests that psychological states such as self-affirmation, felt security, and ego-threat impact openness to criticism, particularly in threatening evaluative contexts. However, it remains unclear whether these same states shape responses to constructive criticism, which combines negative evaluation with benevolent intent. The present study examined whether momentary psychological states that buffer against or induce threat influence responses to constructive criticism within friendships. Participants (N = 204) were randomly assigned to a self-affirmation, felt-security, ego-threat, or control condition before reading two standardized scenarios depicting constructive criticism from a close friend. Participants then completed measures assessing perceived hostility, trust in the feedback and feedback-giver, receptiveness to feedback, and adaptive and maladaptive relational responses. Contrary to predictions, the experimental conditions did not differ significantly across any outcome variables. Across conditions, participants generally perceived the criticism as highly non-hostile and responded with high levels of trust, receptiveness, and adaptive responding. Perceiving the criticism as non-hostile was associated with greater trust in the feedback, which in turn was associated with greater receptiveness and more adaptive relational responding. These findings suggest that when criticism is delivered constructively, the framing and perceived intent of the feedback may play a more important role in shaping responses than recipients' temporary psychological states. More broadly, the results raise the possibility that constructive criticism may not evoke sufficient threat for transient self-threat and threat-buffering states to substantially influence evaluative processing.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 13A

Debono, Ruth Faculty Advisor: Justin Mann

Poetic Medicine: Black Feminist Poetics of Cancer and HIV/AIDS

Public health research indicates disparities in health outcomes along racialized and gendered lines. This includes health inequities for Black femmes who are faced with oppression by virtue of both race and gender. The subjective experiences of Black femmes are lost to the privileging of biomedical knowledge and a reliance on medical systems built on the exploitation of Black bodies. This project analyzes Black women's poetry and creative nonfiction writing about health and wellbeing to examine the experiences of illness not legible to medical systems, drawing comparisons between cancer and HIV/AIDS. I argue that poetry is a tool of personal and collective healing, a space of worldbuilding and an opportunity to illuminate and make communal internal experiences of precarity, and oppression that stem from the individual and individuating experience of illness. Yet, all illnesses are not the same, and so I highlight differences between poetry about cancer and HIV/AIDS. The poetics of cancer often originates from a confrontation with the self through the body, while poetry about HIV/AIDS confronts self-definition and social meaning. Working towards an interrogation of the relationships between normativity, medical systems, and understandings of the self, "Poetic Medicine" shows that personal writing is a tool of self and communal study. Throughout, I elucidate aspects of wellbeing that medical systems fail to palliate (and that they often instead harm), especially for Black femme patients living with disease. Ultimately, I hope to broaden public health practice by infusing scientific and medical processes with the suff of poetry.

Presentation Format: Oral
Time: 12:45-2:15PM
Room: Rock Room

Delapenha, Lauren

Investigating the Effect of Linguistic Labeling on Object Categorization and Memory in 7 and 12-Month-Olds: Evidence from a "Peek-a-Boo" Occlusion Task

Current research suggests that infants can develop language categories as early as six months of age through verbal communication in their environments. However, it is unclear how language labels affect an infant's memory of object categories in more dynamic situations. It is also unknown how persistent language categories are with aging, especially in the newborn-infant stage of development. Therefore, two studies in 12-month and 7-month olds were conducted to investigate if naming objects with distinct language labels with a change-detection task positively impacted their object categorization skills. Stuffed animals were shown with verbal pseudowords (e.g. boff or dax) in the learning phase. In two test trials, a stuffed animal from the learning phase was shown and then covered with a blocker. After removing the blocker, the stuffed animal was either the same as before the blocker was placed or different. The infants' level of surprise, and therefore individuation of the animal as an object category, was determined from their average looking time after the blocker was removed. As predicted, there was a significant increase in average looking time in the distinct naming Within Change condition in 12-month-olds. This indicates that strong object individuation occurred, and that it persisted over the occlusion task. Preliminary findings show that more participants are needed to show statistical significance in the 7-month-old study. We believe that linguistically naming objects can improve an infant's object categorization and memory skills, which may influence how caregivers interact with children to improve their language and cognitive development.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 38A

Desmaras, Juan Faculty Advisor: Matthew Glucksberg

Development of a Fluorescence-Based Endomicroscope System for Quantitative Assessment of Pediatric Aspiration

Aspiration, the accidental inhalation of foreign material into the lungs, is a severe, yet critically understudied, clinical danger for pediatric and neurologically impaired patients that can lead to life-threatening respiratory complications. Current diagnostic methods, such as videofluoroscopic swallow studies, only provide indirect, qualitative assessments. They lack the sensitivity to directly visualize or measure aspirated material at the microscopic level. To address this gap, we present a proof-of-concept fluorescence endomicroscope designed to directly detect and quantify aspirated fluorescent particles (microspheres) within lung tissue. Our system utilizes a specialized optical setup, featuring a broadband LED, wavelength filters, and a high-resolution sensor, to illuminate the tissue and capture the fluorescence of these microscopic particles, acting as a proxy for aspiration severity. We successfully demonstrated that this system reliably visualizes microspheres in lung tissue samples. By applying an image analysis algorithm, we quantified particle density, providing an objective metric of the aspiration burden. Experimental results showed that while thicker, physiologically relevant tissue reduced image contrast due to light scattering, the particles remained clearly identifiable. System resolution easily distinguished individual 10-micrometer particles, and sampling analysis confirmed that this performance is limited by optical physics rather than sensor constraints. These findings establish fluorescence endomicroscopy as a viable method for directly visualizing and measuring microscale aspiration. This platform introduces a vital shift from subjective observation to precise, quantitative measurement, offering a powerful new framework for early diagnosis, severity grading, targeted treatment monitoring in highly vulnerable patient populations, and general aspiration study.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 55A

Dommert, Nick Faculty Advisor: Rosemary Braun

How Bacteria Tell Time: Mathematical Models of the Cyanobacterial Circadian Clock

The cyanobacterial circadian clock is a biochemical process that enables cyanobacteria to predict changes in their environment. Timekeeping of the cyanobacterial clock is accomplished through routine, ATP-dependent cycling between the phosphorylation states of the clock protein KaiC and its binding with mediator proteins KaiA and KaiB. The cyanobacterial clock has been traditionally modeled using ordinary differential equations (ODEs) that produce predictable and periodic trajectories of protein concentration over time. In nature, however, this biochemical process is subject to biological noise, including random fluctuations in environmental conditions, rate constants, and protein concentrations. This project develops "noisy" stochastic models of the cyanobacterial clock to test the validity of the ODE framework under biologically relevant noise conditions and to better analyze sensitivities of this biochemical process. The noise tolerance of each parameter was determined by identifying statistical variation between the trajectory of stochastic simulations and the period of the ODE model. Statistical analysis of the stochastic simulations revealed that the model is highly tolerant to random fluctuation in ATP and the concentration of unphosphorylated KaiC, consistent with the fact that these parameters are most susceptible to noise in a natural environment. Analysis also revealed that the model is sensitive to minor fluctuation in the concentrations of KaiA and sequestered KaiC, a new insight gained through stochastic modeling. These results exhibit how stochastic modeling can validate and refine mathematical models of biochemical processes and reveal underlying sensitivities of a biochemical system to drive future experimentation.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 29A

Dougherty, Ryan Faculty Advisor: Milan Mrksich

High-Throughput Discovery of Trispecific Antibodies for Cancer Therapies Using MegaMolecules.

Immunotherapy is transforming oncology by leveraging antibodies to target specific cancer cell surface receptors, thereby reducing the off-target side effects associated with nonspecific treatments like chemotherapy. Once bound, these antibodies eliminate tumors by blocking growth signals, recruiting immune cells, or delivering toxic payloads. Multispecific antibodies—which bind to multiple receptors simultaneously—offer even greater potential by disrupting several signaling pathways at once, increasing binding strength through enhanced avidity, and increasing selectivity for cancer cells. Modern drug discovery is built on high-throughput screening of large and structurally diverse libraries of candidates to identify those with activity and narrow down millions of possibilities to a few promising leads. However, synthesis of large libraries of trispecific antibodies is a significant challenge. MegaMolecules provide a modular platform that enables rapid, precise assembly of multispecific antibodies. Rather than designing whole, rigid IgG scaffolds, megamolecules utilize high-affinity, covalent enzyme-inhibitor reactions to prepare multivalent antibodies with unprecedented scale and stability. In this project, we utilized the megamolecule architecture to construct a diverse library of 300 trispecific antibodies targeting numerous commonly expressed breast cancer epitopes from a base library of 23 fusion proteins. By successfully synthesizing and characterizing this library, this project demonstrates the unique efficacy of the megamolecule platform to address the prevailing limitations of immunotherapeutic design, establishing a scalable foundation for the discovery of the next generation of multispecific cancer therapies.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 04P

Douglas, Julia Faculty Advisor: Rebecca Ewert

Clean Conscience: Exploring College Students' Enactment of Environmental Beliefs Through Domestic Labor

In the face of climate change and the imminent threat it poses to global human welfare, what explains disparities in individuals' level of concern about environmental issues and subsequent willingness to act on this concern? The feminization of environmental action has been well-documented since the start of the 20th century and is particularly evident in studies of middle-aged adults' household labor dynamics. However, this extensive literature largely overlooks college students, who constitute a population of interest as the next generation of environmental decision-makers. Through semi-structured, in-depth interviews with 21 students at Northwestern University, I endeavored to answer the question of how gender influences the development and enactment of college students' environmental beliefs. I tested three hypotheses regarding feminized environmental action from the aforementioned literature: the maternal responsibility, knowledge support, and economic salience hypotheses. The distinctive characteristics of private, four-year colleges like Northwestern challenge the norms inherent to each hypothesis; such institutions are a site where students live independently and create less-formalized relationships, gain knowledge and analytical skills regardless of their gender, and are encouraged to cultivate an economic mindset. My findings indicate that, despite this unique context, gender socialization remains strongly influential to students' environmental beliefs and behaviors, as evidenced by participants' tendency to align themselves with mainstream gender expectations through their actions as well as the language they used when justifying them. This indicates that shifting norms (gendered and otherwise) could potentially bring about widespread behavioral change-an implication warranting further research and attention from sociologists and policymakers alike.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 62P

Du, Charlie Faculty Advisor: Bahador Bahmani

Probabilistic Neural Operator for Dynamical Systems Under Uncertainty

In everyday life, we encounter many systems that evolve over time, known as dynamical systems. Examples include vibrations in structures, weather patterns, financial markets, and the spread of diseases. Scientists and engineers have studied these systems for many years, but traditional models often struggle to take full advantage of the large amounts of data now available. Such models also rarely tell us how reliable their predictions are. Our research aims to develop a new data-driven modeling approach using neural operators, a modern machine learning technique that learns how entire systems change over time. Unlike standard models, our approach not only predicts future behavior based on past observations but also estimates how confident those predictions are. To achieve this, we incorporate ideas from statistics, such as quantile regression and Bayesian methods, to explicitly account for uncertainty. By doing so, the model can capture both errors in the data (from sensors that record noisy, imperfect measurements) and limitations in the model itself. As a result, our method produces predictions along with confidence intervals, giving users a clearer understanding of possible outcomes and risks. We compare our approach with existing methods that estimate uncertainty. Preliminary results suggest that our approach achieves similar or better predictive performance, while providing more trustworthy confidence intervals with a simpler and more efficient framework. Such reliability is especially important for decision-making in areas like healthcare, climate science, and finance, where uncertainty plays a critical role.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 24P

Du, Kaitlyn Faculty Advisor: Yevgenia Kozorovitskiy

Knockdown of Clock in Ventral Tegmental Area Dopamine Neurons Influences Bipolar-Like Behaviors and Sleep Deprivation Responses in Mice

Bipolar disorder (BD) affects approximately 1 in 150 adults worldwide and is a leading cause of disability, typically emerging in adolescence. BD is characterized by dramatic shifts between manic and depressive mood states, linked to increased and decreased activity of dopamine (DA) neurons, respectively. Sleep and circadian rhythm disruption are also strongly associated with mood cycling and BD. The circadian transcription factor *Clock* has been implicated in BD by GWAS and modulates DA function in the ventral tegmental area (VTA), a major hub of reward-related DA signaling. Previous work shows that *Clock* disruption in mice can produce mania-like phenotypes and alter depression-related behavior. However, effects of knocking down *Clock* in VTA DA neurons are poorly understood, even less so when coupled with acute sleep deprivation. Notably, existing studies with *Clock* manipulations offer contradictory results: global knockdown (KD) resulted in an increased mania-like phenotype characterized by reduced depression-like behavior, but VTA-specific knockdown led to increased depression-like behavior. In preliminary behavioral testing, neonatal VTA *Clock* KD mice showed reduced anxiety-like behavior relative to controls, consistent with an increased mania-like phenotype. After sleep deprivation, KD mice also exhibited lower anxiety levels than controls. At baseline, KD mice displayed more depression-like behaviors. Together, these results suggest that early postnatal reduction of *Clock* in VTA DA neurons can simultaneously reduce anxiety- and increase depression-related behaviors, and modulate behavioral responses to sleep deprivation. Manipulating *Clock* and sleep at different points in development may help elucidate developmental mechanisms linking circadian dysregulation and DA circuitry to BD.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 35P

England, Olivia Faculty Advisor: Robin Bates

The King and Queen of Rock: Little Richard, Religion, and the Stipulations of Black Queer Fame

Little Richard, born Richard Wayne Penniman in 1932, is widely recognized as a key founder of rock and roll. However, the intersections of his Blackness, queerness, and religiosity have received limited attention from scholars. This paper examines why Penniman continually oscillated between embracing and rejecting his queerness across his seventy-year career, arguing that his global stardom amplified criticism from Black and white populations, exacerbating his lifelong struggle with sexuality and faith. Using his 1984 authorized biography, magazine profiles, and television interviews, along with modern-day scholarship on Black queerness, race theory, and music history, the paper traces Penniman's public positions on his queerness from the 1950s until his passing in 2020. Each withdrawal into intensified religiosity is a clear response from Penniman to external pressures. Such stresses included his father's violent rejection, the AIDS epidemic, and declining health near the end of his life. When he returned to openness, it was a display of irrepressible queerness, with the stage as the place where fashion and extravagance could become a language for his identity. The research concludes that Penniman's life is not a narrative of contradictions but rather of the consequences of Black queer people being stuck in the interlocking oppressive systems of white supremacy, homophobia, and the institution of religion. Little Richard's story illuminates dynamics that still shape Black queer communities today, demonstrating that harmful theological and cultural forces remain active and influential.

Presentation Format: Oral
Time: 12:45-2:15PM
Room: Armadillo Room

Espinosa, Kimberly Faculty Advisor: Calvin Liang

Decoding Decisions: Folk Theories of Urgent Care Triage Algorithms from Koreatown (K-Town), Los Angeles

Growing efforts to address algorithmic bias in healthcare have considered increasing data collection methods and model transparency. Yet, this often prioritizes a statistical definition of bias, failing to account for the compounded biases systematic barriers present, particularly for populations who have historically been neglected by the healthcare system. In healthcare, the process in which patients are evaluated and prioritized to receive care, known as triage, is an algorithm which may look different across clinical settings. Understanding how individuals form folk theories (i.e., intuitive and informal theories to explain outcomes of processes and which influence reactions and behavior) about the triage process is significant as it reveals how perceptions of triage algorithms and their potential bias shape access to healthcare through future decision-making.

In this project, based in the neighborhood of Koreatown, Los Angeles, 10 interviews with low-income community members from different family generations and an autoethnography by the author, revealed the formation of specific folk theories when seeking urgent care for themselves and/or for a family member. Folk theories included self-presentation of symptoms, awareness of intersectional identities, and perceptions of resource constraints. Additionally, reflections on personal experiences undergoing triage revealed considerations patients make about leaving the waiting room and in some cases, choosing alternative healthcare provider options. This research underscores the importance of folk theories, ones which do not need to be technically accurate, but which nonetheless play a role in informing patients' and their families' decisions about when, how, and whether to seek future care.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 46P

Estep, Kaleah Faculty Advisor: Zachary Nissen

Post-Colonial Community Perspectives on Cultural Heritage and Archaeology in Corozal, Belize

How people engage with and understand cultural heritage shapes how it is preserved within communities and how it is understood by broader academic audiences. Heritage is typically described as a tangible or intangible property inherited; however, due to the diverse cultural identities within a Post-Colonial context, Belizean cultural heritage challenges top-down models of heritage management and raises new questions about who benefits from defining and curating heritage sites. This paper asks how community members in Corozal, Belize, perceive and engage with cultural heritage in a Post-Colonial context. Using photos of local heritage sites to facilitate discussions and gather community perspectives on heritage and archaeology fosters an inclusive approach to these topics. A unique understanding of Corozal's current state of cultural heritage and archaeology is created by bringing diverse community voices into conversation with existing literature. Many of the issues discussed by community leaders and scholars centered on disseminating information and on the relationships among the community, community leaders, and archaeologists. Local claims are supported by community discussions, revealing complex relationships between people's engagement with sites and their knowledge of the landscapes' long-term history. Therefore, the community's voices provide a better understanding of how to frame archaeology and heritage efforts in the area.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 67P

Fein, Brooke Faculty Advisor: Lilah Shapiro

Antisemitism and Anti-Zionism: A Case Study on Jewish Student perspectives at Northwestern University

This thesis explores how Jewish undergraduate students at Northwestern University understand and navigate the relationship between anti-Zionism and antisemitism. Drawing on 15 semi-structured interviews, I examine how students interpret political discourse, experience campus climate, and make sense of the boundary between legitimate political critique and identity-based harm. This research is important at Northwestern because the university has become a highly visible site of these tensions. Over the past few years, encampments, administrative decisions, and national political attention have made Northwestern an active case study of how these issues unfold. That visibility means students are not only forming their own perspectives, but also reacting to media narratives, federal interventions, and institutional responses that often amplify or distort what they actually experience on the ground. Findings show that students hold a range of perspectives shaped by upbringing, education, and personal connections to Israel. Some maintain a clear distinction between anti-Zionism and antisemitism, while others see them as frequently overlapping, especially when rhetoric generalizes or targets Jewish identity. Antisemitism is often described as indirect, showing up less through explicit incidents and more through discomfort, self-censorship, and uncertainty in campus spaces. Students also express frustration with campus discourse that treats Israel solely as political, overlooking its personal and cultural significance. This disconnect contributes to feeling misunderstood and limits opportunities for nuanced dialogue. By centering Jewish student perspectives when Northwestern is under intense scrutiny, this study highlights how national debates are actually lived on a single campus, and why more thoughtful conversations are needed.

Presentation Format: Oral
Time: 10:45AM-12:15PM
Room: Arch Room

Filipek, Anna Faculty Advisor: Lilian Li

Cardiac Synchrony in Co-Rumination Dyads: Using EKG to Study Interpersonal Dysfunction in Depression

Interpersonal dysfunction, or a reduced capacity to build and maintain social connections, is a significant component of depression. Depressive symptoms, such as withdrawal and disconnection, can be maladaptive, leading to cycles of distress and isolation that further exacerbate these symptoms. One identified marker of interpersonal dysfunction is co-rumination, defined as the excessive discussion of problems and negative emotions within close relationships. The NEAR Lab is currently analyzing data from a co-rumination study, in which friend dyads were recruited to engage in observed naturalistic conversation; in each pair, at least one individual reported difficulties with self-confidence. During these conversations, behavioral, linguistic, and physiological data were collected. The present analysis focuses on the physiological data collected, specifically the cardiac functioning within dyads measured through electrocardiogram (EKG) recordings. Heart rate, heart rate variability (HRV), and measures of cardiac synchrony will be assessed to determine the extent to which co-rumination can be detected physiologically. We hypothesize that dyads will exhibit increased cardiac synchrony, reflected by increased heart rate and HRV alignment. Together, these patterns may suggest a physiological component underlying interpersonal dysfunction and co-rumination. Findings will contribute to a growing understanding of how depression is shaped by social interaction and how individual mental illness symptoms may influence members of one's social network.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 39A

Foong, Ethan

HANDFUL: Dexterous Multi-Step Robot Manipulation with Resource Awareness

Dexterous robot hands offer rich opportunities for multi-step manipulation: for example, grasping an object and then pushing another object, pulling a drawer, or pressing a button. However, most prior work in dexterous manipulation focuses on single, isolated tasks. In contrast, our key insight is that these multi-step manipulation tasks require *resource-aware* grasps where the robot must conserve fingers for later actions rather than simply optimizing an initial whole-hand grasp in isolation. We introduce HANDFUL, a learning framework that treats finger usage as a limited resource and rewards the robot for exploring one- and two-finger grasps that leave fingers and in-hand space available for downstream tasks. These grasps are then efficiently selected and trained for downstream tasks via curriculum-based policy learning We further propose HANDFUL-Bench, a simulation benchmark that introduces sequential dexterous manipulation tasks across multiple second-subtask objectives, including pushing, pulling, and pressing, under a shared grasp + second objective setup. Second objectives include: pushing another object, pressing a button, twisting a knob, pulling a drawer, and grasping a second object. Results show that resource-aware grasp planning improves both success rate and robustness compared to a baseline that greedily optimizes the initial grasp before attempting the second subtasks. We additionally validate our approach on a real dexterous hand. Together, this work establishes resource-aware grasp planning as a key principle for multi-step dexterous manipulation. Supplementary material is available on our website: [handful-dex.github.io] (http://handful-dex.github.io).

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 25P

Fox, Lula Faculty Advisor: Megan Baker

Inafa'maolek Archaeologies: CHamoru heritage resurgence amidst U.S. militarization on Guahan (Guam)

The rise of U.S. militarization on Guam and the passage of federal laws governing cultural resource management and environmental protection have rapidly increased archaeological surveying over the last few decades. This research investigates the role of CHamorus, the Indigenous people of Guam and the Mariana Islands, who work as archaeologists and historic preservationists to steward Indigenous cultural heritage amidst military build-up. These applied anthropologists play a crucial role in protecting CHamoru heritage, working with archaeological materials spanning up to 4,000 years old that connect present day CHamorus to ancient ancestors. Using ethnographic methods, including participant observation and semi-structured interviews, this research examines how CHamoru archaeologists and historic preservationists are incorporating CHamoru knowledge systems throughout their work on their ancestral homelands, focusing on the newly constructed Marine Corps Camp Blaz base. This research maps the current institutional landscape of heritage management on Guam, identifying some of the challenges of CHamoru archaeologists and historic preservationists within their work. For these practitioners, archaeology has been utilized as a site of resistance towards colonial knowledge production on the CHamoru people. Drawing on Indigenous and activist archaeological literature, this research explores how CHamoru practitioners are subsequently navigating Indigenous identity, community, and place-making as they work on specific heritage sites. By centering CHamoru relationships to land and prioritizing community access, CHamoru practitioners in the field are utilizing archaeology as a tool to foster sovereignty through decolonial heritage education, while simultaneously mobilizing to protect Indigenous cultural landscapes amidst violent militarization.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 66P

Freeland, Amelia Faculty Advisor: Magdalena Osburn

Decoding Cave Wall Patterns to Search for Life Elsewhere

Caves on Mars and the Moon suggest that similar environments on Earth can help scientists search for signs of life beyond our planet. One promising avenue of research is focused on "biovermiculations" - worm-like patterns of microbe-rich material that form on cave walls. Most studies of these features come from sulfur-rich caves, where microbes drive their formation through sulfur-based chemistry. This study examines biovermiculations in Mammoth Cave, Kentucky, a limestone cave with different environmental conditions, to better understand how they form and whether they are truly biological in origin. Our hypothesis is that biovermiculations in non-sulfur caves are shaped by active microbial processes. We combined microscopy, spatial mapping, and DNA analysis techniques to analyze these patterns. In Mammoth Cave, biovermiculations appear across five distinct morphologies and are made from materials like soot, mud, and cricket guano. Their form is influenced more by the type and behavior of available material than by nearby patterns. Microscopy revealed motile microorganisms that could actively rearrange material on cave surfaces. We also found that these features have developed over at least the past century, growing into 19-20th century soot deposits but not significantly affecting newer ones (post-1970). Genetic analyses show that the microbial communities within biovermiculations are distinct from other sites and share similar compositions across samples. Many of these microbes are associated with sulfur and iron cycling, as well as methane use, suggesting diverse metabolic activity. These findings help refine how scientists identify signs of life in cave-like environments on other planets.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 47A

Fregene, Paula Faculty Advisor: Yiji Zhang

My Tutor Bot: Integrating Large Language Models into Intelligent Tutoring Systems for Fundamental Programming

This research explores how integrating large language models (LLMs) into intelligent tutoring systems (ITS) can enhance student learning outcomes in fundamental programming courses. Building on findings that combining computer-based and human tutoring is highly effective, this study examines how LLMs can replicate the benefits of human tutors, such as personalized feedback and scaffolding, while offering scalable, adaptive support. To investigate this, I implemented "My Tutor Bot" within Northwestern's Comp Sci 211 across Winter and Spring Quarters 2026, creating a trustworthy, instructor-controlled environment by training the LLM exclusively on Professor Zhang and Hummel's course materials. If a student has any questions beyond the scope of the provided course materials, My Tutor Bot informs them that it cannot respond. My Tutor Bot is also FERPA-compliant and built to protect students' privacy. This ensures a safe space for students to explore content and receive 24/7 adaptive scaffolding tailored to the curriculum. By analyzing chat logs, interviews, and surveys, the research evaluates the impact of targeted AI tutoring on conceptual understanding, student engagement, and educational equity. The results aim to identify best practices for leveraging AI in academic settings and provide insights into how LLM-driven ITS can close equity gaps and improve learning outcomes in STEM fields.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 26P

Fu, Katie Faculty Advisor: Ken Paller

Auditory Processing and Behavioral Responses to Sound During Sleep

Sleep has traditionally been viewed as a state of sensory disconnection, but past studies have found that the brain continues to process external stimuli across sleep stages, with sensory responses maintained during REM sleep. Prior research also found that behavioral responses to auditory lexical-decision tasks can occur during sleep, but most participants do not recall these interactions upon awakening unless they experience lucid dreams. In this study, we investigated when external sounds become incorporated into dreams and whether real-time behavioral responses can occur without conscious awareness. Participants were trained to associate five distinct sounds with specific sniffing patterns. These cues were presented during N2, N3, and REM sleep, followed by awakenings after delays of 0.5, 3, or 10 seconds to collect dream reports and assess recall. Across 78 analyzed awakenings, participants were able to volitionally respond to external stimuli during sleep through sniffing responses. Among these awakenings, 16 total incorporation trials were observed: 5 during N2 sleep, 2 during N3 sleep, and 9 during REM sleep. This suggests that REM may be the most receptive stage for integrating external stimuli into dream content. Overall, these findings show that the sleeping brain can respond to external stimuli in real time. This work has implications for applications in dream research, memory consolidation, and sleep-based interventions.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 87A

Gallistel, Avery Faculty Advisor: Sandra Waxman

Infants Can Reason About Objects Even When They Are Hidden: Evidence from Containment Events

Before infants can speak, hearing an object (e.g., my cat) named influences how they represent it-as an individual ("Misty") or a category member (CAT). These distinct representations affect infants' memory: when an object is named with a distinct name, but not a category name, infants encode that object as a unique individual and distinguish it from other objects. Here, we ask whether naming objects also has downstream consequences for infants' reasoning about the objects in dynamic events.

In two ongoing experiments, 12- and 16-month-olds (planned N=96) participate via Zoom. All infants are full-term and primarily English-exposed. In Experiment 1, infants first see videos of four plush bunnies that differ only in color, and each bunny is labeled with a distinct novel noun. Then, in the test videos, the last-named bunny (blue bunny) is hidden inside a small container and then retrieved after a pause. In the *no-change* event, the same-colored bunny is retrieved. In the *change* event, a different colored bunny (pink bunny) is retrieved. Experiment 2 is identical to Experiment 1, except the four bunnies are first labeled with the same name. We expect infants to look longer at the test event they find unexpected.

Preliminary results revealed that infants looked longer at the change than the no-change event in Experiment 1, suggesting that both 12- and 16-month-olds successfully reasoned about the individual bunnies when they had distinct names. This work extends prior work to reveal new insights about infants' early link between language and cognition.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 31P

Gibson, Sharmel Faculty Advisor: Robert Ward

Method Matters: Including Critical Race Theory in the Academic Tradition of Qualitative Meaning Making

Critical Race Theory (CRT) recognizes race as a social construct. It examines and acknowledges how race is embedded in policies, institutions, and social structures. The current social and political climate asserts that racism shouldn't be interrogated, and the lived experiences of people of color aren't mediated by race. As a result, Critical Race Theory has been under attack and delegitimized as a credible research framework. However, utilizing CRT as a framework and analytical lens in qualitative research allows for rich and diverse stories to be expressed. Specifically analyzing the case study methodology, which is a detailed examination of an event or phenomenon, this study worked to understand how CRT can be used in case studies and inform the narratives that can be communicated. To investigate, I created research outlines, conducted a literature review, and completed annotated bibliographies on CRT and its application in case studies. I discovered that using CRT in case studies isolates and highlights race to uncover the impacts of race and racism on people of color's lives, validating people of color's experiences with racism. As racism continues to affect people of color negatively, understanding how race shapes lived experience is essential for driving meaningful policy and structural reform. This research has direct implications for how race-centered studies and research involving people of color are designed and conducted. Further development of CRT within qualitative research methodologies is warranted to deepen meaning-making and expand the range of stories that academic research can share.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 02A

Globerman, Ari Faculty Advisor: John Armstrong

Cortical Interneurons in the Mouse Model of Fragile X

Autism spectrum disorder (ASD) affects how people communicate, learn, and interact with the world, and it impacts more than 1% of people globally. One of the most common inherited causes of ASD is Fragile X syndrome, which occurs when the *FMR1* gene is turned off. Researchers believe that many ASD symptoms may result from an imbalance between excitatory brain signals, which increase activity, and inhibitory signals, which calm activity down. In Fragile X syndrome, this imbalance may be caused by the loss or reduced function of inhibitory brain cells called interneurons.

This project investigates whether the death or dysfunction of these interneurons in early development contributes to this imbalance. Using a mouse model of Fragile X syndrome, we focus on the somatosensory cortex, the part of the brain that processes sensory information such as touch. Specifically, we study parvalbumin- and somatostatin-containing interneurons, which help regulate brain activity and are critical for normal circuit development.

To track these cells, we use genetically modified mice in which interneurons luminesce red under fluorescent microscopy. Brain tissue is collected at key stages of early postnatal development, sectioned, and imaged to count interneurons and inhibitory synapses across different layers of the cortex. Additional staining methods allow us to measure markers of inhibitory connections.

By comparing Fragile X mice to healthy controls, we aim to determine whether reduced interneuron survival helps explain cortical hyperexcitability in ASD. Identifying when and where these changes occur could guide future therapies designed to restore healthy brain circuit balance in autism.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 24A

Gonzalez, Lisseth

Developing a cFos Atlas to map brain activity

Light plays a critical role in regulating physiological and behavioral processes, yet the neural circuits underlying these effects remain incompletely understood. Intrinsically photosensitive retinal ganglion cells (ipRGCs) contribute to functions such as circadian photoentrainment, mood regulation, and thermoregulation, and project broadly throughout the brain, mapping their downstream signaling pathways is essential for understanding how light impacts brain function. This project aimed to investigate how ipRGCs influence brain-wide activity by developing a c-Fos-based activity atlas. C-Fos, an immediate early gene, was used in order to detect the areas of the brain that are activated upon light exposure. To accomplish this, two groups of mice were used: a control group maintained in darkness and an experimental group exposed to a brief light pulse. Following stimulation, brains were extracted, coronally sectioned, and processed using immunohistochemistry to label c-Fos. Additional markers, including NeuN and ChAT, were used to identify neuronal populations. Fluorescent imaging enabled visualization of activity patterns across brain regions. The resulting dataset provides a foundational framework for constructing a whole-brain c-Fos atlas of light-responsive activity, revealing distinct regions activated in response to ipRGC stimulation. This work establishes a reference map that can be used to compare activity patterns in future studies, including models lacking ipRGC function, to better define the role of these cells in neural circuits. Ultimately, this research advances a systems-level understanding of how light influences brain function and behavior, with broader implications for circadian biology and mood-related disorders.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 19A

Griffin, Madilynn Faculty Advisor: Vijay Mittal

When Silence Speaks: Comparing Conversational Delays in Schizophrenia and Depression

Psychosis is a mental health condition in which an individual exhibits severe impairments in the ability to discern between what is real and what is not. While psychosis is generally known for presenting as hallucinations and delusions, a growing body of literature has also identified speech abnormalities in psychotic patients, characterized by lulls in conversational turn-taking. More specifically, these individuals tend to have extended pauses before answering questions. Our research aims to extend these findings by differentiating delays in response latency between individuals with major depressive disorder (MDD) and those at clinical high-risk (CHR) for psychosis. These findings could be used to assess potential differences in symptomatology in order to better distinguish early-warning signs of psychosis from other mood disorders. Through retrospective analysis of clinical interviews, our team has categorized speech patterns to investigate whether response latency differs significantly between MDD and CHR populations. Observed differences in turn-taking delays may be indicative of distinct underlying mechanisms in social functioning and could support conversational turn-taking as a behavioral biomarker for assessing psychotic risk. This could have major implications in improving early detection of psychosis by enabling timely intervention and thereby optimizing long-term clinical outcomes.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 86A

Guallpa, Amy Faculty Advisor: Loretta Li

PRL2 and the PTEN/PI3K/AKT signaling pathways in T-ALL

T-cell Acute Lymphoblastic Leukemia (T-ALL) is an aggressive blood cancer that comprises 10-13% of leukemia diseases in children. Certain signaling pathways, such as the PI3K/PTEN/AKT/mTOR signaling pathway in these T-ALL cancer cells are switched on and promote the progression of the disease. Many of these pathways are controlled by oncogenes such as PRL2 (Phosphatase of Regenerating Liver 2), which are proteins that promote unregulated cell division and growth that are characteristic of cancer cells. This project aims to determine if PRL2 promotes cancer growth by disabling the PTEN tumor suppressor and therefore activating the PI3K/PTEN/AKT/mTOR signaling pathway. To investigate this signaling pathway, we sequenced for different conditional mouse models to observe the interactions of these proteins: PRL2 -/- (KO), PTEN WT; PRL2 -/- (KO), PTEN +/- (F/+); PRL2 +/+ (WT), PTEN WT; PRL2 +/+ (WT), PTEN +/- (F/+). By isolating and analyzing the DNA sequences offetal mouse livers, we created four different groups of mice DNA with different genetic combinations, including mice that lacked the PRL2 protein (knockout mice, KO). Creating these conditional groups of mice allows us to isolate the mechanisms of PRL2 and how it influences the spread and severity of disease. Knockout of PRL2 allows us to determine if PTEN regains function and reduces overall disease progression in mice and the mechanisms behind it. We hope our findings will help us determine a new way to prevent cancer growth through targeted therapies, leading to better survival rates and treatments for pediatric T-ALL cases.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 87P

Gupta, Rishi Faculty Advisor: Jason Brickner

How Crm1 Brings Genes to the Nuclear Edge-and the Proteins That Help It

Cells regulate gene activity not only through molecular signals but also by physically organizing DNA within the nucleus. Recent research suggests that genes positioned near structures called nuclear pores can become more active. A protein called Crm1, best known for transporting molecules out of the nucleus, also helps bring genes to these locations, acting as a docking factor that influences gene activity. However, the additional factors required for this process remain unknown. Crm1 is also frequently overactive in human cancers, but the mechanisms underlying its role in disease are not fully understood.

This project aims to identify proteins that cooperate with Crm1 in controlling gene positioning and activity. Using a yeast model system, mutations are introduced throughout the genome and compared between normal cells and cells with elevated levels of Crm1. Mutations that reduce the harmful effects of excess Crm1 are analyzed to identify candidate proteins involved in this pathway.

This work aims to uncover the network of factors that enable Crm1 to position genes at the nuclear periphery, providing new insight into how spatial organization influences gene expression. Understanding this mechanism may help explain how disruptions in nuclear organization contribute to diseases such as cancer.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 85P

Hancock, Taylor Faculty Advisor: Robin Bates

Black Players, White Schools: How integration reshaped Southern college football

The landscape of college football has long been shaped by unequal access to power, resources, and visibility, yet one sector of programs remains consistently overlooked: historically Black colleges and universities (HBCUs). This paper challenges the common narrative of integration as a straightforward success story by revealing how it reproduced new forms of inequality within the sport. Understanding this dynamic not only reframes the history and temporality of college football but also helps explain the persistent disparities in funding, exposure, and institutional power that continue to shape the game today. In my research, I argue that the integration of college football in the 1960s and 1970s, particularly in the South, did not benefit the Black community; it actually harmed it. Integration reshaped the sport in ways that pulled talent, money, and attention away from historically Black colleges and universities and concentrated them towards predominantly white institutions (PWIs). Drawing on a combination of academic sports scholarship and primary sources, including Black newspapers and collegiate publications, my research uses a chronological and comparative analysis to trace how integration unfolded across regions and institutions. By examining key case studies, including the rise of HBCU programs, the structure of segregated PWI football, and pivotal moments such as the integration of major Southern powerhouses, the paper shows how success by integrated teams prompted PWIs to recruit Black athletes for competitive advantage rather than moral progress. This shift concentrated visibility and resources within already dominant programs, weakening the institutional and cultural centrality of HBCU football.

Presentation Format: Oral
Time: 10:45AM-12:15PM
Room: Rock Room

Hirschle, Jack Faculty Advisor: Ken Paller

Novel Strategies for Assessing Awareness in the Borderlands between Sleep and Wake

Our study aims to exploit lucid dreaming, the state of being aware of dreaming while in a dream, to investigate the threshold between wake and sleep. The goal of the study is for napping participants to provide signals about their experience in response to auditory cues during sleep-onset, the period when the body transitions into sleep. Part of the challenge for participants is to retain memory of their experiences after awakening. Electrodes for monitoring sleep physiology are applied to participants' scalp and face. Prior to a 90-minute nap, participants listened to a 3-beep cue while cultivating mental clarity of being aware of the present moment. These cues were presented again during sleep to reactivate that mental clarity, so that participants could signal about their subjective state. They were instructed to frown to indicate feeling asleep or smile to indicate feeling awake. Participants were awoken periodically to provide a report of their experience and complete a standard set of questions. EEG data were analyzed to determine participants' sleep stage. Data collection is ongoing, and so far 62 trials have been completed on 5 participants. Out of these 62 trials, in 7 (11%), an awake participant reported feeling asleep and in 12 (19%), a sleeping participant reported feeling awake. This provides evidence for the prevalence of sleep-state misperception, in which participants incorrectly perceive whether they are asleep or awake. These results open the door to asking more about why awareness does not always line up with objective sleep state.

Presentation Format: Oral
Time: 10:45AM-12:15PM
Room: Arch Room

Hou, Abby Faculty Advisor: Nivedita Arora

Functionalizing Yarns with Dip Coating for Smart Knitted Haptic Tactile Interfaces

The human skin is lined with several mechanoreceptors that are constantly converting things like pressure, vibration and textures into tactile signals that our brain can interpret. Replicating a small portion of this sensitivity into everyday materials would make sensing possible without being intrusive. Clothes are almost all knitted and universally needed by everyone, which makes it the perfect choice to begin embedding sensing into what we use. However, the process is limited by the material as there are no commercially available conductive yarns that are machine knittable, flexible, and insulated in a way that can be useful for research and prototyping. This project addresses this problem by developing a low-cost dip coating process that produces custom insulated conductive yarns compatible with knitting machines. We studied the effects that varying mechanical, chemical, and environmental parameters had on coating uniformity, thickness, and electrical isolation. By characterizing this behavior, we have developed a fabrication process that meets all requirements for being integrated into everyday knitted items. The initial results of these custom yarns have been tested in a proper knitting machine which proved the abrasion resistance was enough to maintain proper electrical insulation after being used to knit. These knitted items were used to create capacitive sensing pressure arrays that mimic mechanoreceptors and triboelectric nanogenerators for low power embedded systems. These results demonstrate that by solving the yarn fabrication problem, a new class of embedded smart knitted textiles applications can be created, from haptic feedback interfaces to pose tracking for AI databases.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 71A

Huang, Claire Faculty Advisor: Jeffrey Lopez

Optimization of Carbon Black in Conversion Cathodes: Investigation of Additive Impurities

The growing environmental concerns associated with fossil fuels have driven demand for sustainable energy alternatives, the adoption of electric vehicles, and the decarbonization of the electric grid. In order to support this transition, there is an increasing need for more energy-dense lithium-ion batteries (LIBs) to support electrification. A typical battery consists of three main components: the anode, cathode, and electrolyte, with the cathode composed of an cathode active material, conductive additive, and binder. While conventional cathodes rely on cobalt- or nickel-based intercalation materials, these face supply risks and are nearing their theoretical capacity limits. Conversion cathodes are a promising alternative which offers 200-300% higher energy density than traditional intercalation cathodes. Here, we study FeF2 conversion cathodes, which have been extensively studied due to their high energy density, low cost, and mineral abundance. However, due to low electronic conductivity, conversion cathodes require ~30% conductive additive, whereas intercalation cathodes typically achieve ~90% active mass loading. To overcome this, my research involved optimizing carbon black conductive additives by investigating their composition, electrochemical performance, and impurities. Commercial and impurity-rich carbon blacks were analyzed using galvanostatic cycling and inductively coupled plasma optical emission spectroscopy (ICP-OES) to investigate the effect of impurity concentration on cycling performance. It was demonstrated that Li-ion half-cells fabricated using sulfur-rich carbon black samples suffered a loss of long-term cycle stability. These results suggest that controlling impurity chemistry in conductive additives is crucial for the overall performance of Li-ion batteries.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 48A

Huang, Joyce Faculty Advisor: Ann Bradlow

Say what again? Investigating the intelligibility of Formula 1 driver speech between situations, contexts, and language backgrounds

This study investigated how background noise and speaker characteristics affect intelligibility in real-world contexts, specifically by using naturally occurring speech samples from Formula 1 drivers in different race-related situations. Thus, this study extends controlled laboratory-based research on speech-in-noise recognition to a naturally occurring, high-stakes communicative situation.

We compiled a corpus of 630 utterances from 21 Formula 1 drivers (9 first-language English speakers, 12 second-language English speakers) across three acoustic environments: studio interviews (minimal noise), paddock interviews (moderate noise), and race communications (high noise). Utterances from 4 first-language English speakers and 4 second-language English speakers then were selected to form the experimental stimulus set. 51 first-language English-speaking listeners (ages 22–81) transcribed 96 experimental utterances (8 speakers × 3 noise conditions × 4 utterances) in a web-based task. Participants were randomly assigned to either see contextual question prompts while listening to each utterance or transcribe utterances without contextual support. Transcriptions were scored using Autoscore, an automated scoring software (Borrie et al., 2019).

Results revealed substantial effects of background noise, speaker native language status, and contextual support on intelligibility. Intelligibility increased progressively from race to paddock to studio conditions. Participants correctly transcribed utterances from first-language English speakers at a higher rate than second-language English speakers. Contextual prompts improved intelligibility, with benefits increasing as noise levels increased.

These findings reveal the compounded communication challenges in Formula 1 and, by extension, other real-world speech communication situations where speakers must communicate in their second language while contending with extreme acoustic degradation during high-pressure situations.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 75P

Ikirezi, Vanessa

AI-Assisted Validation of Transfer Data in Economic Research

Largescale economic research increasingly relies on structured datasets compiled from hundreds of academic papers. However, extracting and validating this information remains a major challenge due to inconsistent reporting formats, ambiguous variable definitions, and the need for manual interpretation. These issues introduce errors and significantly slow down the research process, particularly in metaanalyses of global poverty interventions. Accurate extraction of transfer amounts is especially important, yet papers report values in local currency (LCU), nominal USD (NUSD), or purchasing power parity (PPP), making manual validation timeconsuming and prone to human error.

This project addresses these challenges by developing an AIassisted system to audit key financial variables in cash transfer studies. Using structured prompting and automated workflows, the system extracts planned transfer amounts directly from research papers and compares them to preexisting values stored in a centralized dataset. For any selected variable (xfer_amt_pln_lcu, xfer_amt_pln_nusd, or xfer_amt_pln_ppp), the pipeline generates four subcolumns: the original spreadsheet value, the AIextracted value, a match/mismatch assessment, and the supporting quote from the paper. This structure enables systematic validation across studies and reduces reliance on manual verification.

Developing this system required integrating AI coding agents, resolving Google authentication issues, and debugging cases where Codex returned incorrect values. Testing in the metadata audit sandbox demonstrates that the system reliably identifies agreement and disagreement between sources, highlighting both data inconsistencies and limitations in automated interpretation.

By automating a previously manual and errorprone process, this project strengthens the reliability of transfer amount metadata and contributes to GPRL's broader effort to build scalable, AIassisted validation tools for development economics research.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 74A

Im, Vincent Faculty Advisor: Eyal Kimchi

Effects of Systemic Inflammation on Neuronal Activity in Transgenic Alzheimer's Mice using c-fos

Understanding the interplay between Alzheimer's Disease (AD) pathology and systemic inflammation is crucial for elucidating the mechanisms underlying acute disorders of cognition such as delirium. We therefore employed a multifaceted approach to investigate the effects of systemic inflammation on neuronal activity in brain slices in a transgenic AD mouse model (5xFAD and TauP301S)

We studied 5xFAD transgenic mice overexpressing amyloid, TauP301S transgenic mice, with their Tau tangles, and their respective wild-type (WT) siblings. Lipopolysaccharides (LPS) from E. coli (0.5 mg/kg) induced systemic inflammation, while a control group received a saline injection. To quantify activity in mouse brains, we used c-Fos, an immediate early gene, as the marker for neuronal activity in brain tissue. We performed immunohistochemistry on brain slices using the DAB reaction to analyze c-Fos+ cells in multiple regions of interest involved in delirium pathophysiology.

Preliminary home-cage behavioral analysis revealed and validated that systemic inflammation suppressed spontaneous locomotion in these mice, disrupting typical circadian rhythms and being more evident in dark periods. The imaged slices for the experimental groups are currently being analyzed via QuPath - a quantitative pathology software.

Preliminary data analysis have revealed significant effects of substance, especially within cortical areas. Further statistical analysis is ongoing to evaluate potential differential sensitivities between 5xFAD, TauP301S and WT mice to these manipulations.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 20A

Jacob, Natalie Faculty Advisor: Andrew Jacobson

Fingerprinting Carbon Capture: Tracing Enhanced Rock Weathering Through Soil with Strontium Isotopes and Alkalinity

Limiting global climate change requires not only reducing greenhouse gas emissions but also verifying carbon dioxide removal (CDR) strategies at statistically significant scales. Enhanced rock weathering (ERW) is a CDR strategy that accelerates the dissolution of silicate minerals in soils, converting atmospheric CO2 into stable aqueous alkalinity. As interest in ERW grows, accurately verifying carbon capture and distinguishing silicate-derived products from other sources has become a central challenge. Alkalinity measurements are widely used to quantify weathering-driven CO2 removal, but they are not source-specific and may reflect inputs from carbonates, fertilizers, or background soil processes. Current data show that basalt-amended systems produce elevated and stable alkalinity, with mean values of 28,940 3,020 mg/L compared to 25,480 2,870 mg/L in controls over a 13-month storage period, indicating minimal temporal variability; however, these measurements alone cannot confirm silicate-derived carbon capture. Combining radiogenic strontium isotope ratios (87Sr/86Sr) measured through a thermal ionization mass spectrometer with alkalinity measurements enables source-specific tracing of ERW processes through soil profiles. Depth-resolved analyses may reveal that basalt-amended soils exhibit distinct unradiogenic isotopic signatures alongside increased alkalinity, indicating mobilization of silicate-derived weathering products. These findings demonstrate that integrating isotopic tracers with aqueous chemistry improves ERW verification by linking geochemical signals directly to mineral sources, strengthening carbon accounting, and supporting the deployment of scalable soil-based climate solutions.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 29P

Jeon, Jay Faculty Advisor: Darren Gergle

Centralizing Search: How AI-Generated Summaries on Search Engine Results Pages Reshape User Engagement and Perception

The integration of generative AI into search interfaces is shifting user attention from multi-source exploration toward reliance on a single synthesized response. This study examines how interface type (AI summary vs. traditional link-based) influences landing page engagement, article access patterns, and perceptual outcomes, including credibility, trust, and satisfaction, and whether these effects vary by search topic (health vs. pop culture). Using a 2 x 2 between-subjects design with 48 undergraduate participants, we find that AI summary interfaces significantly increase time spent on the landing page-by 3.5 times in the health domain and by 2.1 times in the pop culture domain-suggesting a shift toward more centralized attention. However, effects on downstream behaviors and perceptual outcomes are limited. Article access patterns are driven more by task domain than by interface type, with pop culture participants accessing more sources than health participants across conditions. We find no clear evidence that interface type influences perceived credibility or trust. At the same time, health content is rated as more credible than pop culture content regardless of interface, consistent with domain-based epistemic expectations. For satisfaction, a significant interaction emerges: traditional interfaces produce substantial variation across topics, whereas AI interfaces reduce variability, yielding more consistent - though not necessarily higher - user experiences. Taken together, these findings suggest that AI summaries centralize attention and standardize user experience, shifting search from exploratory engagement toward more uniform modes of information consumption.

Presentation Format: Oral
Time: 12:45-2:15PM
Room: Arch Room

Jia, Victor Faculty Advisor: Jeffrey Richards

Electrically Tunable Structural Color in Photonic Crystal Films of PEDOT:PSS-SiO2 for Electrochromic Devices

Electrochromic devices (ECDs) are devices that change color and optical properties under an applied voltage and have widespread uses in smart windows, displays, and wearables. Typically, ECDs function through an electrochemical redox reaction, in which a change in voltage induces the transport of ions between the cathode and anode. This, in turn, changes the material's light absorption properties by transporting ion into or out of the electrochromic layer resulting in either a colored or bleached state that is sustained without the need for continuous power. One major challenge in the development of ECDs is the limitation of color contrast in these devices, providing only shades of a single color instead of a full, continuous visible spectrum. This restricted color range limits the versatility of ECDs in applications that require rich color displays. An approach to overcome this limitation explores the use of structural color, in which color arises from the interaction of light with the nanoscale structures of particles. Structural color results from optical phenomena such as interference, diffraction, and scattering from nanostructures, allowing the color of light to be tuned by adjusting structural parameters such as interparticle spacing. Therefore, integrating structural color mechanisms into electrochromic architecture offers a promising pathway toward ECDs with a broader color palette. By incorporating nanostructures whose particle spacing can be electrically modulated into ECDs, it becomes possible to dynamically tune coloration across the continuous visible spectrum. Such systems have potential in expanding the functionality of ECDs for next-generation adaptive displays and optical technologies.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 06P

Jourdes Ferdman, Paloma Faculty Advisor: Lilah Shapiro

The Creeping Return of Antiquated Views on Womanhood: How Gen-Z Women Experience the Socio-Political Climate

This study aims to understand the impacts of the current socio-political climate on Gen-Z women's emotions, identity, and interpersonal dynamics. Focusing this study on Northwestern's campus ensures that accurate patterns and results can be concluded while keeping as many variables as possible consistent. To examine the impact of the socio-political climate on young women, I conducted 15 semi-structured and open-ended interviews with individuals who identified as female, were eligible to vote in the 2024 election, and were upperclassmen at Northwestern University. Findings revealed significant trends in the influence of the socio-political on young women's every day and personal life. Young women cited feeling distressed about the overall future of the world, specifically in the regression of gender equity and inclusion. Feelings of emotional suppression were also common and deliberately used to create mental distance from the political landscape. Many young women described feeling solidarity among themselves, insinuating that they felt more confident and comfortable around other women. In some instances, women had distorted self-perceptions of themselves, created by a heightened hyper-awareness of social perception. Finally, participants expressed negative sentiment and difficulty in connection with those of opposing ideologies. Many described a physical and emotional gap that has been created between young men and women, emphasizing the difficulty for women to form close and intimate relationships with men. Future directions for this research should focus on comparing the emotional, identity forming, and interpersonal impacts the socio-political climate has had for Gen-Z men.

Presentation Format: Oral
Time: 10:45AM-12:15PM
Room: Rock Room

Jung, Eugene Faculty Advisor: Derk Joester

Automated X-ray Analysis of Ice Formation in Drosophila Larvae

In Earth's coldest environments, organisms survive sub-zero temperatures by producing antifreeze proteins (AFPs) that control ice formation and prevent cellular damage. The snowfly *Chionea alexandriana* expresses a hypothetical anti-freeze protein, CaAFP, but its mechanism of ice binding and crystal growth regulation remains unclear. We developed an X-ray diffraction (XRD) assay to probe the impact of CaAFP expressed in *Drosophila melanogaster* larvae on the freezing point. Our objective was to determine the impact of CaAFP on crystallographic parameters of ice crystals as a first step of determining its mechanism. To facilitate the analysis of XRD patterns (280-240 K, 1K intervals), we developed an automated Python pipeline to simultaneously fit the (100), (002), and (101) reflections with Pseudo-Voigt profiles. Peak parameters were extracted separately from the multi-peak fits and pooled across temperatures within each sample group prior to statistical comparison. No significant differences in peak positions or amplitudes were observed, indicating unchanged lattice parameters. However, all three reflections showed broader full width half maximum (FWHM) in CaAFP-expressing samples. The FWHM of the (100) reflection in CaAFP-expressing larvae (FWHM1 = 0.181) was significantly larger than in WT larvae (p = 3.36 10-7) consistent with decrease in crystallite size or an increase in lattice strain, or a combination thereof.

Our findings reveal that CaAFP directly reshapes the crystallographic parameters of ice crystals formed *in vivo*, supporting direct ice binding mechanism. The approach further establishes a quantitative framework for studying ice-protein interactions and advances cryopreservation and the design of bioinspired antifreeze materials.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 30P

Kaka, Taera Faculty Advisor: Alissa Chung

Exploring the Relationship between Adverse Childhood Experiences and the Development of Coping Strategies in Young Adults

Adverse Childhood Experiences (ACEs) can be categorised as potentially traumatic incidents that occur before the age of 17. These are well established as predictors of long term negative outcomes. However, little is known about the processes through which ACEs lead to these outcomes, or why individuals with similar environmental (ACEs and Positive Childhood Experiences (PCEs)) and intrinsic factors (personality traits) may have vastly different trajectories. No study has integrated these factors into a single model, and therefore the current study shifts focus from outcomes to mechanisms, examining coping as a key pathway linking early experiences to later outcomes. Participants (N=378) completed measures of ACEs, coping behaviour, PCEs, and personality traits. Regression analyses showed that ACEs positively predicted all coping styles including avoidant (maladaptive), emotion-focused, and problem-focused (generally adaptive) coping. Coping increased at low-moderate ACE levels and plateaued at higher ACE exposure, which suggests a nonlinear trend of cumulative risk. PCEs were associated with greater adaptive coping and lower avoidant coping. Extraversion and conscientiousness were linked to adaptive coping, while neuroticism was linked to maladaptive coping. In cumulative risk models, ACEs explained additional variance beyond personality in avoidant emotional and problem-focused coping, while PCEs contributed to adaptive coping. No interaction effects were observed. PCEs and personality appear to be independently associated with more adaptive coping. These findings suggest that ACEs influence the overall organisation of coping mechanisms and meaningfully contribute to their development along with critical factors of positive experiences and personality that could inform future interventions.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 12A

Kales, Naomi Faculty Advisor: Joshua Ewen

Understanding Causes of Altered Imitation in Autism: A Novel Hypothesis

Imitation is important for learning motor and social skills, both of which are often altered in autistic individuals. Prior work suggests that autistic children have more difficulty imitating gestures when multiple joints move at the same time, but it is not clear whether these differences are driven by coordinating multiple movement elements versus overall motor complexity. This study tested whether breaking gestures into step-by-step parts differentially improves imitation in autism and examined the roles of motor and non-motor processes. Typically developing (n = 70) and autistic (n = 54) children (ages 8-12) completed two experimental tasks. In Experiment 1, they imitated 20 novel gestures shown either all at once or step-by-step, and performance was scored based on element-wise accuracy. In Experiment 2, they completed a delayed-match-to-sample judgment task to measure non-motor processing. Step-by-step presentation improved imitation and reduced group differences (condition diagnosis interaction, $t(114) = 2.1$, $p = .042$), but a gap between groups remained. In the judgment task, there was no effect of presentation style, though autistic children still differed from neurotypical children. Mediation analysis showed that non-motor processes explained about 34% of the group difference in imitation. Overall, breaking gestures into steps reduces imitation differences, but does not fully close the gap, suggesting that interventions which scaffold actions into sequential components may improve performance but should also directly target motor coordination. These results also suggest that motor processes are a primary contributor to imitation differences in autism and may be an important target for intervention.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 66A

Kalkowski, Lea Faculty Advisor: Lilah Shapiro

Interpreting Justice: How Legal Practitioners Conceptualize Fairness, Justice, and their Role in the Changing Illinois Criminal Record Relief Landscape

Among Illinois's 12.7 million residents, approximately 3 million have a criminal record, which can restrict access to employment, housing, and educational opportunities. The Clean Slate Act represents a significant shift in how the state approaches criminal record relief. Little is known about how legal professionals will navigate this new landscape, not only in terms of procedure, but in conceptions of their own roles within the system. This thesis draws on seven semi-structured interviews to examine how legal aid lawyers make sense of justice, fairness, and professional responsibility within Illinois's evolving expungement and sealing system, engaging in analysis through open coding and a Grounded Theory approach. Attorneys' conceptions of justice emerge through three interconnected themes, each reflecting a core tension they must navigate. First, lawyers articulate abstract ideals of justice while simultaneously confronting a criminal legal system they view as inherently complex and overly punitive. Second, they experience criminal record relief as a transformative but imperfect remedy to the system. Third, lawyers consistently frame their work in moral terms, describing an obligation to clients, a responsibility to advocate, and a commitment to educating clients, even as policy change brings both hope and skepticism. These findings contribute to a more nuanced understanding of how policy reforms are implemented on the ground, examining how well-intentioned legislation can produce unintended consequences or inconsistencies in practice. Lawyers are responsible for interpreting and navigating these policies, so academic analysis of their experiences is critical to inform more effective and equitable future reforms.

Presentation Format: Oral
Time: 12:45-2:15PM
Room: Rock Room

Kang, Andrew Faculty Advisor: Kelley Lab Office

Repurposing ALOX5 Inhibition for Psoriasis Treatment

Psoriasis is a common autoimmune, inflammatory disorder affecting 2-3% of the global population. Plaque psoriasis, the most common manifestation of the disease, is linked with soreness, joint pains, and dry, itchy patches of skin. While the disease's pathogenesis has been largely attributed to IL-17 secretion by T-helper 17 (Th17) cells to amplify keratinocyte-mediated skin inflammation, the underlying intracellular signaling mechanisms within keratinocytes remain incompletely defined. This gap in mechanistic understanding has constrained therapeutic development, with current treatments largely limited to costly systemic biologics that can carry significant side effects. To identify novel regulators of keratinocyte responses to IL-17, we performed a genome-wide CRISPR knockout screen in primary human epidermal keratinocytes, focusing on genetic regulators of IL-17 receptor A (IL17RA), a central node in psoriatic inflammation. This screen identified arachidonate 5-lipoxygenase (ALOX5) as a top candidate. We demonstrate that ALOX5 functions as a lipid mediator that stabilizes IL17RA expression at the cell surface, thereby amplifying IL-17-driven inflammatory signaling. Importantly, targeting ALOX5 with Zileuton, an FDA-approved drug currently used to treat asthma, showed strong inhibitory effects towards IL17RA in keratinocytes. Zileuton also demonstrated potent therapeutic effects when applied topically to a mouse model of imiquimod-induced psoriasis, performing comparably to existing FDA-approved anti-IL17RA antibody treatments for plaque psoriasis. Overall, topical Zileuton suppressed pathogenic Th17 responses, promoted anti-inflammatory macrophage polarization, and normalized epidermal hyperproliferation. Together, these findings identify ALOX5 as a previously unrecognized regulator of the IL-17 signaling axis and highlight the potential of repurposing Zileuton as a targeted therapeutic strategy for psoriasis.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 80A

Kaplan, Yanira Faculty Advisor: Alissa Chung

Gender Differences in Self-Perceptions of ADHD Diagnosis: Identity, Relief, and Stigma

Despite its prevalence, there is little research examining how individuals make sense of their Attention-Deficit/Hyperactivity Disorder (ADHD) diagnoses. The purpose of this study was to explore the relationship between gender and self-perceptions of receiving an ADHD diagnosis through two domains: positive appraisals (self-understanding and sense of relief) and negative appraisals (stigma). Age of diagnosis was also examined as a potential explanatory factor. A sample of 364 young adults from the U.S. with self-reported formal ADHD diagnoses was surveyed through Prolific on their experiences following diagnosis. The results indicated that women were significantly more likely than men to experience self-understanding and relief as a result of getting diagnosed. Age at diagnosis was partially related to this, with later diagnosed individuals reporting higher levels of self-understanding and relief, but gender differences remained even when age was considered. There were no significant gender differences observed with stigma, though individuals diagnosed before their adolescence reported higher levels of stigma. These findings underscore the importance of considering gender and diagnostic timing when exploring the psychological impact of an ADHD diagnosis. On a broader scale, these results reinforce the need to address gender disparities in ADHD diagnosis, potentially driven by diagnostic criteria and expectations that center on a prototypical, male presentation. Differences in appraisal may reflect the greater distress experienced by girls and women as a result of years going undiagnosed.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 88A

Kharrazi, Ella Faculty Advisor: Erica Hartmann

Identification of Host Ranges, Receptors, and Defense Genes For the JG024 Phage in Clinical Isolates of Pseudomonas Aeruginosa (PA)

Pseudomonas aeruginosa (PA) is a major cause of hospital-acquired infection and can rapidly develop antibiotic resistance. Bacteriophage (phage) therapy offers a promising and highly specific alternative treatment. However, PA possesses diverse and complex anti-phage defense systems, many of which remain uncharacterized.

Understanding the mechanisms by which PA resists phage is critical to developing effective phage therapies. Identifying specific genes and receptors responsible for phage susceptibility or resistance would allow targeted strategies to enhance phage therapy success. In this experiment, we investigate phage-host receptor interactions and use cloning-based approaches to introduce candidate defense genes to uncover these mechanisms.

To explore phage defense mechanisms in PA, we first determined the host range of 215 PA clinical isolates against the lytic phage JG024.

For a phage to lyse its target, it must first bind bacterial surface receptors. The O11 and O3 lipopolysaccharide antigens and the type IV pili component PilA5 were identified as candidates, showing statistical correlations with JG024 susceptibility in bioinformatic analyses. Preliminary results suggest a positive correlation between the O11 antigen and JG024 susceptibility. To determine whether these receptors mediate phage binding, qPCR is being used to quantify JG024 binding to isolates expressing each receptor.

Bioinformatic analysis also identified genes statistically correlated with phage susceptibility across isolates, from which a candidate defense gene was selected for introduction into a phage-susceptible PA reference strain. This gene did not confer phage resistance; however, ongoing qPCR binding assays and defense gene investigation aim to inform future strategies for optimizing phage therapy against PA.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 03P

Kim, Michael Faculty Advisor: Matthew Glucksberg

Development of a Fluorescence-Based Endomicroscope System for Quantitative Assessment of Pediatric Aspiration

Aspiration, the accidental inhalation of foreign material into the lungs, is a severe, yet critically understudied, clinical danger for pediatric and neurologically impaired patients that can lead to life-threatening respiratory complications. Current diagnostic methods, such as videofluoroscopic swallow studies, only provide indirect, qualitative assessments. They lack the sensitivity to directly visualize or measure aspirated material at the microscopic level. To address this gap, we present a proof-of-concept fluorescence endomicroscope designed to directly detect and quantify aspirated fluorescent particles (microspheres) within lung tissue. Our system utilizes a specialized optical setup, featuring a broadband LED, wavelength filters, and a high-resolution sensor, to illuminate the tissue and capture the fluorescence of these microscopic particles, acting as a proxy for aspiration severity. We successfully demonstrated that this system reliably visualizes microspheres in lung tissue samples. By applying an image analysis algorithm, we quantified particle density, providing an objective metric of the aspiration burden. Experimental results showed that while thicker, physiologically relevant tissue reduced image contrast due to light scattering, the particles remained clearly identifiable. System resolution easily distinguished individual 10-micrometer particles, and sampling analysis confirmed that this performance is limited by optical physics rather than sensor constraints. These findings establish fluorescence endomicroscopy as a viable method for directly visualizing and measuring microscale aspiration. This platform introduces a vital shift from subjective observation to precise, quantitative measurement, offering a powerful new framework for early diagnosis, severity grading, targeted treatment monitoring in highly vulnerable patient populations, and general aspiration study.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 55A

Kim, Travis Faculty Advisor: Bharath Chandrasekaran

Feedback-based Training and Musicality Teach People to Detect Deepfake Speech

Recent advancements in AI text-to-speech systems have enabled the creation of high-quality deepfake voices increasingly indistinguishable from real voices. These deepfakes have been used to impersonate and defraud people, yet few studies have looked at ways to teach people to detect deepfakes. This study sought to determine whether feedback-based teaching paradigms and musicality confer improvements in deepfake detection. 50 participants (of either gender) listened to sentences produced either by a human speaker or an AI-generated voice clone before judging each sentence as real or fake. After judging these sentences for ~10 minutes, participants underwent a ~20 minute training block where they received feedback (correct/incorrect) based on their answers, before returning to a post-training block identical to the pre-training one. Participants also underwent a musical skills test. Behaviorally, participants demonstrated poor discriminatory sensitivity prior to the training block and improved sensitivity after the training, revealing that even short feedback-oriented training procedures can be used to teach deepfake detection, contrasting recent research saying that short training sessions do not lead to improvements in deepfake detection. Additionally, musicality and rhythm-based skills were correlated with deepfake detection and larger improvements in that detection after training. These findings establish a framework for effective deepfake detection pedagogy, a step towards reducing the societal threats posed by deepfake voice fraud. Understanding that musicality in specific leads some listeners to detect deepfakes better than others reveals tractable perceptual mechanisms that can be leveraged to design further research into deepfake detection training interventions.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 30A

Kim, Paul

Baseball's Highest Stakes Situation and a Decision Error Three Decades Deep

It's the bottom of the ninth inning, two outs, bases loaded. The conventional instinct in baseball is to swing at a hitter's count. This paper argues that instinct has been wrong for at least three decades and that the sport has not corrected it. With bases loaded and two outs, a walk simultaneously scores a run and preserves the identical threat for the next batter, who then faces the same high-leverage situation from a fresh count. This self-reinforcing property means that patience compounds across consecutive plate appearances in a way that attacking any individual pitch cannot match. This paper formalizes this using an augmented Markov Decision Process applied across 192 possible game states. The model recommends taking the pitch at 10 of 12 count states in a tied game and 11 of 12 when trailing by one run. Against 35 years of Major League Baseball pitch by pitch data, the gap between optimal and observed behavior is large and entirely unidirectional. Batters over-swing at every non-two-strike count, at costs reaching 19 percentage points of win probability at the single most expensive state. Not one of the 48 count-and-deficit combinations in the data shows under-swinging. The gap has not narrowed from 1990 to 2024, surviving the full analytics revolution. The finding connects to the behavioral economics literature on action bias and suggests that trained motor responses, unlike deliberate decisions, resist strategic correction.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 64A

Kim, Michael Faculty Advisor: Matthew Glucksberg

Application of Red Fluorescent PLGA Microspheres for Quantitative Assessment of Pediatric Aspiration

Pediatric aspiration is a clinically significant condition that can lead to chronic lung injury, infection, and long-term respiratory complications, yet current diagnostic methods such as videofluoroscopic swallow study (VFSS) and fiberoptic endoscopic evaluation of swallowing (FEES) remain largely qualitative and do not quantify aspiration severity. This project aimed to address this gap by developing a microsphere-based approach to enable quantitative assessment of aspiration. Red fluorescent poly(lactic-co-glycolic acid) (PLGA) microspheres were used as surrogate tracers to model aspirated material due to their biocompatibility, controlled size, and improved detectability compared to free dyes. The use of red fluorescence was selected to reduce interference from natural tissue autofluorescence in the airway. The system was evaluated across multiple experimental models, including fixed pig trachea, fresh murine lung and tracheal tissue, and a New Zealand rabbit airway perfusion model to simulate aspiration under physiologically relevant conditions. Known concentrations of microspheres were introduced to assess detectability and quantifiability across conditions. Results demonstrated that microspheres could be reliably detected and quantified in fresh tissue, with higher concentrations producing clear, countable signals and improved signal-to-background contrast. These findings support the feasibility of using microspheres as quantitative tracers for aspiration and establish a foundation for correlating particle distribution with aspiration severity. This approach has the potential to improve diagnostic precision, enable objective severity grading, and support better clinical decision-making in the management of pediatric aspiration.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 78A

Kim, Joshua Faculty Advisor: Gemma Carvill

Targeting poison exon splicing to modulate CHD2 protein production

Neurodevelopmental disorders (NDDs) can be caused by changes in certain genes. One of these genes, CHD2, helps control DNA packaging and gene activation. Harmful changes in CHD2 reduce the amount of CHD2 protein the body makes, which can lead to conditions such as epilepsy, intellectual disability, and autism. There are currently no treatments that directly address the underlying cause, and traditional gene therapy is difficult because CHD2 is too large to target with standard methods. CHD2 contains a short poison exon, a segment of RNA that, if included in the final transcript, causes the RNA to be destroyed before it can make protein. This lowers the overall amount of CHD2 protein. We propose that reducing inclusion of this poison exon could naturally increase CHD2 levels. To test this idea, we will use engineered cells that express a modified protein fused to a splicing factor. Guide RNAs can direct this complex to the poison exon to change how the exon is spliced into the transcript. We will measure and sequence CHD2 RNA to determine whether targeting the poison exon decreases its inclusion and increases overall transcript levels. We aim to demonstrate that adjusting splicing can raise CHD2 protein levels, and that splice-based approaches may offer a promising method of treating disorders caused by genes that cannot be targeted with conventional gene therapy.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 52P

Kissel, Simon Faculty Advisor: Erica Hartmann

Linking niche diversity to phylogenetic relatedness in freshwater bacteria assemblies

Aquatic bacteria are critical to the health of lake ecosystems because they aid in the cycling of nutrients and removal of pollutants. Current methods to study these bacteria rely heavily on taxonomic classification. Taxonomic classification works well for understanding environmental roles of higher organisms, for example plants are primary producers and carnivorous animals are predators. However, these classifications are less useful for bacteria, whose ecological role may be more variable, because they do not directly reflect the function of these bacteria in lake ecosystems. To better understand the relationship between taxonomic classification and ecological function in aquatic bacteria, we use two decades of publicly available bacterial DNA sequencing data from Lake Mendota in Madison, WI. By analyzing all the genes present in these environmental samples, we compare functional gene diversity patterns in 10 groups of bacteria. Results indicate that groups have variable diversity levels, meaning some bacterial groups may perform a wider array of ecological functions than others despite having the same taxonomic classification. Specific motility and phage-related genes show differential presence patterns across different groups, which present a possible explanation for this variation. Abstract mathematical analysis called topological data analysis further indicates complicated dynamics in bacterial diversity. Together, these results demonstrate that genetically identified groups have different patterns in functional adaptability, indicating a need for revision in the current taxonomy model for one that accounts for the entire genome. Future efforts will be needed to validate these data across a broader array of environments and with more accurate sequencing.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 11P

Knupp, Savannah Faculty Advisor: Richard Zinbarg

How Trait-Mindfulness and Working Alliance Predict Treatment Engagement in an Adolescent Mindfulness Study

As various interventions for mental health conditions have been developed, engagement with such treatments and their homework between sessions has been linked to numerous key outcomes, including reductions in symptoms of depression and anxiety, and signs of overall therapeutic change. The aspects of trait-mindfulness and client-rated working alliance have been shown to predict such engagement in an intervention, but gaps still exist, especially with regards to examining adolescent populations and for more accessible interventions that do not necessarily involve a professional clinician. The present research thus utilizes data from the Youth Mindful Awareness Program (YMAP) study that delivers two evidence-based interventions, one emphasizing mindfulness and the other focusing on expressive writing. This project also uses trained undergraduate coaches and focuses on adolescent populations rated as being at higher risk for developing depression and anxiety. In the current study, the variables of trait-mindfulness, measured at baseline, and working alliance, measured at the midpoint of the study, are tested to see if they predict engagement in the YMAP study across both conditions. Here, engagement is defined as completion of the assigned between-session practices for the 9 weeks of the study. Results of a multiple regression analysis showed a non-significant relationship between these variables and engagement. Recommendations are given for future directions and studies that can possibly better represent the effects of working alliance and trait-mindfulness and contribute to existing theories.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 14A

Kraidy, Maya Faculty Advisor: Matthew Goldrick

Signs of Change: A Cross-Cultural Exploration of Minority Language Revitalization through Linguistic Landscape

Global language loss is occurring at an unprecedented rate, with profound cultural, intellectual, and social consequences. Language revitalization seeks to counter historical and ongoing forces of marginalization, yet its success is often difficult to measure in practice. This study examines the visibility of minority and Indigenous languages in multilingual, post-colonial urban contexts through a mixed-methods approach, combining semi-structured interviews with ethnographic observation and photographic documentation of public signage. The analysis of the linguistic landscape of Wales, South Africa, Singapore, the Philippines, and New Zealand provides a tangible metric for assessing the material presence of these languages, while interviews contextualize patterns of visibility within local policy frameworks, community attitudes, and sociolinguistic realities. Findings reveal how public signage reflects and reinforces language hierarchies, highlights discrepancies between revitalization ideals and implementation, and offers insight into the role of Indigenous and local knowledge in shaping language visibility. By linking linguistic theory with material and social evidence, this study contributes to understanding the interplay between language policy, public space, and efforts toward linguistic and cultural sustainability.

Presentation Format: Oral
Time: 10:45AM-12:15PM
Room: Armadillo Room

Kumar, Anusha Faculty Advisor: Elham Hoominfar

From Scarcity to Solidarity: Mutual Aid as a Pathway to Food Sovereignty and Urban Resilience in Chennai, India

In the Indian state of Tamil Nadu, 23.1% of the population is undernourished, according to estimates from the Global Hunger Index, despite the state's robust welfare programs. In an urban area like Chennai, food insecurity is compounded by poverty, gender inequality, and caste apartheid. Additionally, extreme weather driven by climate change has caused localized food shortages that disproportionately affect marginalized groups. In the face of these overlapping inequalities, mutual aid may empower local people to reclaim their right to food produced through ecologically sustainable means. Mutual aid refers to a collective exchange of resources between community members for reciprocal benefit. This presentation investigates how mutual aid networks in Chennai organize to create pathways to food sovereignty and urban resilience. Through interviews with members of grassroots organizations, including a climate justice non-profit, an urban farming initiative, and a food re-distribution network, this study unpacks how mutual aid can be understood in a South Asian context, specifically in relation to caste apartheid and gender inequality. This study further examines how these networks engage with notions of reciprocity while challenging or reproducing charitable logics. Women's self-help groups (SHGs) and grassroots organizations were found to be significant models of mutual aid in Chennai. These organizations engage in the local production and distribution of nourishing food. Such structures support the creation of green spaces and the reduction of food waste. These systems of community care operate in democratic ways that empower lower-income women and center caste-oppressed and tribal communities in aid distribution efforts.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 10A

Kunhart, Aubrey Faculty Advisor: Thomas Meade

Synthesis of Cyanine-labeled Peptides for AuNP Labeling and Lysosomal Enzyme Detection

Lysosomal enzymes, known as cathepsins, facilitate the controlled degradation and recycling of cellular proteins, which is necessary to maintain cellular balance and function. In many cancers, cathepsins become overactive and begin degrading components of the extracellular matrix, the structural network that keeps cells in place. This breakdown enables tumor cells to invade neighboring tissue and form malignant growths. Detecting cathepsin activity in real time is therefore important for diagnosis, monitoring disease progression, and evaluating therapeutic response. Fluorescent imaging probes can serve as indicators of enzyme activity, as their emission can be altered depending on their incorporation into a molecule. However, conventional probes emit light continuously, regardless of whether the target enzyme is active, which produces high background fluorescence and makes it difficult to distinguish activity from noise. This project aims to develop an enzyme-activated ("turn-on") fluorescence probe by attaching specially designed cyanine dyes to gold nanoparticles (AuNPs) through a cathepsin-cleavable peptide. When this peptide is connected to the AuNP, the dye will be held close to the gold surface, where its fluorescence is quenched (appears "off"). When cathepsin cuts the dye-AuNP linker, the dye is released and its fluorescence intensity increases. The goal of this project is to synthesize and evaluate a small library of these cyanine dye constructs, compare how structural variations of linkers influence performance, and determine which probe designs produce a strong and reliable fluorescence activation signal.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 83A

Kurt, Kiera Faculty Advisor: Kelly Bachta

Isolation and Characterization of Seven Novel Psuedomonas aeruginosa Bacteriophages

Pseudomonas aeruginosa (*Pa*)is a Gram-negative bacterium responsible for many hospital acquired infections that are particularly difficult to treat due to high levels of antibiotic resistance. As the number of multidrug-resistant strains continues to rise, interest in alternative forms of treatment, like bacteriophage (phage) therapy, has been renewed. Phage therapy involves the use of bacteriophages, or viruses that infect bacteria, to target and kill the pathogen of interest. However, due to the overwhelming abundance and diversity of phages, further characterization is needed for reliable clinical use. In this study, seven novel bacteriophages (p001-p007) were isolated from wastewater that were active against multidrug-resistant *Pa*. After enrichment and purification, the host range of each phage, as well as that of three previously isolated phages (e.g. JG024, phiKZ), was tested using a panel of antibiotic-resistant clinical isolates (n=51). For each phage-strain combination, efficiency of plating (EOP) was calculated by comparing plaque formation on the test strain to the strain from which each phage was isolated. Host range varied widely among phages, with one phage infecting only one additional strain (n=2) while another infected almost half (n=25) of all stains tested. The turbidity of plaques did not correlate with EOP, suggesting that plaque morphology is not a dependable indicator of productive infection. To further characterize each original phage, genomes were sequenced and transmission electron microscopy was used to examine phage morphology. Overall, this study expands on current understanding of phage diversity and provides preliminary data for future therapeutic applications.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 84P

Lahlou-Nabil, Rayan Faculty Advisor: Rana Raddawi

Improving Medical Training to Address Health Literacy Gaps in Middle Eastern North African Americans

Middle Eastern and North African (MENA) Americans are currently classified as White in U.S. datasets, which obscures meaningful health disparities and limits the ability to study their specific healthcare needs. Health literacy (HL), defined as the ability to understand and use health information, is a critical determinant of health outcomes. This study aims to evaluate differences in HL between MENA Americans and White individuals and to highlight the need for more accurate demographic classification.

We conducted a retrospective analysis using data from the NIH All of Us database. Health literacy was measured using the BRIEF-3 scale, where scores 9 indicate inadequate HL. Participants were stratified by race, education, income, and age. Statistical analyses included one-way ANOVA to assess differences across subgroups (p & 0.05 considered significant) and to evaluate overall differences between racial groups.

Results showed that HL confidence increased with higher education, income, and age for both groups. However, White participants consistently reported higher HL scores than MENA participants across all subgroups. These differences were statistically significant, with strong interactions between race and education, income, and age. Distributional analysis further confirmed persistent disparities, with MENA individuals more likely to have lower HL scores.

These findings demonstrate a clear disparity in health literacy that is currently masked by racial misclassification. Recognizing MENA as a distinct demographic group is essential for improving data accuracy, informing targeted interventions, and promoting health equity in medical training which is the primary focus of this study.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 42P

Lam, Sabrina Faculty Advisor: Wendi Gardner

The Role of Self-Concept Clarity in Decision-Making Under Time Pressure and Groupthink

Companies often face high-stakes decisions in which uncertainty, time pressure, and hierarchical structures can shape how individuals contribute to group outcomes. In such contexts, groupthink can lead individuals to align with authority figures, reducing overall decision quality. Although situational pressures are known to influence conformity, less is known about how individual differences such as self-concept clarity (SCC) affect willingness to voice independent opinions. Prior research on SCC has largely focused on individual rather than group decision-making. This study examines how SCC interacts with groupthink and time pressure to influence decision-making and argument quality. Participants (N = 425) were randomly assigned to one of three conditions: high time pressure with groupthink, low time pressure with groupthink, or low time pressure without groupthink. Participants participated in a simulated decision stimuli regarding whether to recall a defective medical device. We found that groupthink significantly influenced decision outcomes: participants were more likely to align with the CEO's recommendation under groupthink regardless of time pressure ($x2(1, 285) = 64.66$, p & 0.001). Groupthink also reduced argument strength (v = 0.133, p = 0.006). Although SCC did not have a direct effect on argument strength (v = 0.002, p = 0.963), it significantly moderated responses under groupthink conditions (v = 0.116, p = 0.016), such that individuals with higher SCC produced stronger arguments when social pressure was present. These findings suggest that while groupthink drives conformity, self-concept clarity functions as a psychological buffer that supports independent judgment under social pressure.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 15A

Landsly, Emma Faculty Advisor: Andrew Prigge

Juvenile Regulatory T Cells Exhibit Decreased Capacity to Suppress Inflammation During Severe Influenza Pneumonia

Healthy children are more likely than adults to develop severe viral pneumonia, which can lead to long-term impairments in cardiopulmonary function. Our group demonstrated that juvenile mice exhibit prolonged inflammation following influenza A virus (IAV) infection compared to adults, despite equivalent viral clearance. In adults, regulatory T cells (Tregs) suppress inflammation and promote tissue repair, in part by suppressing conventional T cell (Tcon) proliferation. We hypothesize that juvenile Treg reparative function is impaired during IAV infection, contributing to prolonged inflammation and aberrant lung repair. To test this, the suppressive capacities of Tregs from adult and juvenile mice were compared using lymphocyte suppression assays. Juvenile Foxp3DTR mice were infected with a sublethal titer of IAV and treated with diphtheria toxin to deplete Tregs. Bronchoalveolar lavage fluid (BALF) IL-6 and Amphiregulin (AREG) were quantified using an enzyme-linked immunosorbent assay, and lung immune cells were enumerated using flow cytometry. Lungs were assessed by H&E and immunofluorescent microscopy staining for lung injury and repair markers. Juvenile Tregs suppressed Tcon proliferation less effectively than adult Tregs. Treg-sufficient and Treg-deficient juvenile mice failed to reach the growth potential of uninfected mice. Depleting Tregs did not alter BALF IL-6, neutrophil, or AREG levels. Both Treg-sufficient and Treg-deficient mice exhibited severe injury, Krt5+ regions, and Krt8hi cells consistent with abnormal alveolar architecture and repair. These findings suggest juvenile Tregs are insufficient to mitigate inflammation and promote repair during severe viral pneumonia. Further understanding of juvenile Treg function may inform strategies to improve pediatric recovery from severe respiratory infection.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 10P

Lange, Joshua Faculty Advisor: Hao Li

Neurotensin circuit regulation under approach-avoidance conflict

In nature, seeking rewards is essential for survival, but rarely comes without risk. Food, shelter, and social connection are all important rewards that can require risk taking. How the brain decides when a reward is worth seeking is an essential question that underlies important disorders such as anxiety and substance use disorders (SUDs). This project investigates how Neurotensin (NT), a neuropeptide highly expressed within the paraventricular nucleus of the thalamus (PVT), modulates reward and aversive behavior. To study this, our lab developed the Approach-Avoidance Conflict (Conflict) Task in which mice are trained to seek a sucrose reward while receiving progressively stronger foot shocks, an environment that weighs a decision between desire and risk. We used video-based scoring to quantify mouse risk assessment behaviors, reactivity to foot-shocks, and freezing. In addition, projections from NT neurons located in the PVT to nearby regions were mapped in the mouse brain. Through this, we've established that both male and female mice respond similarly to the Conflict Task. In addition, mapping has allowed us to establish that NT neurons in the PVT project to nearby regions, including the central amygdala (CeA) which has been established as a high neurotensin receptor 1 (NTSR1) region. This work helped to design a preclinical behavioral model in mice to understand risk averse reward seeking, a behavior that when maladaptive can contribute to psychiatric disorders. Looking forward, this Conflict Task can also be used to study other potential neural targets and their role in orchestrating behavior.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 16P

Lanning, Mackenzie Faculty Advisor: SonBinh Nguyen

Graphene Oxide as an Environmentally Friendly Alternative to PFAS and Plastic in Paper Products

Water- and oil- resistant barrier coatings are crucial for food-packaging materials, yet many commercial products rely on plastics and polyfluoroalkyl substances (PFAS), which pose significant risks to both human health and the environment, to create these properties. As PFAS and plastics are being banned in many countries, our lab has been focused on finding alternatives to these materials, especially for paper coatings that can be broadly used across *all* food-packaging materials, given the renewable nature of paper.In this study, we investigated a graphene oxide (GO)-based coating formulation designed to increase the water resistance and mechanical strength of recycled Kraft paper commonly used in grocery bags. Using water-contact angle measurement, tensile testing, and macroscopic absorption procedures, we have demonstrated that GO-coated recycled paper has significantly increased water resistance and mechanical strengths relative to uncoated controls. The combination of these two properties could greatly improve the resilience of consumer food-packaging products (e.g., paper grocery bags) under conditions where they typically fail, such as exposure to moisture and water. Our results indicate that GO-based coatings have the potential to extend the functional lifetime of recycled paper products, reduce product failures, and decrease the amount of material needed to achieve proper durability. As GO can be made from agricultural waste, GO-based barrier coatings represent a promising, environmentally sustainable alternative for the next-generation of food-packaging applications. Showcasing this project as an oral presentation will allow me to more clearly communicate the broader environmental significance and the urgency of developing sustainable alternatives, while engaging directly with a wider audience about building a greener future for food-packaging materials.

Presentation Format: Oral
Time: 10:45AM-12:15PM
Room: Lake Room

Lao, Ethan Faculty Advisor: Neha Kamat

Optimization of Lipid Nanoparticle Formulation for Large Plasmid DNA Delivery

Lipid nanoparticles (LNPs) offer a non-viral platform for the delivery of nucleic acids, seen predominantly during COVID-19 vaccine development. The vaccine used messenger ribonucleic acid (mRNA), the most common cargo used in LNPs due to their small size and high loading efficiency. LNPs can also be loaded with other cargo, including plasmid DNA (pDNA). pDNA is cheaper and easier to purify than mRNA, can carry multiple genes in a plasmid construct, and enables promoter-specific cell targeting. Large pDNA plasmids can encode genes, CRISPR-Cas9 components, and cell-specific promoters into one construct, but as they get larger, encapsulation efficiency drops. LNPs are formed through mixing pDNA with the lipids needed for the LNP structure using a microfluidic mixer to control the formation flow rate. We have reason to believe that the lipids involved and formation flow rate impact large pDNA encapsulation. Thus, there is a need for methods to efficiently load large plasmids into LNPs. First, the pDNA LNP model's ability to encapsulate pDNA using a smaller plasmid, CMV EGFP (4.5 kbp), was verified with varying formation flow rates. The LNP lipid composition was chosen based on a SM-102 formation reported in literature. LNP size and polydispersity was measured using a Malvern Zetasizer. A PicoGreen plate reader assay was used to find the pDNA loading efficiency. The pDNA LNPs ranged in size from 60-100 nm depending on formulation flow rate. Encapsulation efficiency varied between 87-97% throughout all flow rates. A DNA gel also confirmed no leakage of pDNA from the LNPs.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 83P

Latifova, Leyla Faculty Advisor: Simone Campanoni

Persistent Translation Validation

With the emerging uses of AI to automate various software engineering tasks, one such use case is integration of models into compilation pipelines. Due to intrinsic stochastic properties of LLMs, the correctness of program transformations no longer relies on time-tested compilers, turning them very untrustworthy. Validating those transformations increases reliability, but not without a cost; such a computation takes considerable amount of time.

There is however redundancy as the program goes through a chain of transformations. As an example, in the vast majority of programs, a pass through a language model would preserve the program's memory backbone; it is also the hardest constituent to verify. Identifying such redundancies would greatly reduce the problem size, and we therefore propose to break down programs into persistent and non-persistent components, with only the need validate the latter with each new transformation. Therefore, this work enables AI transformation passes to be integrated into compilation pipelines without an overwhelming overhead.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 78P

Latz, Jason Faculty Advisor: Almaz Mesghina

Romance and Report Cards: The Effect of Romantic Relationship Commitment on Perceived Academic Performance

Romantic relationships and academics play significant roles in college students' lives. However, researchers disagree whether romantic relationships can facilitate or impede a student's perceived academic success, and most research focuses on GPA, rather than student-reported measures, as a proxy for academic performance. We sought to determine, first, whether the presence of a romantic relationship predicts perceived academic performance, and second, whether the relative seriousness of one's romantic relationship predicts perceived academic performance. Undergraduate students (N = 78) were surveyed on their romantic relationship status, relationship seriousness, and academic performance through a series of questionnaires. Results indicated that neither the presence of a romantic relationship nor the seriousness of that relationship significantly affected perceived academic performance. Post hoc, we found a significant positive correlation between sleep duration and academic performance, meaning students who reported more hours of sleep felt they performed better academically. A subsequent linear regression analysis (controlling for sleep) suggested that sleep may mediate the relationship between romantic involvement and academic outcomes. Our findings show that students in romantic relationships are not necessarily academically disadvantaged, although students who sleep longer durations are likely academically advantaged. These insights help students balance academic and interpersonal commitments to fulfill their personal goals.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 16A

Lee, Caden Faculty Advisor: Samuel Stupp

Investigating the effects of bioactive peptides on peptide amphiphile nanofibers on an organoid spinal cord injury model

Spinal cord injury often results in lifelong sensory and motor dysfunction due to the poor regenerative capacity of the central nervous system and the formation of an inhibitory glial scar in the injury site that causes degeneration of white and gray matter. A leading cause for this is the increase of chondroitin sulfate proteoglycans (CSPGs) surrounding the cystic cavity which upon binding to protein tyrosine phosphatase (PTP) receptors (PTPσ) inhibits regeneration. Interestingly, this roadblock to recovery has been reversed by a membrane-permeable peptide that deactivates PTPσ. Despite this therapeutic potential, the short systemic circulation of these peptides poses a significant clinical burden, requiring daily injections for months after injury. To prolong the bioactive effects of this peptide, a novel system has been developed in which these cell-permeating peptide "sliders" associate dynamically with peptide amphiphile supramolecular nanofibers through electrostatic interactions. This system forms an injectable hydrogel that provides both mechanical support and biological potency to support neuronal growth by prolonging the effects of PTPσ-targeting peptides. Here, we investigate the biological potential and mechanisms of the "sliders" in a clinically translational 3D human organoid model. Through this study, it was discovered that treatment with "sliders" significantly enhanced axonal growth in the sites of injury. It was also discovered that delivery of peptides by peptide amphiphiles were highly specific to the site of injury and showed high levels of cellular internalization. These findings demonstrate the significant potential of the novel "sliders" for future *in vivo*, and ultimately therapeutic applications.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 28A

Lee, Chloe Faculty Advisor: Andrew Miri

Investigating Motor Thalamus Encoding of Forelimb Muscle Activity in Mice

The motor thalamus (Mthal) is a key brain structure important for motor learning and motor execution. Mthal influences movement by projecting to the primary motor cortex (M1), a region that plays a primary role in driving muscle activity to generate movement. While evidence suggests that Mthal provides "high-level" direction to M1 that specifies the identity of an upcoming movement (e.g. a reach or a pull), it may also convey "low-level" direction, encoding patterns of muscle activity during movement execution similar to M1. Due to a lack of simultaneous neural and muscle activity recordings, the extent to which Mthal conveys high versus low-level information to M1 is unclear.

To address this gap, the Miri lab collected simultaneous Mthal, M1, and muscle activity recordings in mice performing a naturalistic climbing task. Using these recordings, I sought to compare the amount of muscle predictive information conveyed in both Mthal and M1. To do so, I fit ridge regression models, a regularized form of linear regression, to predict muscle activity from past neural activity in either M1 or Mthal. When muscle activity was predicted using equal amounts of neurons in M1 and Mthal, I found that Mthal was similarly predictive of muscle activity as M1.

This suggests that Mthal may convey substantially more low-level muscle-activity commands than previously expected. This challenges the current view that Mthal primarily encodes high-level movement and offers new insight into how the brain generates the patterns of muscle activity that underlie movement.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 73A

Lee, Ethan Faculty Advisor: Nozomu Takata

Engineering Vascularized Spinal Cord Organoids through Bioactive Peptide Amphiphile-Mediated Vascular Cell Integration

Organoids are revolutionary miniature human tissue models made from pluripotent stem cells that replicate the native tissue structures and functions of human organs. However, organoids that lack a vascular system for gas and nutritional exchange will eventually experience necrosis or poor differentiation in the interior as they grow larger in size. To address this limit, this project focuses on engineering vascularized spinal cord organoids (SCOs) by integrating vascular cells using bioactive peptide amphiphiles (PAs). In this study, preliminary trials were conducted to determine the optimal concentration and type of bioactive PAs, designed to promote vascular growth and proliferation. Histology markers for proliferation, morphology, and vascular receptor activation determined that the ideal treatment conditions consist of micelle-forming angiogenic PAs and fiber-forming diluent PAs. Following these trials, a 3D co-culture system was established by introducing vascular cells to SCOs. To optimize vascular cell integration and penetration into the organoid core, three distinct PA treatment timelines were examined: two days before vascular cell introduction, simultaneous application with vascular cells, and delayed treatment following co-culture. These co-cultures were maintained for 10 days to monitor the rate of vascular cell integration and proliferation. Results were evaluated through immunostaining and imaging to analyze vascular cell penetration and vascular receptor activation within the SCOs. Our experimental strategies would provide useful insights into vascular engineering in human spinal cord organoids by leveraging the potent bioactivity of PAs. These improved vascular organoid models will facilitate future studies of spinal cord injuries and the development of regenerative therapeutic strategies.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 27A

Lee, Grace Faculty Advisor: Alissa Chung

What We Say to Ourselves: Inner Speech, Neuroticism, and Psychological Flexibility

This study examined how neuroticism and psychological flexibility relate to the frequency and content of inner speech, with a focus on clarifying whether these relationships follow a moderation or mediation model. Understanding these links is important because inner speech is closely tied to emotion regulation and mental health, yet the mechanisms connecting personality traits to inner speech remain unclear. A demographically representative sample of adults in the United States was recruited through Prolific. Participants completed self-report questionnaires measuring neuroticism, rumination, psychological flexibility, and different aspects of inner speech, including total frequency and content. We tested both moderation and mediation models to evaluate how these variables are connected. The results did not support a moderation model in which neuroticism changes the effects of inner speech. Instead, findings supported a mediation model: higher neuroticism predicted greater rumination, which in turn was associated with increased total inner speech. There was no evidence for a "for better or worse" effect based on level of neuroticism. Psychological flexibility was not related to total inner speech but was associated with greater positive regulatory inner speech and was only indirectly related to lower rumination through its negative correlation with neuroticism. These findings suggest that total inner speech is largely driven by rumination, while positive regulatory speech reflects a separate process linked to psychological flexibility. This distinction helps refine current models of inner speech and points to different targets for future research and interventions aimed at improving emotional well-being.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 11A

Lee, Jennifer Faculty Advisor: Bin Jiang

Modeling Peripheral Artery Disease with Engineered Skeletal Muscle Tissue

Peripheral artery disease (PAD) is characterized by arterial blockages that reduce blood flow to the limbs causing nutrient and oxygen deficiencies in skeletal muscles. Of the 200 million people affected worldwide, half experience muscle loss, tissue damage, and declines in mobility and quality of life.

PAD research primarily relies on 2D cell cultures, which cannot replicate tissue-level behavior. Engineered muscle tissues (EMTs) overcome this by combining myoblast cells with a hydrogel to recreate the cellular organization and environment of native skeletal muscle tissue, enabling tissue-level observations. Since EMTs can be grown using various hydrogel compositions, we aimed to identify conditions that provided a straightforward and reproducible setup.

EMTs were generated on PDMS posts, to simulate muscle attachment between bones, using two hydrogel formulations: (1) 3 U/mg thrombin + 4 mg/mL fibrinogen + 5 mg/mL Matrigel (2) 4.1 mg/mL rat tail type I collagen + 10 mg/mL Matrigel. Surface treatments of posts were also tested to determine effects on cell adhesion and viability.

None of these conditions produced EMTs attached to the PDMS posts, although the collagen-based hydrogel successfully assembled as a thin sheet, and was the simplest and most consistent to work with. Future work will involve optimizing PDMS functionalization, characterizing the makeup of the collagen hydrogel using histology, and adding fibroblast cells to improve tissue formation. Optimization of the EMT generation process enables observations that more closely reflect PAD pathology in humans without the cost, time constraints, and ethical concerns associated with animal models.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 17P

Leon, Sydney Faculty Advisor: Laurie Shannon

Playing the Supplicant: The Usurpation of Flattery in Shakespeare's Richard III

In William Shakespeare's *Richard III*, the villain, Richard, employs flattery as his method to usurp the throne and rule as a tyrant. Richard's manipulation tactic, flattery, traditionally thought of as a monarch's weakness, is unexpected given the teachings of classical and medieval scholars like Plutarch and Erasmus. Flattery, which according to these thinkers was a vice monarchs were instructed to avoid, becomes a tool for Richard to exploit. My thesis explains why Richard's utilization of flattery, while initially a source of strength in Richard's ascension to the throne, is actually a sign of weakness and instability in his status as a ruler, and in his kingdom more broadly. I analyzed classical, medieval, and modern scholars to understand the significance of flattery within kingship, and more specifically, amongst Shakespeare's monarchs. After close-reading two scenes that, when compared, most capture the shift in flattery as Richard's strength to a symbol of his weakness, I found that it is Richard's status as a usurper that forces him to resort to flattery as a means to obtain the throne. Richard's illegitimate rise to power allows flattery to act as a strength while he works to secure the throne because his ascent is already based on murder and falsities. However, once in power, this unstable foundation of manipulation causes flattery to be an ill-equipped substitute for the paranoia and death that Richard has caused. Framing flattery as Richard's weakness, rather than strength, allows for Richard's tyranny to be more than just a series of murders.

Presentation Format: Oral
Time: 10:45AM-12:15PM
Room: Armadillo Room

Leon, Carlos Faculty Advisor: John Rogers

Wearable Sweat Sensor for Chronic Kidney Disease Biomarkers

Chronic Kidney Disease (CKD) is a global health burden that affects hundreds of millions of people and strains healthcare systems with rising treatment costs. As the condition advances, clinical severity increases exponentially, yet many individuals remain undiagnosed due to the invasive nature and high cost of traditional blood-based screening. This barrier is particularly evident in low-resource environments where limited clinical infrastructure prevents early intervention. Consequently, there is an urgent need for a cost-efficient, non-invasive early screening tool to identify health changes.

Our research compared the concentrations of urea and creatinine biomarkers in sweat and blood using laboratory standards (LC-MS) to establish sweat as a reliable substitute for monitoring kidney health. Next, we developed a wearable colorimetric sensor featuring an on-demand, local sweat stimulation feature and two distinct detection pathways: a pH- compensated system for urea and a spectrophotometric system for creatinine. The device utilizes colorimetric assays for qualitative screening via the naked eye, while color analysis enables quantitative detection of biomarker concentrations.

Quantitative validation showed strong linear trends between sweat and blood biomarkers ($R^2 = 0.89$ for creatinine; $R^2 = 0.79$ for urea). This data demonstrates great potential in classifying different stages of CKD, which is vital for early screening. Future work will focus on optimizing colorimetric assay performance and designing user-friendly device form factors for on-human validation to evaluate sensor stability. This innovation aims to empower patients with autonomy and provide accessible screening to improve the quality of life for those at risk of CKD.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 18P

Lev, Emmy Faculty Advisor: Amy Paller

Sensory Neuron and Mast Cell Interaction in Itch Responses of Recessive Dystrophic Epidermolysis Bullosa (RDEB)

Recessive Dystrophic Epidermolysis Bullosa (RDEB) is an inherited blistering disorder that is characterized by inflammation and severe itch. Our preliminary data showed increased mast cells from the immune system and a specific type of sensory neuron called Mas-related G-protein-coupled receptor D (MrgprD). While MrgprD-expressing neurons are typically involved in itch-related behaviors, their role in RDEB has not been well studied. In normal skin, MrgprD+nerves communicate with mast cells through a molecule called glutamate to reduce their activation, thus reducing excessive skin inflammation. However, in injured tissue, glutamate has been shown to increase inflammatory signaling in mast cells. We hypothesize that RDEB promotes increased MrgprD+ nerve density in the outermost layer of the skin and enhances MrgprD+ interaction with mast cells, contributing to itch and pain. To study this, we used an RDEB mouse model to visualize the nerves and examined their proximity to mast cells in skin and dorsal root ganglia (DRGs), a cluster of neurons responsible for relaying sensory information to the brain. Tissues were stained to identify activated mast cells and imaged for quantitative analysis. This analysis showed a significant 18-fold increase in the density of MrgprD+nerves in the outermost layer of the skin, and these nerves were located significantly closer to mast cells. Additionally, RDEB DRGs had an significant increase in mast cell specific mRNA levels. These results suggest the increase in MrgprD$^+$ density and number of mast cells may contribute to the persistent inflammation in RDEB.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 14P

Levine, Lea Faculty Advisor: Doris Warriner

And Here I Am!" Chronotopic Memories in a Holocaust Survivor's Autobiographical Narratives

At another family gathering I felt a relative's hands on my cheeks. In awe, he uttered "*pontosan ugy nez ki az arcod, mint a dednagyanyade*." This Hungarian phrase roughly means that my face looks exactly like my great-grandmother's. I was named after her son and my grandfather, Dr. Leslie Koltai. Along with his name, I inherited recordings of his story. Born in Karcag, a small town in Eastern Hungary, Koltai survived the Holocaust, then became an anti-fascism journalist during the Hungarian Revolution. Later, fearing persecution, he escaped the country and settled in the United States. This thesis analyzes how Holocaust testimonies change over time and how these changes inform self-understanding. Drawing on Bakhtin's theory of the chronotope, a literary device that attends to "space-time" in narrative, I analyze how Koltai's recollections illuminate not only his lived experiences but also processes of sense-making with particular attention to the mobilization of time as a resource. Findings show that temporality serves as a resource for understanding the past in relation to the present and advances a narrator's understanding of the possibilities of the imagined future. However, key aspects of the narrator's understanding of his life experience do not change and are consistent across both tapes. This work demonstrates the importance of recording survivors' second and third testimonials and the value of attending to the holistic nature of the experiences recounted. Future research directions might involve analyzing multiple testimonials from survivors of conflict to show how these experiences are understood by future generations.

Presentation Format: Oral
Time: 12:45-2:15PM
Room: Armadillo Room

Lin, Steve Faculty Advisor: Xiaomin Bao

Potential Protectors of Genomic Stability in Keratinocytes

Cutaneous squamous cell carcinoma (cSCC), which originates from keratinocytes in the epidermis, is characterized by the highest mutation rate of any cancer, making it difficult to identify cancer-driving mutations, and develop targeted treatments. Microsatellite instability (MSI) contributes to genomic instability and mutation. Microsatellites are regions in the DNA where a 1-6 base pair sequence repeats many times. During cell division, errors are more common at these sites, but are fixed by DNA repair proteins like MSH6. ACSS2 could potentially preserve genomic stability in keratinocytes, since decreased ACSS2 expression was observed in skin cancer cells. ACSS2 is an enzyme that catalyzes the conversion of acetate to acetyl-CoA, to provide a source of acetyl groups for post translational modifications (PTMs) to proteins which alter their functions. MSH6 is a protein that forms a complex which is stabilized by acetylation. To study if ACSS2 maintains microsatellite stability by stabilizing MSH6, shRNA was used to knock down expression of ACSS2. The DNA was then extracted, amplified, and loaded into gels to observe irregular sized DNA fragments, which would indicate MSI. A plasmid, a circular piece of DNA, is also being developed that uses fluorescence to signal the presence of MSI. Fluorescent imaging of keratinocytes infected with the reporter plasmid confirm presence of the fluorescence as expected, but it has yet to be evaluated whether it is due to the presence of MSI. Findings to this project would reveal potential pathways to genomic instability in cSCC, providing a foundation for research for targeted therapies.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 89P

Liu, Xinyu Faculty Advisor: George Schatz

Simulating the Phase Behavior of Polydisperse DNA-Nanoparticle Systems

Nanoparticles that are surface-functionalized with DNA strands can act as programmable atom equivalents (PAEs), where the nanoparticle core behaves analogously to an atom and DNA hybridization defines the interparticle interactions. By manipulating PAE sizes and DNA sequences, the PAEs can form crystals with controllable lattice symmetry and nanoparticle spacing. The assembly of polydisperse PAE systems into a single phase has proven to be challenging due to the thermodynamic instability caused by size disparity. Recent experimental attempts have shown some success in single-phase crystallization by introducing flexible spacers within the DNAs to make the effective hydrodynamic radii more similar. However, it is unclear what the limits of the flexible spacers are in overcoming the effects of size disparity. Understanding the phase behavior requires systematic and quantitative analysis, which is extremely challenging experimentally due to the large number of systems that need to be synthesized and investigated.

In this work, we construct a phase diagram of polydisperse PAE systems using molecular dynamics (MD) simulations with two different models: a colloidal model and a coarse-grained DNA model. These models can provide crucial thermodynamic information with relatively low computational cost. The colloidal model is used to determine the boundaries of the phase diagram, while the coarse-grained model is used to explain any discrepancies with the experimental data and provide a detailed understanding of the thermodynamics of the interactions. In doing so, we explain the effects of nanoparticle core diameter, spacer flexibility, and nanoparticle type on crystallization, and provide guidelines for experimental synthesis.

Presentation Format: Oral
Time: 12:45-2:15PM
Room: Lake Room

Long, Kumi Faculty Advisor: Adrian Rodriguez Contreras

A Rat Model of Neonatal Encephalopathy: Exploring the Effects of Oxygen Deficiency on Central Auditory Function

Prematurely born infants commonly experience oxygen deficiency due to episodes of asphyxiation before, during, or shortly after birth. This contributes to neonatal encephalopathy, a condition of neurological dysfunction that is linked to hearing loss and other communication impairments. However, the underlying mechanisms of these deficits and their developmental trajectories are poorly understood. Our lab is investigating the impact of oxygen deficiency on auditory function depending on age of exposure and oxygen reduction magnitude. We used auditory brainstem response (ABR) testing to evaluate 21-day-old rat pups who were exposed to 10 minutes of anoxic (&1% oxygen) or normative (~21% oxygen) treatment on two consecutive days. Another group of pups was exposed to 11% oxygen to assess the effects of milder oxygen reduction. ABRs with 4, 8, 16, and 32 kHz tone burst stimuli were processed using an automated R-script and peak I-III latencies were compared between and within groups. When pups were exposed to anoxia at birth and postnatal day 1, 100% survived and preliminary analysis revealed delayed latencies at 8 kHz in comparison to the control and 11% oxygen groups. However, when pups were exposed at postnatal day 6, 100% died and ABRs were unobtainable. These results suggest that a greater magnitude of oxygen reduction amplifies the detrimental impacts of oxygen deficiency on neurological and auditory processes, but further experimentation is needed to understand the role of exposure age. Our findings may also clarify the implications of preterm births on hearing and communication development.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 54A

Long, Zach Faculty Advisor: Dayne Swearer

Dilute Plasmonic Copper Alloys: Leveraging Surface Engineering for Enhanced Catalysis

Modern catalysis is increasingly focused on making chemical processes more sustainable, including using light as an energy source. Dilute plasmonic alloys are a promising class of materials for this purpose. These nanoparticles smaller than the wavelength of light combine a light-responsive metal with small amounts of a catalytically active metal, allowing them to both absorb light and drive chemical reactions. Systems such as copper–platinum and copper–rhodium have shown promise for propane dehydrogenation, an important reaction that produces propylene, a key industrial chemical. However, replacing expensive metals with more abundant, non-precious alternatives remains a major challenge.

Here, we study copper–cobalt nanoparticles as a low-cost alternative. We synthesize 2–3 nm particles and show that their surface composition changes depending on the surrounding gas and temperature. Using solid-state infrared spectroscopy (Diffuse Reflectance Infrared Fourier Transform Spectroscopy), we find that hydrogen pushes cobalt below the surface, while carbon monoxide at 150 °C and oxygen at 350 °C pull cobalt to the surface, properties that are consistent with theoretical properties of cobalt, and are accompanied by an initial enhancement in catalytic rates with surface cobalt. These shifts are reversible and stable over time.

These surface changes are also linked to changes in catalytic activity, suggesting that the active sites can be tuned by controlling reaction conditions. This work demonstrates that copper–cobalt nanoparticles can serve as a model system for designing catalysts with adjustable active sites. More broadly, it highlights the potential of earth-abundant materials for efficient and controllable catalytic processes.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 20P

Louis, Sarah Faculty Advisor: katrina quisumbing king

Territorial Membership: U.S. Naturalization Law and Defining Who Can Be American, 1897-1938

This study investigates how federal judges decided on eligibility for citizenship for people in US territories and what rationales they employed to justify their decisions. This work expands the criteria of Ian Haney-López's book, *White by Law*, in which he investigates how race is legally defined through naturalization cases. Haney-López limits his study to cases where race is the sole deciding factor for citizenship as opposed to territorial status, procedural complications, or jurisdiction. However, expanding his criteria to consider how judges reasoned through both territorial status and race is important because it highlights how legal racial boundaries in the United States were reconfigured as the U.S. empire expanded. To identify patterns in legal reasoning, I read and coded decisions from eighteen naturalization cases and compared my codes to those of another coder. We then compared the case decisions and justifications with the territorial status of the petitioner. Our analysis revealed that when there is an ambiguous territorial status and no legal precedent, judges speculate the intentions Congress had when writing the naturalization acts. Furthermore, stemming from contradictory acts of Congress, judges scrutinized particular jurisdictions such as the Philippines. This study has implications for the sociolegal understanding of how citizenship status and eligibility is understood in the United States empire. Our study suggests that definitions of race, territorial status and naturalization eligibility that were legally restrictive created systems of racially discriminatory exclusion from naturalization, or discriminatory social ideas of who could be American for the petitioners studied.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 05A

Lowenthal-Wojcik, Payton Faculty Advisor: Mesmin Destin

First-Year Transition Programs: an X Factor for the FGLI College Experience

In PhD candidate Rebecca Lawrence's study, she explores the relationship between participation in first-year transition programs (FYTPs) and mental well-being among first-generation/low-income (FGLI) students of color. She interviewed a group of students who participated in a three-week FYTP at a highly selective, predominantly white institution (PWI) and a non-FYTP control group. The FYTP aims to help FGLI students acclimate to university life both socially and academically through mini-courses, mentoring, and resource workshops. This study addresses persistent feelings of alienation among FGLI students and examines how the FYTP supports belonging and mental health outcomes. As an Emerging Scholar, I cleaned transcripts, conducted thematic coding, and wrote summary memos to identify trends in student experiences. These trends would inform FYTP coordinators of areas for improvement while also encouraging other institutions to implement bridge programs for incoming students. Our findings show how, compared to the non-FYTP control group, participants reported stronger support systems, a more critical reception to general orientation, and improved academic adjustment. This work reveals the importance of comprehensive FYTPs for FGLI students. By fostering social networks in which FGLI students connect with each other and preparing students academically, FYTPs may soften the potential whiplash of transitioning to college. The findings suggest that implementing FGLI bridge programs and continuing support for students even after the program's completion, may improve equity and student outcomes across institutions.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 04A

Lu, Michelle Faculty Advisor: Almaz Mesghina

Romance and Report Cards: The Effect of Romantic Relationship Commitment on Perceived Academic Performance

Romantic relationships and academics play significant roles in college students' lives. However, researchers disagree whether romantic relationships can facilitate or impede a student's perceived academic success, and most research focuses on GPA, rather than student-reported measures, as a proxy for academic performance. We sought to determine, first, whether the presence of a romantic relationship predicts perceived academic performance, and second, whether the relative seriousness of one's romantic relationship predicts perceived academic performance. Undergraduate students (N = 78) were surveyed on their romantic relationship status, relationship seriousness, and academic performance through a series of questionnaires. Results indicated that neither the presence of a romantic relationship nor the seriousness of that relationship significantly affected perceived academic performance. Post hoc, we found a significant positive correlation between sleep duration and academic performance, meaning students who reported more hours of sleep felt they performed better academically. A subsequent linear regression analysis (controlling for sleep) suggested that sleep may mediate the relationship between romantic involvement and academic outcomes. Our findings show that students in romantic relationships are not necessarily academically disadvantaged, although students who sleep longer durations are likely academically advantaged. These insights help students balance academic and interpersonal commitments to fulfill their personal goals.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 16A

Ma, Katelyn Faculty Advisor: Debra Weese-Mayer

Toward Autonomous Breathing: Predicting CO2 in Children with CCHS Using Wearable Sensors

Breathing is something that most people rarely think about, seamlessly adapting to changes in sleep, exercise, or everyday movement. But for children with Congenital Central Hypoventilation Syndrome (CCHS), this automatic control is impaired, requiring reliance on assisted ventilation to breathe.

Advances such as phrenic nerve stimulation have enabled patients to live longer and more active lives by restoring diaphragm function. However, these systems still depend on manual, clinician-driven adjustments, limiting their ability to respond in real time to changing metabolic demands.

This project investigates whether a patient's ventilatory needs can be anticipated before physiological signals reflect these changes. Transcutaneous CO2 (tcCO2), a key clinical measure of ventilation, is inherently delayed due to physiological and diffusion constraints, restricting its use in real-time control.

To address this, we developed a predictive framework using noninvasive wearable signals. We built upon an existing sensing platform and designed an experimental protocol to capture interactions between ventilation and activity. Using synchronized expired CO2 (etCO2) and accelerometry data, we trained a machine learning model (CNN to biLSTM) to learn these physiological patterns.

The model was able the identify the delay between signals and predict future CO2 levels with mean absolute errors of 0.55-0.58 mmHg. These results demonstrate the feasibility of predicting clinically meaningful CO2 dynamics from wearable data, providing a foundation for closed-loop, anticipatory control of phrenic nerve stimulation. Future work will focus on building a large, diverse physiological dataset across individuals, enabling personalized and adaptive respiratory support across age, activity level, and patient variability.

Presentation Format: Oral
Time: 12:45-2:15PM
Room: Lake Room

Macy, Annika Faculty Advisor: David Uttal

Living and Learning: Exploring the influence of place on students' communication about climate change across time during a local field trip

Local environmental education is important to discussions about the future of our climate, which can be overwhelming for students. As the climate changes, every community will encounter unique challenges affecting their lives, so we should incorporate relevant environmental information into educational content. Thus, this project explored place-based learning in an informal context through a study of fifth and sixth grade students' reflections on a field trip about marine ecosystem change in the Gulf of Maine.

To investigate how place influences students' discussion of environmental phenomena, transcript coding was used to categorize and quantify the words students used. Independent-samples t-tests and one-way ANOVAs were used to compare word types between school locations and across three school years. A subset of transcripts was also considered qualitatively to identify key themes students discussed.

Word use varied significantly more between coastal and inland schools than between urban and rural schools. Inland groups spoke more frequently and used more data analysis and marine species words. Students who used emotional language also used more data analysis terms.

Equipping middle school students with the tools to understand and communicate scientific concepts may better prepare the next generation to address specific local effects of climate change. Furthermore, by understanding smaller-scale concepts, students can better recognize their role in the broad scope of climate change. This research shows the potential benefits of considering student identity and place on learning about the environment in informal settings.

Presentation Format: Oral
Time: 10:45AM-12:15PM
Room: Arch Room

Magiera, Faith Faculty Advisor: Jeffrey Masten

Renaissance Mythological Afterlives: Transgressing Allusionary Bounds of the Lovers/Victims of Jove's Desire

Helen, the face who launched a thousand ships, is a figure who emerges in Homer but endures into the English Renaissance. Shakespeare, Marlowe, Jonson, and their English Renaissance drama contemporaries overflow with allusions to female mythological figures. Renaissance drama's classical connection has provoked scholarship for decades, but the comparative study of allusions to mythological women, and particularly the human female targets of Jove's (Jupiter's) violent and deceptive desire, is an under-researched topic that I addressed with my comparative literary analysis of allusions to four lovers/victims of Jove: Danae, Europa, Leda, and Semele. Lovers/victims is a hybrid term that I coined to encompass the divergent, paradoxical characterizations these female figures take up in classical mythology as lovers of Jove and also victims of his violating, omnipotent desire. Through reading the story of the lovers/victims in classical corpus in tandem with the allusions to the lovers/victims across a broad sample of Renaissance drama, I found that female characters in Renaissance drama engage in a conflict-ridden conversation with these classical allusions that embody patriarchal expectations of female embodiment. This point of conflict leads to the emergence of a different form of female subjectivity that emphasizes transformation, both of speech and embodiment. This project presents a new framework for viewing female subjectivity in Renaissance drama that acknowledges these classical allusions not as mythological background noise, but as a foundational place where female embodiment emerges differently in literature and culture itself through a dialectical conversation with the classically-constrained past.

Presentation Format: Oral
Time: 10:45AM-12:15PM
Room: Armadillo Room

Man, Antonio Faculty Advisor: William Dichtel

Elucidation of Urethane Exchange Mechanism with DABCO

Thermoset polyurethanes (PUs) are commonly used for applications such as insulation and seat cushions. However, these polymers contain chemical cross-links that make them difficult to recycle through traditional recycling methods. To allow for PU reprocessing in a feasible and scalable manner, dynamic cross-links have been introduced to thermoset PUs, forming PU covalent adaptable networks (CANs). Previous works have found that residual catalysts from the production of commercial PUs, such as 1,4-diazabicyclo[2.2.2]octane (DABCO), can catalyze dynamic urethane exchange reactions, thereby creating PU CANs directly from post-consumer waste. By understanding the mechanism of DABCO-catalyzed urethane exchange, we can gain control over the bond exchange reactions and optimize reprocessing for different PU products. To investigate how DABCO-catalyzed PU CANs undergo bond exchange, I conducted a small molecule study which modeled urethane exchange in the presence of DABCO and excess alcohol. When a small amount of alcohol was added, increasing the concentration of alcohol accelerated the rate of DABCO-catalyzed urethane exchange. However, at higher free alcohol concentrations, the catalytic activity of DABCO was inhibited. I observed an initial period with a slow reaction rate, suggesting that an activated catalyst is formed. I found that urea, a side product created from the reaction between urethane and trace water, caused the reaction to exhibit behaviors consistent with the observed initial period. Since commercial PUs contain a wide variety of urethane, urea, and alcohol groups, I hope to apply these fundamental findings to better reprocess post-consumer PU waste.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 84A

Mateu-Asbury, Anya Faculty Advisor: Jen Munson

Hunger and Learning: How Food Deserts and Food Insecurity Shape Academic Success

In Chicago, one in five households with children experiences food insecurity, and over half a million residents live in neighborhoods designated as food deserts (Gallagher, 2006; Greater Chicago Food Depository). This mixed-methods honors thesis investigates the relationship between food access and academic success among school-age children and college students. Quantitative analyses use Chicago Public Schools data from 2018-2019 and U.S. Department of Agriculture food access data to test whether low access to nutritious food predicts academic success. Linear regression models explore associations between food desert prevalence and academic indicators, controlling for poverty and racial composition across neighborhoods. Complementing these analyses, ten in-depth interviews with Northwestern students who have experienced food insecurity provide qualitative insights into the lived experiences of student hunger. Interviews reveal how food insecurity affects energy levels, emotional wellbeing, and concentration, and how intersecting identities-such as race, class, and immigrant status-shape these experiences. Results are expected to demonstrate that residing in areas of low food access correlates with lower rates academic success and that students' narratives underscore the long-term educational consequences of food insecurity. By integrating statistical and narrative data, this study illuminates how structural inequalities in food access reinforce educational disparities and highlights the urgent need for policies addressing both nutritional and academic support for food-insecure youth.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 08A

McAllister-Nevins, Olivia Faculty Advisor: Leslie Harris

Race and Real Estate: Nathan William MacChesney and the Consequential Establishment of Northwestern's Streeterville Campus

This project examines the history of Northwestern University's medical schools' move to Streeterville in the 1920s and the enduring consequences for both the city's landscape and the institution's development. The relocation from the South Side was not merely logistical, but strategic, positioning Northwestern as a foundational presence in what became some of the most valuable real estate in Chicago. As the catalyst of this move, Nathan William MacChesney, Northwestern Board of Trustees member and prominent Chicago lawyer, is the research focus. Using archives from the McCormick Library and the Chicago History Museum, this study uses Board of Trustees meeting minutes, correspondence involving the Chicago campus, published writings and addresses of MacChesney, and Sanborn fire insurance maps to trace the physical transformation of the Near North Side. These sources reveal MacChesney's extensive influence in the relocation and in shaping the surrounding area. MacChesney's role extended beyond university planning into developing restrictive covenants, private legal agreements used to enforce racial segregation, and eugenic thought and criminological institutions. Archival evidence indicates likely self-dealing in his promotion of University land purchases adjacent to his own property. The recent opening of a new Northwestern Medicine clinic in Bronzeville marks a return to the South Side, providing a contemporary moment to reflect on the origins and implications of this earlier move. By situating Northwestern's expansion within these broader systems, this project contributes to conversations about how institutions reckon with complex histories, highlighting the entanglement of educational growth, urban development, and racial inequality.

Presentation Format: Oral
Time: 12:45-2:15PM
Room: Rock Room

McNiel, Jack Faculty Advisor: Matthew Kay

Projecting the Public: A Framework for Constructing Empirical Visualization Studies of Broader Audiences

Since the early 2000s, researchers have called for extending studies of how people engage with visualizations to broader, non-expert audiences, yet existing frameworks for designing such studies were built largely for domain experts. Broader-audience work currently represents only 2.1% of recent visualization publications, and studying these audiences poses distinct challenges that existing methodology fails to address, raising the question: What conceptual framework can help researchers design broader-audience visualization studies while anticipating the realities of doing so? To address this gap, we systematically reviewed 73 visualization research papers published between 2020 and 2024 across four premier venues, then conducted 15 semi-structured interviews with authors of such studies, asking researchers to describe their study design decisions, aspirations, and tradeoffs. Reflexive thematic analysis of transcripts revealed a persistent pattern of aspirations versus practical reality. From this, we derived two outputs: a three-part framework for constructing empirical visualization studies, namely (1) casting an audience into a study sample, (2) constructing an experimental setting to study participants in, and (3) capturing how audiences respond and engage, alongside a detailed account of researcher-described gaps between aspiration and reality for each element; e.g., broader recruitment created design tensions, while realistic settings introduced unexpected tradeoffs. Unlike prior frameworks built for specific and controlled domains, our framework addresses the challenges that broader-audience research demands, supporting a visualization research community that designs for diversity of experience and expertise, meets audiences in real-world contexts, and remains transparent about the practical limits of any individual study.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 72A

Meguerditchian, Sarine Faculty Advisor: Elizabeth Hurd

Recognition without Responsiveness: Comparing the Turkish-Armenian and Armenian-American Communities

This thesis examines how classificatory systems shape political possibilities for ethnic minorities by comparing Armenian minorities in the United States and Turkey. Drawing on constructivist theories put forth by Rogers Brubaker, Ian Hacking, and Michel Foucault, this project analyzes how distinct regimes - American multiculturalism and Turkish nationalism - produce convergent outcomes of political marginalization for the Armenian ethnic community. This comparative study is based on 14 semi-structured interviews with American-Armenians conducted over Zoom with participants from across the U.S. and 14 interviews with Turkish-Armenians conducted in-person in Istanbul. Interviews were supplemented with ethnographic observations, secondary sources in Armenian studies, and archival research. In the U.S., Armenian-American legal classification as "white" enabled integration and institutional success for early Armenian immigrants, but renders current Armenian-American foreign policy claims as narrow ethnic interest demands. This limits policy success beyond symbolic recognition. In Turkey, Armenians are officially recognized as a religious minority under the Treaty of Lausanne, allowing for institutions to remain alive and functioning. However, suppression of ethnic identity and the criminalization of genocide discourse constrain open political advocacy. Across both contexts, recognition operates less as a form of political responsiveness than as a mechanism that limits political voice. These findings challenge the assumption that recognition leads to accountability, showing that classificatory regimes can shape whether states aptly respond to the minority communities they claim to represent.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 60P

Menon, Anagha Faculty Advisor: Katie Insel

Basal Body Temperature as a Noninvasive Marker of Menstrual Onset and Hormonal Dynamics in Adolescents

Pubertal hormone surges play a crucial role in adolescent brain development; however, few neuroimaging studies account for neuroendocrine factors in their experimental design and analyses. This is largely due to methodological gaps related to reliability, cost, and invasiveness of traditional approaches. Basal body temperature has recently emerged as a potential proxy for hormonal shifts in adult menstrual cycles, yet this approach has not yet been validated in adolescents. This study investigates whether continuous body temperature tracking can serve as a noninvasive marker of hormonal fluctuations in adolescent females.

Data were drawn from a longitudinal neuroimaging study conducted by the Child and Adolescent Translational Science lab in adolescents (N = 4; F; age = 14.9 1.3 years). Nightly distal body temperature was recorded via Oura Ring, and saliva samples were collected over 5-7 weeks and analyzed for estradiol, progesterone, and testosterone concentrations. This study investigates whether fluctuations in temperature over time predict the onset of menstruation compared to self-reported dates, and whether a correlation exists between physiological and hormonal signals.

Preliminary analysis revealed that hysteresis-based temperature modeling accurately predicted menses onset in an initial participant. Further analyses will investigate whether predictive accuracy can be generalized across participants of varying ages and menstrual cycle regularity.

These findings suggest the possibility for continuous temperature tracking in offering a noninvasive and scalable method for capturing pubertal hormone dynamics, opening new pathways for ecologically valid measurements of adolescent development in endocrine research.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 79A

Messenger, Gabriella Faculty Advisor: Arvind Krishna

Hyperparameter Tuning in Causal ML Models: The Bayesian vs. Frequentist Approach

Machine learning's predictive power and flexibility have driven its integration with econometric methods, giving rise to the field of causal machine learning (causal ML), which focuses on estimating treatment effects in complex, high-dimensional settings. Despite promising advances, empirical applications of causal ML remain constrained by underexplored challenges in model configuration, particularly hyperparameter tuning, which is frequently ad hoc and inadequately studied. This thesis investigates whether Bayesian hyperparameter tuning outperforms standard automatic tuning methods in causal ML applications, using the X-learner as a framework for estimating conditional average treatment effects (CATEs). The analysis distinguishes two sequential objectives: a tuning stage, in which hyperparameters are selected to minimize cross-validated predictive error, and an evaluation stage, in which models are assessed on the accuracy of treatment effect estimation. The central question is whether predictive gains achieved through Bayesian tuning in the first stage translate into improved causal estimation in the second. Results confirm that Bayesian tuning yields consistent predictive improvements over automatic methods, consistent with the broader ML literature. However, evidence that these gains translate to more accurate treatment effect estimation is lacking, revealing a disconnect between the tuning and evaluation stages. This disconnect reflects a fundamental misalignment between the internal processes that guide ML model development and the primary objective of causal inference frameworks. As ML methods become more deeply integrated with econometric frameworks, aligning internal model metrics with causal estimation objectives emerges as a critical direction for future research.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 73P

Mews, Evelyn Faculty Advisor: Mark Aparece

Quantification, Spatial Analysis, and Source Attribution of Polynuclear Aromatic Hydrocarbons (PAHs) in Chicago Soils

Polynuclear aromatic hydrocarbons (PAHs) are a carcinogenic class of persistent organic pollutants found ubiquitously in soils. While a background survey of PAHs in the soil of the city of Chicago was conducted two decades ago, it lacked analysis of emission sources and human health risks, had poor resolution, and does not reflect the modernization of the city. This study assesses contemporary PAH contamination levels and distribution in Chicago's public green spaces to determine current sources of contamination and associated exposure risks. Seventy-six soil samples were collected from 22 different community areas and analyzed via microwave-assisted extraction and gas chromatography-mass spectrometry to quantify 16 PAH congeners. Results reveal that 97% of samples exceed residential soil screening levels set by the EPA for at least one carcinogenic PAH congener. Significant spatial gradients show samples adjacent to major arterial roadways have higher concentrations of PAHs. Source diagnostic ratios identify fossil fuel combustion, mainly from vehicular traffic, as the dominant source of contamination across the city. These findings suggest that, though modern vehicular traffic poses health risks to the citizens of Chicago through its atmospheric emissions of PAHs, the soil of Chicago's parks and public spaces acts as a contaminant sink and additional avenue for human exposure. These results provide an updated baseline of PAH contamination and demonstrate how industrial and vehicular fuel combustion contribute to environmental contamination and ultimately affect human health.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 37A

Milukas, Ava Faculty Advisor: Nicholas Hagar

Potential of AI Research Tool NotebookLM in Journalism

For journalists, answers to investigative questions are often hidden in extensive archives, poorly photocopied or complicated by archaic language. Archives can provide context for current topics; however, the process of digging through archives is time consuming and tedious, especially when journalists have to break urgent news. NotebookLM, powered by Google Gemini, is a closed-source AI tool that conducts note-taking and research. Contrary to open models like ChatGPT, NotebookLM only produces responses from the sources it's given, like publication archives. This study investigates the potential of using NotebookLM to maximize research efficiency for journalists. Using 20th-century issues of *The Times*, the study simulates common background research tasks in journalism- building timelines of events, summarizing detailed stories, and tracking the appearances of people and organizations across archives- to test how well NotebookLM can extract precise information from complex archival content. This examination reveals that while the AI tool can create coherent overviews and paraphrased summaries of stories, inaccurate or "hallucinated" outputs of data are an ongoing challenge with NotebookLM. Despite these accuracy issues, NotebookLM's ability to denote underlying sources does enable relatively quick fact-checking of AI responses, so the tool may still be useful to save time when processing large archives of information. When directing urgent and in-depth investigations, journalists must feel that they can use an AI tool in confidence. These findings from a research assistant in the School of Communication's Computational Journalism Lab are a first step toward understanding how journalists can best incorporate AI into their professional practices.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 31A

Min, Justin Faculty Advisor: Eyal Kimchi

Evaluating Human Pose Estimation Models in a Hospitalized Patient Population

Monitoring activity in patients hospitalized on neurological services can be crucial for timely detection of conditions such as falls, delirium, and epilepsy, but remains labor-intensive. Continuous video-based monitoring using pose estimation models offers a scalable alternative to optimize clinical assessments. Model performance in clinical populations, where variable postures and complex environments pose unique challenges, remains unexplored.

Three state-of-the-art pre-trained pose estimation models, YOLOv11, OpenPose, and MMPose, were evaluated in hospitalized patients undergoing video-EEG. Each model generated coordinates and confidence scores for facial and body keypoints. From each video, ten frames were randomly selected and manually annotated (17 common keypoints) as ground truth and used to define model-specific confidence thresholds. Sensitivity, specificity, balanced accuracy, and ROC curves were computed. Keypoint accuracy was calculated using Euclidean distance between ground truth and model detections.

We analyzed 46 videos from 24 hospitalized patients with diverse racial backgrounds. OpenPose showed the highest AUC in 52% of cases (24/46) and highest overall balanced accuracy (BA) of 72% (53% sensitivity, 89% specificity). OpenPose performed better for facial detection (BA 77%, 64%, 90%), while MMPose performed better for upper body (BA 74%, 62%, 85%) and lower body (BA 66%, 48%, 82%). Distance errors were smaller for facial keypoints than for other keypoints.

State-of-the-art computer vision models face challenges when applied to clinical videos. Accuracy was highest for facial keypoints and lowest for lower body keypoints. Enhancing performance for neurological inpatients may require custom models trained on clinical datasets or novel, potentially combined, approaches.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 33P

Mocan, Lara Faculty Advisor: Neha Kamat

Bispecific Lipid Vesicles for Controlled Cargo Release and Enhanced Immunotherapy

Natural Killer (NK) cells are part of our immune system and mediate antibody-dependent cellular cytotoxicity through CD16 (FcghRIII), allowing clearance of cancer cells. However, upon activation, CD16 is cleaved from the NK cell surface via the ADAM17 metalloprotease, reducing cytotoxic activity. Inhibition of ADAM17 via TNF Protease Inhibitor 2 (TAPI-2) has been shown to block shedding of CD16 and enhance NK cell activity. However, as ADAM17 is present on many cell types, this approach lacks cellular specificity. We hypothesized that bispecific lipid vesicles encapsulating TAPI-2 could enable activation-dependent local inhibition of CD16 cleavage in addition to increasing cancer cell lysis. Vesicles were engineered with surface-conjugated CD16 and HER2 antibodies via SNAP and Halo tags to promote simultaneous engagement of NK cells and HER2+ cancer cells. We proposed that NK cell activation through CD16 engagement leads to perforin release, inducing vesicle membrane permeability and localized TAPI-2 release, therefore inhibiting ADAM17-mediated CD16 shedding. Using flow cytometry, we evaluated CD16 surface expression on NK cells following receptor activation in the presence of soluble TAPI-2 or TAPI-2 encapsulated bispecific vesicles. TAPI-2 presence significantly reduced CD16 shedding relative to controls. Ongoing studies are assessing the impact of TAPI-2 encapsulated bispecific vesicles on cancer cell lysis. This research introduces a vesicular platform to enhance NK cell cytotoxicity and activity, as well as increase cancer cell lysis, by coupling cancer-specific activation and localized inhibition of CD16 cleavage, proposing a novel synthetic biology strategy to enhance NK cell-mediated cancer cell lysis.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 26A

Modhera, Roman Faculty Advisor: Frank Limbrock

How Children's Health Insurance Reduces Crime: The Role of Behavioral Health Services in Medicaid

The United States spends nearly $300 billion annually on the criminal justice system and $140 billion on children's Medicaid. Understanding whether childhood health insurance can reduce adulthood crime - and how - could reshape how policymakers think about both programs. This thesis investigates the mechanisms by which childhood Mediacid eligibility reduces adult criminal behavior. Using variation in state Medicaid eligibility rules across birth cohorts from 1981 to 1984, I estimate the effect of childhood health insurance access on adult arrest rates from FBI data. I then test three possible channels - improved education, reduced financial stress, and better access to behavioral health services - across national datasets covering education, income, mental health, and substance abuse treatment. Each additional year of childhood Medicaid eligibility substantially reduces adult drug and violent arrest rates, with the strongest effects emerging from eligibility during adolescence. Education and financial stress channels show largely insignificant results. Instead, the evidence points to behavioral health as the primary pathway where adolescent substance abuse treatment admissions roughly triple with each additional year of eligibility. These findings suggest that Medicaid reduces crime not by improving schooling or financial outcomes, but by connecting at-risk youth to mental health and substance abuse services during critical developmental windows. This points to Medicaid as both a health investment and a long-run public safety investment.

Presentation Format: Oral
Time: 12:45-2:15PM
Room: Rock Room

Nagy, Alex Faculty Advisor: Nicolette Bruner

State Supreme Court Selection Mechanisms and Dissent

Partisan judicial selection reforms occurring across states raise concerns among scholars over judicial independence, yet existing measures that rely on dissent rates as an expression of judicial independence inadequately capture *how* justices dissent. This study employs mixed-methods to examine the relationship between selection mechanisms, dissent frequency, and the rhetorical content of dissents across state supreme courts. A synthetic control analysis of North Carolina's 2018 transition to partisan elections finds no reliable effect on dissent frequency. Yet a computational textual analysis of 72,480 state supreme court dissents between 1965 and 2019 finds that rhetoric varies across selection mechanisms: partisan-elected courts associated with the most ideologically concordant language, followed by nonpartisan-elected courts, retention courts, and appointment systems. Court composition partly mediates this pattern, as partisan elections that homogenize benches correspond with less ideologically discrepant rhetoric. These findings raise questions about whether mechanisms promoting ideological unity or diversity better foster judicial independence.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 65A

Nareddy, Harika Faculty Advisor: Christian Petersen

Potential New Regulators of Body Axis Formation in a Regenerating Marine Worm

The marine worm *Hofstenia miamia* has the ability to fully regenerate from virtually any injury, making it a powerful model organism for studying regeneration mechanisms. An important prerequisite for this regenerative ability is the maintenance of signaling gradients that act as a body-wide coordinate system, constantly informing cells of their location within the organism. This coordinate system is essential for measuring the extent of missing tissue following injury and determining which tissues must be regenerated. Though the Wnt signaling gradient is known to govern anterior-posterior (AP) patterning, recently the gene *isthmin-like* (*isml*) has been shown to play a role in AP patterning and may be a novel regulator of Wnt signaling.

To further investigate *isml* function, we used RNA interference (RNAi) to knock down *isml*, finding that this produced a "double-tail" phenotype in regenerating worms. This is similar to RNAi of *notum*, a known Wnt antagonist. Additionally, fluorescent in situ hybridization (FISH) allowed us to visualize *isml* mRNA expression, revealing that it is concentrated in the tail tip of the animal. Double FISH of Wnt and *isml* revealed colocalization in some cells, suggesting that *isml* may regulate the Wnt signaling gradient. Additionally, AlphaFold predicts that human Isthmin and Notum proteins may bind. This suggests that *isml* may regulate Wnt signaling through enhancement of Notum, and this mechanism may be conserved in humans. Hyperactive Wnt signaling promotes formation of human cancers. If Isthmin proteins are found to be novel regulators of the Wnt pathway, they could be manipulated for therapeutic purposes.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 76A

Nartey, Michelle

Style Guide: Guiding the Institution and Beyond

The need for updated, student-centered style guides in journalism is increasingly important as reporting standards evolve across platforms, identities, and subject areas; under the guidance of my mentor, Elise K. De Los Santos, this project considers how institutional guides, such as the Medill Stylebook, can better reflect the broader ecosystem of journalistic standards rather than function as static, standalone resources. As an emerging scholar, I approached this question by developing a structured database of existing style guides, organizing them into three categories: affinity-specific, medium-specific (such as photography and writing), and general or topic-specific guides. I analyzed each guide for accessibility, public availability, and relevance to current industry practices, while also documenting how guides reference and build upon one another. This process revealed that no style guide operates independently; instead, they are deeply interconnected, frequently drawing from shared language and frameworks, suggesting that institutional guides benefit from aligning with this larger body of work. The project also resulted in a comprehensive, organized record of these relationships and a system for tracking updates over time. Taken together, this approach points toward a more sustainable way of maintaining style guides as evolving documents, while making them more accessible and usable for students who may not engage directly with multiple external sources, and reinforcing the role of institutional frameworks within a continuously developing field.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 03A

Neal, Elise Faculty Advisor: Magdalena Osburn

Greenland's earliest ecosystems after the last glaciation: Younger Dryas and early Holocene leaf wax analysis

The Greenland Ice Sheet is shrinking due to anthropogenic climate change, which may dramatically alter ecosystem dynamics, especially in the Northern Hemisphere. During the last deglaciation 12,900 years ago, rapid melting of the Laurentide ice sheet abruptly reversed Earth's warming trend and returned northern regions to near-glacial conditions for over a thousand years. This event, termed the Younger Dryas (YD), is known for its severe temperature decrease, but less is known about its impacts on vegetation and hydrology. My project studies a Greenland lake sediment core spanning the last 13,000 years to analyze the environmental impacts of YD cooling. Lipid biomarkers from leaf waxes are well-preserved in lake sediments throughout time and are therefore used as records of the surrounding landscape. Leaf wax compositions are linked to vegetation source, productivity, and hydrology. Using these properties as paleoproxies, results show a sudden shift to an arid climate at the start of the YD, which stabilizes quickly to higher precipitation conditions after the YD termination. In contrast, regional vegetation productivity is low throughout the YD and is not reestablished until ~800 years post-YD, indicating a gradual landscape development to suitable growing conditions. This ~10.9 kya mark also displays a drastic vegetation turnover similar to a nearby lake which exhibits a shift from herb tundra to a more diverse dwarf shrub heath comparable to modern landscapes. Understanding these differing impacts and recovery timescales from the YD is crucial for contextualizing this extreme climate event in the view of current ice sheet loss.

Presentation Format: Oral
Time: 10:45AM-12:15PM
Room: Lake Room

Neuhart, Evan Faculty Advisor: Matthew Glucksberg

Application of Red Fluorescent PLGA Microspheres for Quantitative Assessment of Pediatric Aspiration

Pediatric aspiration is a clinically significant condition that can lead to chronic lung injury, infection, and long-term respiratory complications, yet current diagnostic methods such as videofluoroscopic swallow study (VFSS) and fiberoptic endoscopic evaluation of swallowing (FEES) remain largely qualitative and do not quantify aspiration severity. This project aimed to address this gap by developing a microsphere-based approach to enable quantitative assessment of aspiration. Red fluorescent poly(lactic-co-glycolic acid) (PLGA) microspheres were used as surrogate tracers to model aspirated material due to their biocompatibility, controlled size, and improved detectability compared to free dyes. The use of red fluorescence was selected to reduce interference from natural tissue autofluorescence in the airway. The system was evaluated across multiple experimental models, including fixed pig trachea, fresh murine lung and tracheal tissue, and a New Zealand rabbit airway perfusion model to simulate aspiration under physiologically relevant conditions. Known concentrations of microspheres were introduced to assess detectability and quantifiability across conditions. Results demonstrated that microspheres could be reliably detected and quantified in fresh tissue, with higher concentrations producing clear, countable signals and improved signal-to-background contrast. These findings support the feasibility of using microspheres as quantitative tracers for aspiration and establish a foundation for correlating particle distribution with aspiration severity. This approach has the potential to improve diagnostic precision, enable objective severity grading, and support better clinical decision-making in the management of pediatric aspiration.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 78A

Nguyen, Kien Faculty Advisor: Christian Malapit

Nitrogen-nitrogen bond formation via electrochemistry

Many clinically approved drugs contain N-N bonds. Yet, directly bonding two electron-rich nitrogen atoms defies the conventional reactivity pattern because most reactions take place between an electron-rich molecule (nucleophile) and an electron-deficient one (electrophile). For that reason, existing strategies for N-N bond formation often involve the formation of transient electron-deficient nitrogen intermediates. However, these approaches generally require at least one planar, sp2-hybridized nitrogen source. N-N coupling between two non-planar, sp3-hybridized nitrogen atoms is rare, with few reported examples and limited structural diversity. Additionally, these systems often undergo competing decomposition due to weak adjacent carbon-hydrogen bonds. This project employs electrochemistry to remove an electron (oxidize) from the nitrogen atom of a bicyclic compound known as quinuclidine, generating an sp3-hybridized nitrogen radical. Notably, quinuclidine radical avoids the typical decomposition pathway due to its structural constraints. The intermediate is also electrophilic, thereby overcoming the issue with polarity mismatch and reacting readily with nucleophilic nitrogen partners to form N-N bonds. Under the optimized electrochemical condition, oxidized quinuclidine couples with its electron-rich, unoxidized counterpart to produce a doubly positively charged N-N bonded dimer in high yield. This product can be further diversified through ring-opening reactions with oxygen-, nitrogen-, and sulfur-based reagents, providing access to a diverse set of highly functionalized molecules. Overall, this work provides a new route to medicinally relevant N(sp3)-N(sp3) bonded structures that have previously been difficult to synthesize.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 81P

Nogin, Katya Faculty Advisor: Sadie Wignall

Chemically Induced Dynein Relocalization in C. elegans Disrupts Faithful Meiosis

The formation of healthy egg (reproductive) cells depends on a special type of cell division called meiosis. During this process, cells build a structure called the spindle to separate chromosomes. Unlike most cells, egg cells lack typical organizing centers for this structure, making it less clear how the spindle forms and functions properly. This project focuses on dynein, a motor protein that helps move components within the cells along microtubules, which serve as tracks for chromosome movement. Because dynein is found in multiple parts of the spindle, we asked whether its function depends on where it is located. To test this, we use a system in the model organism *C. elegans* that allows us to artificially move dynein to a specific region, the spindle poles.

We found that relocating dynein to the poles caused noticeable defects in cell division. In particular, cells showed misaligned chromosomes and failures in a key step called polar body extrusion. We also performed an embryonic lethality assay to assess the downstream effects of these defects and found significantly reduced embryo viability following dynein relocalization.

Our findings suggest that dynein plays different roles depending on its position within the cell, and that disrupting its normal location interferes with proper cell division. By better understanding how proteins like dynein function in egg cells, this research may provide insight into the causes of chromosomal abnormalities and reproductive challenges.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 86P

Nourbakhsh, Mitra Faculty Advisor: Julia Behrman

"A snack is not a meal": Culture, Identity, and Dietary Change in Malawi

From KFC outlets in cities to freshly-fried potatoes in rural markets, Malawians have ever-increasing access to Westernized food options. Dominant frameworks for understanding dietary change predict an inevitable shift toward processed, unhealthy diets driven by income growth and urbanization. Without accounting for cultural nuances, however, such frameworks fail to capture uneven adoption of non-traditional ways of eating. Drawing on qualitative interviews, informal survey data and a historical analysis of Malawian food systems, this thesis proposes a four-part framework for understanding the social, cultural, and psychological dimensions through which Malawians navigate new food choices: financial pragmatism, competing heuristics of health and hygiene, the tension between tradition and novelty, and the weight of social perception.

Central to this argument is the finding that a culturally specific distinction between a "snack" and a "real meal" functions as a mechanism of cultural appropriation, allowing Malawians to integrate Western food into their lives while maintaining a linguistic and conceptual separation from traditional eating habits. Furthermore, interviews reveal hygiene concerns are circumvented through brand-name heuristics and community-built online approval; health knowledge does not straightforwardly determine behavior; and status functions as an external enforcement mechanism which, unlike tradition, does not uniformly push against Western food but cuts in contradictory directions depending on context, generation, and the type of food involved. Together, these findings challenge homogenization narratives and call for food policy that engages not only with nutritional and economic dimensions, but with the cultural logic of eating.

Presentation Format: Oral
Time: 12:45-2:15PM
Room: Arch Room

Ogburn, Lily Faculty Advisor: Amy Stanley

Women Between Worlds: American Wives as Cultural Ambassadors in Meiji Japan

This research examines the role of American women in Meiji Japan as participants in cross-cultural exchange during a period of rapid political and social transformation. The research takes the form of a history thesis focused on Emma Wigmore, Mary Crawford Fraser, and Mary McNeill Fenollosa, who all traveled to Japan as the wives of foreign diplomats and hired professionals. It argues that expatriate wives functioned as informal, two-way cultural ambassadors who mediated between Japanese and American societies. Drawing on letters, diaries, and published writings as the primary source materials, this study foregrounds the experiences of three women to show how they interpreted Japanese culture for audiences at home while simultaneously introducing Euro-American customs into the spaces they occupied in Japan.

The thesis situates these experiences within the broader context of Japan's efforts to renegotiate unequal treaties and assert its status as a modern nation in the late nineteenth century. As foreign diplomats, educators, and technical experts established communities in cities such as Yokohama and Tokyo, women became embedded in the social and domestic networks that sustained cross-cultural interaction in Japan. They engaged with Japanese people and culture by hosting gatherings, forming friendships, traveling, collecting material goods, and writing. This thesis demonstrates that by the end of the 1890s, these women were actively immersing themselves in Japanese society and producing knowledge about the country through published writings. This study expands the scope of diplomatic history during the Meiji Period and underscores the importance of women's experiences during this era.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 37P

Oishi, Megumi

Unleashed Horror in the Television: Examining Misrepresentations of the Insanity Plea in Contemporary Film and Television

Existing scholarship establishes that erroneous portrayals of the defense in widely disseminated film and television misconstrues audiences' understanding of its legal reality. The crucial gap, however, is that scholars do not address *how* such misrepresentations manifest in contemporary film and television. By close reading *Primal Fear* (1996), *The Practice* (2003), and *CSI: Crime Scene Investigation* (2011), this project aims to answer the following questions: (1) how does contemporary film and television perpetuate misconceptions of the insanity plea, and (2) how do such misrepresentations alter public perceptions of the legitimacy of the insanity plea? I found that the selected films and series, through visual and semantic representations, reflect three paradigmatic misconceptions of the insanity plea. My project locates instances where contemporary film and television misrepresent the insanity plea, ultimately contributing to broader scholarly conversations on the role of film and television in perpetuating misunderstandings of the law.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 59P

Oliveira, Elena Faculty Advisor: Jennifer Jones

From the Atlantic to the Lakeshore: Practicing Samba and Constructing Brasilidade in Chicago

This sociological study explores how samba practitioners in Chicago teach, discuss, and perform samba as an Afro-Brazilian music and dance practice, and how this relates to race, nationalism, and identity in Brazil. While there is extensive scholarship on samba in Brazil, there is much less on samba outside Brazil, and in U.S. Latine studies and immigration studies Brazilians are understudied. There are a few older studies on one Chicago samba group but none on the current small, passionate, and interconnected scene. Motivated by my experiences attending Chicago samba events and interest in racialization, diaspora, and music related to Latin Americans in the U.S., including Brazilians, this study relies on semi-structured interviews with samba teachers and leaders from 7 organizations, as well as participant observations of samba classes and events. A transnational approach ties the data's emergent themes to ethnographies on samba in Brazil, as well as postcolonial and *Movimento Negro* scholarship. This study is ongoing, but early analysis indicates that a few key Afro-Brazilian leaders trained the majority of Chicago's samba dance teachers, resulting in a number of programs that forefront Afro-Brazilian history in their pedagogy, with some exceptions. They recruit new participants, primarily non-Brazilian and some Brazilians, through appeals to broad interest in Latin dance and festivities. To varying extents, samba organizations in Chicago perform samba as a form of Black diasporic resistance Chicagoans can join or learn about, as familiar national culture Brazilian immigrants can reconnect with, or as spectacle non-Brazilians can consume.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 72P

Olson, Zellie Faculty Advisor: Rayvon Fouche

Storytelling & Video Editing for Protest Tradition: The NBA Bubble Podcast

In the summer of 2020, the NBA created a biosecure "bubble" at Disney World to finish its season amidst a global pandemic. Though often glazed over as a health, athletic, and financial success, this project examines the bubble's reckoning with the pandemic and the Black Lives Matter movement to reevaluate the practice of protest in sports. Drawing on news coverage, athlete statements, social media, and academic scholarship on race and sports media, this podcast investigates three questions: whether the bubble's geographic and economic isolation amplified or constrained political action; whether the two-day work stoppage constituted genuine protest or institutional management of dissent; and whether the negotiated outcome, expanded voting access, represented a meaningful resolution or a deflection from the core demand of ending police violence. The podcast further examines the tension between athletes' economic privilege and their inextricable Blackness, asking whether proximity to wealth severs or deepens connections within communities most harmed by racial violence. Our contributions included reviewing, selecting, and sequencing footage and audio; annotating raw footage; editing video clips using Premiere Pro and Descript; eliminating redundancies; creating promotional short-form content; and balancing humor and tragedy to shape an impactful narrative arc. Effort was dedicated to making the content accessible for all listeners to amplify the impact of sports' role in politics and calls for justice.

This podcast asks who controlled the bubble, and to what end, challenging listeners to reflect on their own social values, expectations of athletes, and the opportunities for progress spurred by protest in sports.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 32A

Ostrander, Julia Faculty Advisor: Jonathan Rivnay

Using Bio-compatible Conductive Hydrogels for On-Demand Drug Delivery

Liposomes, which are tiny spherical capsules made of the same material as cell membranes, have emerged as a promising platform for delivering chemotherapy drugs directly to solid tumors like breast and ovarian cancers. By encapsulating the drugs in these biocompatible spheres, researchers have made progress in protecting them from rapid clearance in the blood, something that has long limited the effectiveness of cancer treatments. However, the drugs are still cleared away from the tumor site too quickly to provide localized, long-lasting delivery. Hydrogels, which are water-swollen polymer networks similar in texture to soft tissue, offer a promising next step as encapsulated drugs are released slowly as the hydrogel is broken down by the body. Since the drug release rate depends on the mechanical properties of the hydrogel it is difficult to optimize. Our proposed solution to this issue involves creating conductive hydrogels that could release the encapsulated drugs on demand when an electrical signal is applied. This would allow for adaptive drug delivery, optimized to the patient's needs.

To build this platform, we incorporated a conductive polymer called S-PEDOT into a sodium alginate hydrogel. We have already confirmed that our hydrogels have mechanical properties comparable to biological tissue and identified an optimal polymer composition that maximizes conductivity. The next step is loading these gels with drug-filled liposomes and demonstrating voltage-triggered release. This platform, which has not been previously reported in the literature, could ultimately enable precise, on-demand delivery of cancer drugs directly to solid tumor sites.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 25A

Park, Elizabeth Faculty Advisor: David Uttal

Foreclosure to Financialization: Mapping Institutional Investor Penetration in Single-Family Housing Markets After 2008

For many aspiring first-time homebuyers, the single-family home has shifted from a pathway to stability to a restricted financial asset, shaped in part by the post-2008 rise of institutional investors in the single-family rental market. This project examines how large-scale investor ownership has reshaped homeownership access, investigating whether it reflects a structural shift or a cyclical response to post-crisis conditions. Drawing on CoreLogic property-level data, the study constructs a longitudinal spatial dataset across three metropolitan areas [Atlanta, Charlotte, and Jacksonville] using kernel density estimation, spatial autocorrelation analysis, and demographic overlays to trace changes from a pre-investor baseline (2009) through a pre-COVID snapshot (2019) to the present (2025). Findings indicate statistically significant spatial clustering in Atlanta (Global Moran's $I = 0.57$, $p < 0.001$) and Charlotte ($I = 0.32$, $p < 0.001$), with Charlotte's clustering nearly doubling between 2019 and 2025. In Atlanta, 18 census tracts remained investor hotspots across both years. In Charlotte, high-investor tracts have median incomes 48% lower and Black population shares five times greater than low-investor tracts, consistent with investor targeting of historically distressed, minority-majority neighborhoods. Together, these patterns suggest institutional investor activity represents a durable structural transformation rather than a temporary crisis response. By mapping institutional portfolio intersections with historically accessible ownership opportunities, this project contributes to debates about financialization, intergenerational wealth-building, and housing equity. The evidence suggests institutional investors have reshaped the single-family market by concentrating ownership in distressed communities, reducing first-time buyer access to homeownership.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 07A

Park, Emily Faculty Advisor: Savio Chan

Mapping Movement Circuits: Projection-Specific Connectivity of Npas1+ Neurons in the Globus Pallidus Externa

The globus pallidus externa (GPe) is a region of the basal ganglia that regulates movement through genetically diverse neuron populations. Traditional models of basal ganglia circuitry treat the GPe as a relay within the indirect pathway through the subthalamic nucleus (STN), often overlooking the diversity of GPe neuron subtypes and their broader projection targets. However, emerging evidence has identified distinct GPe neuron populations based on molecular identity, axonal targets, and behavioral impacts, suggesting that the GPe's contributions to motor control extends beyond a single relay pathway. Among GPe neurons, Npas1+ cells constitute approximately 30% of the population and extend projections beyond classical basal ganglia pathways. Despite this, the organization and downstream targeting across motor-related brain regions remain poorly characterized.

We hypothesized that Npas1+ GPe neurons form a distributed projection network with target-specific connectivity that modulates precise motor and action selection circuits. To test this, we used an intersectional genetic approach in Npas1-Flp;Ai14 mice combined with viral tracing to monosynaptically label neurons that receive Npas1+ GPe input. This cell-type specific strategy enabled identification of downstream targets through histological analysis. We observed projections to multiple motor-related regions, including the dorsal striatum, cortex, GPi, substantia nigra (pars reticulata and compacta), subthalamic nucleus, pedunculopontine nucleus, and olivary nuclei.

Together, these findings demonstrate that Npas1+ GPe neurons participate in a distributed network rather than a single STN-mediated pathway, refining models of basal ganglia function and providing insight into how disruptions in GPe connectivity contribute to motor dysfunction in disorders such as Parkinson's disease.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 53P

Parra, Valentina Faculty Advisor: Alyssa Garcia

@DarkHumor: Race-Blindness, Racism, and Racialization in Mexican Digital Discourse

The ideology of mestizaje, which touts the development of an exceptional "raceless" race through centuries of racial mixture in Latin America, is a core component of the Mexican identity. However, racism remains a central facet of Mexico's landscape, perpetuating the erasure and subjugation of the Black, Indigenous, and Asian presence and experiences in the country's national identity. Despite claiming unity, mestizaje sustains the race-blind conditions needed to ignore, deny, and disavow racism in Mexico. The public nature of social media can provide important insights into the process of racialization by providing an accessible view of popular reactions to racism. Digital satirical environments are rife with racist language, made permissible under the guise of dark humor. The novelty of digital social network research, coupled with the recency of the 2024 presidential elections, signal an opportunity to address understudied phenomena. I focus on a racist political post by popular Mexican meme account @EsDeMamador on X (previously Twitter) on the day of the election. I utilize an intersectional framework to analyze an array of 56 pre-selected (sub)replies to the original post. Drawing from literature on ethics in the digital humanities, Critical and Computer-Mediated Discourse Analysis, post-colonialism, and Critical Race Theory, I illuminate how non-racial categories (i.e., class, political party, religion) are weaponized as instruments of racialization. These reflect, refract, and advance exclusionary national imaginaries. My findings accentuate how white supremacy, anti-Blackness, and anti-Indigeneity continue to permeate daily interactions in Mexico today, despite the country's alignment with race-blind ideologies.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 45P

Patel, Anay Faculty Advisor: Keara Lane

A Deep Learning-Based Segmentation Model for Epithelial Cells Infected with Listeria monocytogenes

Accurate detection of objects from microscopy images, otherwise known as segmentation, is a necessary and limiting first step in quantifying cellular responses. In recent years, this process has been automated using tools such as Cellpose, which rely on neural networks trained on images with cytoplasmic or whole-cell fluorescence signal to segment cell borders. However, in multiplexed imaging approaches, this is challenging as there may not be enough channels to dedicate to segmentation. My project aims to evaluate whether I could use the biosensor signals, which vary between the nucleus and the cytoplasm, to develop a model to identify the borders of epithelial cells infected with the bacterial pathogen *Listeria monocytogenes*. To do this, I developed a computational pipeline to integrate the generalized segmentation model Cellpose-SAM with biosensor images from infected cells. As expected, basic Cellpose models were found to be unable to identify the borders of epithelial cells, preferring to segment the nucleus. I trained a model on manually annotated biosensor images and evaluated it using a standard metric, Intersection over Union. The trained model was able to identify the varied cytoplasmic signals from the biosensors, allowing for better segmentation of epithelial cell borders. To evaluate the utility of the model, I compared the results with those obtained using the Voronoi diagram, a commonly employed method for identifying cell borders in an epithelium. These results demonstrate the effectiveness of utilizing deep learning models for the segmentation of biosensor data, the identification of infected cells, and the quantification of bacterial load.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 67A

Pena Figueroa, Sara Faculty Advisor: Sally Nuamah

Experiences of Racial Mismatch and Relationship to Racial Belonging

The U.S. public school student body is projected to racially diversify over time, but these diverse students will be met with a rather homogenous teaching staff. Since 2011, the share of each racial and ethnic group in the public-school teaching force has not changed by more than 2 percentage points with 80% of teachers identifying as White. Research has identified the important role that teachers play in the lives of students, but scholars have lacked in the examination of how students of color experience their education led by mainly White teachers. With adolescence being an important life stage for racial and ethnic identity development, it is imperative that we understand the potential role that racial dynamics may play in identity development, specifically on racial, ethnic, and cultural belonging. The current research aimed to investigate how college students of color describe their experience with white teachers in high school and how, if at all, they describe the relationship between the experience of racial mismatch and their racial belonging. A sample of ten Northwestern first-year students of color participated in one-hour long interviews where they answered questions regarding their racial and ethnic identity, public school experience, and student-teacher relationship. Emerging findings demonstrate that participants did not articulate a White teacher's race as playing a role in their racial belonging, but many participants were able to express a desire for a teacher of color. This suggests the importance of diversifying the teaching field to fulfill students' desire of culturally connecting to their educators.

Presentation Format: Oral
Time: 10:45AM-12:15PM
Room: Arch Room

Platnick, Elijah Faculty Advisor: Istvan Kovacs

Modeling and Balance of Signed Directed Social Networks

My project looks at balance in directed, signed social networks. In a signed, directed network, connections between nodes can go either direction and can be positive or negative. In the context of this project, nodes are people, and connections can be measures of friendship or enmity. My research uses five datasets. As an example, one dataset is from Pardus, a video game where users can mark each other as friends or enemies (these need not be reciprocated). My project develops better randomization methods to better understand the structure of such networks by creating definitions of balanced triangles, meaning determining whether a given a set of three nodes and their connections is balanced. Some of the current literature suggests status theory as a possible definition of balance. We argue status theory has difficulty accounting for reciprocated and conflicting relationships (in the context of a trust network, reciprocated means both trust/distrust each other. Conflicting means trust one way but distrust the other).

To test a new theory of balance, my project uses a randomized version of each network. To randomize the network my project uses a fixed topology randomization using a framework based on the maximum entropy principle. The number of nodes is the same and the edges are between the same nodes, and only the sign is randomized. This process only changes a very small number of the edges and preserves key aspects of the original network. We find the data is well aligned with two proposed definitions of balance.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 75A

Pothuru, Annabel Faculty Advisor: Carole LaBonne

A Pattern Emerges: Investigating Snail Family Transcription Factors' Role in Ectodermal Gene Regulation Networks

A crucial step in embryonic development is the specification of ectodermal cells into the three fates of the neural crest (NC), central nervous system (CNS) and skin (epidermis). The NC forms the peripheral nervous system & craniofacial skeleton and develops between the CNS and epidermis. These fates are separated into regions with defined boundaries, but disruptions can cause severe developmental defects. A complex regulatory network controls this specification but is incompletely understood; Snail family transcription factors (Snai 1/2) are good candidates for investigation of specification, as they regulate NC expression patterns at certain stages. This study investigated the hypothesis that Snail factors regulate gene expression during ectodermal fate determination to separate the three initially overlapping regions. Using *Xenopus laevis* embryos, Snail activity was manipulated through knockdown and overexpression approaches through injections of dominant-negative constructs and Snai1/2 mRNA respectively. Microinjections were conducted at the two-cell stage to target one side of the embryo, maintaining the other as an internal control. *In situ* hybridization probes targeted fate-specific marker genes *foxd3* (NC), *sox2/3* (CNS), and *epik* (epidermis) to visualize their expression and compare changes in expression patterns caused by the manipulations. *epik* results were inconsistent but informed later methods. Together, NC and CNS results indicated that Snail transcription factors define boundaries between the ectodermal fates by regulating genetic expression to simultaneously reduce specification into CNS and promote specification into NC. Ongoing work builds off these results, exploring how Snail factors regulate specification of cells directly in the boundaries between ectodermal fates.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 33A

Pratasenia, Almira Faculty Advisor: Daniel Richard Foltz

Modeling Histone Chaperone Interactions at the Replication Fork to Elucidate Mechanisms of Genomic Stability

The diploid human genome comprises over 6 billion base pairs, totaling over 2 meters of DNA, that must be compacted to fit within the cell nucleus. To do this, DNA is wrapped around proteins called histones, forming chromatin, which must be transiently disassembled to allow the replication machinery access to the DNA, and subsequently reassembled to preserve genome stability. Histone chaperone proteins facilitate this process by escorting histones to correct locations during DNA replication and chromatin reorganization.

These chaperones operate in a dynamic network with other chaperones and proteins at the replication fork. However, due to the complexity of this process, few physical interactions within this network have been found. This project aims to identify novel interactions among histone chaperones and replication fork proteins using computational structural modeling.

We are employing the AlphaFold3.0 AI-based protein structure prediction tool on Northwestern University's QUEST high-performance computing system to generate high-confidence 3D protein complex structures. Models are assessed using the interface predicted template modeling (ipTM) score and a sliding-window assay to evaluate interaction stability. Highly confident predictions will be further analyzed using the PROtein binDIng enerGY (PRODIGY) analysis tool. Ultimately, predicted structures will be validated by coimmunoprecipitation in HEK293T cells and protein-binding assays with recombinant proteins.

This ongoing work is expected to identify previously uncharacterized interactions within the histone chaperone network. These findings would provide a better understanding of how histones are relocated during DNA replication, offering insight into mechanisms underlying genome instability in certain diseases, such as cancer.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 77A

Qin, Danielle Faculty Advisor: Shelby Hatch

Evaluation of Heavy Metal Contamination and Social Correlates in Chicago Community Areas

Heavy metals are high-density elements that pose a significant toxicity risk to humans at elevated concentrations. Although naturally present in soil at low levels, human activity has contributed to heavy metal accumulation in urban areas, often disproportionately targeting marginalized communities. Since the level of heavy metal contamination in Chicago requires further exploration to improve evaluations of public health, this project aimed to quantify heavy metal contamination across Chicago's 77 community areas and assess its relationship with socioeconomic and racial demographics. We employed community-based methods to comprehensively survey areas of high human engagement with soil, such as playgrounds, and obtained soil samples from 60 community areas. Concentrations of heavy metals were quantified with Inductively Coupled Plasma-Optical Emission Spectroscopy. Using R and ArcGIS software, these results were geospatially integrated with demographic data from the U.S. Census Bureau and heavy metal regulations from the Environmental Protection Agency (EPA). Holistically, our findings suggested that lower-income and majority non-white Chicago neighborhoods have higher concentrations of heavy metal contamination. Notably, community areas, such as Garfield Park and Brighton Park, consistently exhibited contamination levels that violated EPA thresholds for the majority of metals that were analyzed. Some heavy metals, including arsenic and chromium, exceeded EPA limits across all sampled locations. By elucidating heavy metal contamination patterns, our research highlights communities that are at risk and better informs future environmental policy decisions. Importantly, we emphasized collaborating with communities rather than acting as outside experts, which allows our work to empower communities and support local residents.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 77P

Rahmani, Rayan Faculty Advisor: Mehul Raval

The Costs and Burden of In-Person Follow-Up After Pediatric Appendectomy

Appendectomy for uncomplicated appendicitis is the most common emergent surgery in pediatric patients, with roughly 70,000 annual cases nationally. Postoperative care typically includes an in-person follow-up visit, though telemedicine alternatives are being actively explored. The travel-related burdens of this practice remain poorly characterized. We retrospectively evaluated patients who underwent uncomplicated appendectomy at a single institution in 2020 to quantify time, cost, and environmental impacts of in-person follow-up. Patient zip codes were mapped to population-weighted centroids to estimate round-trip driving time and distance. Vehicle operating costs and carbon emissions were calculated from driving data using established national averages. Wage loss for one caregiver was calculated using the average Chicago metropolitan area wage and total time burden, which included a 30-minute allowance for parking, check-in, and waiting. Costs were inflation-adjusted to 2026 US dollars. Among 302 children undergoing uncomplicated appendectomy, the median round-trip drive time for in-person follow-up was 36.3 minutes (IQR 29.5–73.2), and the median estimated wage loss per family was $40.42 (IQR $36.27–$62.91). In aggregate, follow-up visits were associated with 273 total hours of driving and 9,899 miles driven, resulting in an estimated $7,176 in vehicle operating costs and $15,504 in wage loss. Carbon emissions attributable to driving totaled 3,519 kg CO2. In-person follow-up after uncomplicated pediatric appendectomy imposes substantial time, financial, and environmental burdens without proven clinical benefit. These findings underscore the need for comparative effectiveness research evaluating telemedicine-based vs. in-person postoperative follow-up strategies.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 44A

Rai, Aashna Faculty Advisor: Peter Civetta

Theatre as Refuge: Facilitating Identity, Agency, and Belonging through Youth Migration Narratives at C:NTACT and Albany Park Theater Project

While migration research has addressed trauma, policy, and cultural integration, there remains a critical gap in understanding how the arts-theatre, in particular-can support migrant youth in navigating identity and belonging. Theatre offers unique affordances: it is embodied, collaborative, and rooted in storytelling, allowing young people to develop a sense of self and share narratives on their own terms. My thesis examines the pedagogical work of two youth theatre companies-C:NTACT in Copenhagen and Albany Park Theater Project in Chicago-to answer the following question: how do their devising processes employ identity-based theatre-making techniques to cultivate personal agency and social belonging among marginalized migrant youth, and what elements of these practices can be sustainably adapted to support refugee youth across diverse sociopolitical environments? Negotiations between individual and collective storytelling, institutional sustainability, and methods of audience engagement emerge as sites of key pedagogical insights-which are then applied to a preliminary model for the use of artistic techniques in refugee education spaces.

Presentation Format: Oral
Time: 12:45-2:15
Room: Armadillo Room

Raman, Maya Faculty Advisor: Gabriela Caraveo Piso

Investigating the Role of CARHSP1 in Stress Granule Formation and Metabolic Energy Production in Astrocytes

The brain often shows signs of stress before neurodegenerative diseases develop. My project explores how cells in the brain that directly support neurons, called astrocytes, respond to that stress and help protect neurons, particularly in conditions like dementia with Lewy bodies. While harmful protein aggregates are commonly linked to these diseases, growing evidence suggests the real issue may be a breakdown in how cells manage energy and internal balance. I am currently investigating the role of a protein called CARHSP1 in helping astrocytes respond to stress by producing energy for nearby neurons.

To do this, I study cells grown under different nutrient conditions and modify CARHSP1 to observe how it affects two key processes: the formation of "stress granules," which store genetic instructions during stress, and the production of lactate, an important energy source. I also analyze which of the cell's genetic materials are stored during stress and test these findings in brain cells from mice. Finally, I simulate stress signals to see how astrocytes respond in real time.

I expect to find that CARHSP1 plays an important role in regulating how astrocytes manage energy during stress. Understanding this process could reveal new ways to support brain cells and slow the progression of neurodegenerative diseases, offering insight into potential future treatments.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 21A

Ramos, Victoria Faculty Advisor: Almaz Mesghina

Misdiagnosis and Overdiagnosis of Learning and Behavioral Disorders in Black Students

Students across America with a variety of behavioral and learning disorders are able to succeed academically, socially, and professionally in their adult lives. Accurate diagnoses, treatment plans, and support from teachers and psychologists is crucial to a student's progression. However, there is a significant gap in accurate diagnostic and treatment services between Black students and White students in the Kindergarten-12th grade school system. Black students are often misdiagnosed or overdiagnosed with learning and behavioral disorders, therefore, holding them back from reaching their true academic potential. Examining how cultural biases embedded within Western general psychology have contributed to the inequitable diagnosis and treatment of Black students is essential for understanding the disparities they face in the American educational system. Moreover, this paper will adopt a cultural psychology approach to explore the underlying factors driving the overdiagnosis and misdiagnosis of Black students.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 68P

Reddy, Aman Faculty Advisor: Prakash Jayabalan

A Deep Learning Approach to Minimum Medial and Lateral Joint Space Width Measurement

Knee osteoarthritis (OA) is the most common joint disease worldwide and is characterized by progressive cartilage loss and joint space narrowing (JSN). Minimum joint space width (mJSW), the distance between femur and tibia, is a key radiographic indicator of OA progression and is central to diagnostic tools such as the Kellgren-Lawrence grading scale. However, JSW measurement is traditionally performed manually, making it time-consuming, observer-dependent, and impractical for large-scale studies or routine clinical use.

This project presents a deep learning-based tool for objective, automated mJSW measurement from knee radiographs. The pipeline consists of a 5-class 2D UNet segmentation model trained to recognize medial and lateral compartments of bilateral posterior-anterior (PA) knee radiographs. Post-processing software measures the average vertical distance between the medial and lateral condyles using the predicted segmentation masks. Using these measurements and metadata, mJSW is calculated in millimeters. Testing and training sets were sourced from the OAI dataset (125 and 300 images).

Comparison of the segmentation model against a validation set revealed congruency between predicted and annotated masks, with agreement exceeding 87% across all classes. Evaluation on a 100-image set showed that 92% of predictions were within 1mm of manual measurements. The average mJSW across the test set, when stratified by OARSI JSN grade, demonstrated a strong inverse correlation (r=-0.91).

This model shows significant potential in accurately predicting mJSW. This would be of marked benefit to clinicians. Future work will refine the model and explore its ability to capture trends across Kellgren-Lawrence grades.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 61A

Reitz, Natalie

Platelets drive immune suppression and glioblastoma growth in a sex-dependent manner via PAR4 signaling

Glioblastoma (GBM) is the most common primary malignant brain tumor, with outcomes shaped by biological sex differences that influence disease progression, immune function, and interactions within the tumor microenvironment (TME). Although platelets are increasingly recognized as active participants in tumor biology, their contribution to sex-specific immune regulation in GBM has not yet been defined. We identify platelet PAR4 signaling as a central mechanism linking sex hormones to antitumor immunity. To investigate the role of platelets in the immunosuppressive TME, we investigated human platelet function, genetic and pharmacologic perturbation of PAR4, hormone manipulation, and immune profiling of tumor-infiltrating cells. We found that GBM patients display elevated platelet activation driven by protease-activated receptor 4 (PAR4). In vivo, inhibition of PAR4 extended survival in females but not males, and we have shown this benefit is estrogen dependent.. Mechanistically, PAR4 inhibition increases CD8+ T-cell infiltration into the TME, establishing a platelet-driven, sex specific immune response. We further show that estrogen receptor v physically interacts with PAR4 to amplify calcium signaling in platelets, revealing a previously unrecognized hormone-receptor crosstalk that shapes platelet behavior in tumors. PAR4-activated platelets suppress CD8+ T-cell function within the TME, and depletion of CD8+ T cells eliminates both the heightened platelet reactivity induced by GBM and the survival advantage conferred by PAR4 inhibition. Together, these findings identify platelet PAR4 signaling as a regulator of GBM progression and reveal sex-dependent immune mechanisms as critical determinants of therapeutic response and offering therapeutic targets for sex-dependent GBM treatment.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 49A

Relyea, Emma Faculty Advisor: Simone Ispa-Landa

Performing Equity, Missing Change: Exploring Teacher Perspectives on "Accountability Theater" in a Diverse Suburban High School

Faced with pressures from the local school board and state laws to reduce racial disparities in student discipline, Franklin High sought to improve both its discipline metrics and student-realized outcomes. However, only the former was meaningfully addressed. This study examines how the school engaged in "accountability theater," defined as performative compliance that satisfies external demands without producing substantive change.

Although these policies were intended to promote equity, they often backfired when teachers interpreted them as superficial or disconnected from classroom realities. For example, teachers were told to stop writing referrals without clear disciplinary alternatives, leaving them feeling unsupported and uncertain about how to manage their classrooms. As a result, teachers either reluctantly complied with administrative directives or ignored them.

This study asks: When and how did teachers recognize the accountability pressures facing Franklin High? How did their perceptions of the school's initiatives shape trust in leadership and colleagues? Did teachers observe meaningful improvements for students?

Drawing on ethnographic interviews with 44 teachers from 2019 to 2024, three main findings emerge. First, teachers varied in awareness of the external pressures driving these changes. Second, most teachers agreed racial disparities in discipline were a real and important issue, driving a want for change and an observed lack of improvement. Third, many viewed administrative responses as disconnected from classroom realities, leading to diminished trust in leadership.

Accountability efforts must align with the day-to-day realities teachers face in their classrooms in order to enact lasting change for students, schools, and communities.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 44P

Richardson, Samuel Faculty Advisor: Chad Mirkin

Extension of DNA Dendrimers through the Hybridization Chain Reaction

Cancerous tumors suppress the immune system, making it harder for the body to fight them and kill cancer cells. One promising approach to reverse this is by activating the cGAS-STING pathway. This pathway triggers an immune response when a protein called cGAS detects foreign DNA inside a cell, allowing immune cells to kill cancer cells. Longer DNA activates this more effectively, but getting DNA into cells is difficult. DNA dendrimers, branched, tree-like molecules, have been shown to improve the delivery of DNA into cells at once. However, making dendrimers with branches long enough to reliably trigger the immune alarm is difficult, as demonstrated by low yields. This study explored a solution using a process called the hybridization chain reaction (HCR). HCR is like a molecular chain reaction where a short "starter" piece of DNA (Initiator DNA) causes two coiled DNA molecules to unfold and link together repeatedly, producing long DNA chains. Two strategies for combining HCR with DNA dendrimers were tested. The first, building dendrimers from one of the coiled DNA components, proved synthetically impossible due to the complexity of those components. The second strategy, attaching the short, simple Initiator DNA to the dendrimer, produced the desired dendrimer with an acceptable 11% yield. We confirmed that dendrimers carrying multiple starters produced significantly longer DNA chains than Initiator DNA alone. This demonstrates that DNA dendrimers can drive the chain reaction more powerfully than linear DNA, enabling the creation of larger immune-activating DNA structures than was previously possible.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 17A

Rissmiller, Stephanie Faculty Advisor: Eleanor O'Rourke

"It's Okay Sometimes": Insights into Students' Nuanced Evaluations of Their Programming Ability While Using LLMs

Amid concerns that computer science students are over-relying on large language models (LLMs), this paper seeks to explore how students perceive and assess their programming ability while coding with an LLM. Student self-assessments of their programming ability can help us understand how students experience learning, providing insight into how we can support the development of introductory instruction that integrates LLMs while fostering student confidence, learning, and self-efficacy. We conducted two survey studies, totaling 52 participants, to investigate how students assess programming ability while coding with an LLM. Our first (qualitative) survey allowed us to identify a set of moments during which students may evaluate programming ability while coding with an LLM, such as checking code correctness with an LLM. Our second (quantitative) survey investigated how students self-assessed their own abilities when they encountered these newly-defined moments as well as moments identified in prior research before LLMs were widely accessible. Our findings revealed parallels between student self-assessment data gathered prior to and after the introduction of LLMs. They also indicated that students think critically about LLM usage while programming. We ask whether student self-assessments based on our newly defined moments are accurate judgements of their programming ability and whether the accuracy of these judgments varies depending on student interests and goals. We call on researchers and practitioners to question what programming skills and LLM use cases suit the interests and needs of their students and to encourage the development of those skills and behaviors as a means of supporting student self-efficacy.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 56P

Rizzo, Elissa Faculty Advisor: Nicolette Bruner

An Uncertain Peace: Armed Groups, Civilian Disputes, and the Limits of State Presence in Postwar Colombia

Colombia's 2016 Peace Accords, signed between the Revolutionary Armed Forces of Colombia (FARC) and the Colombian government, ended the Western Hemisphere's longest civil war. However, uncertainty remains about the continued presence of armed groups and whether civilians turn to the state or other actors to resolve disputes. Scholarship on Colombian postwar governance often focuses on top-down provision of goods rather than bottom-up civilian experiences. This study draws on survey and interview data collected across 20 Colombian municipalities and roughly 60 communities, using regression analysis to identify patterns in dispute resolution behavior and thematic analysis of interviews to explain these patterns. In locations where armed groups remained after the 2016 Accords, residents were three times more likely to say they "don't know" whether they would use state mechanisms to resolve disputes than those in communities without armed group presence. This finding suggests that the boundary between state and non-state authority has become unclear, leaving civilians unsure who to turn to for dispute resolution. In communities exposed to armed groups during the civil war, residents were 50-60% less likely to use state mechanisms today than those with no wartime armed group presence, reflecting distrust rooted in wartime experiences of state violence and abandonment. This research contributes to post-conflict governance debates by demonstrating that the presence of armed groups continues to fragment civilian engagement with state institutions. These findings reveal a gap in Colombia's 2016 Peace Accords, which left unresolved questions about armed group dynamics and civilian trust in the state.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 68A

Robertson, Hannah Faculty Advisor: Nathan Gianneschi

Proteomimetic Polymer for Neuron-Targeted Delivery.

Protein-protein interactions (PPIs) play essential roles in cellular function and are implicated in many neurodegenerative disorders. In neurodegenerative diseases, some of these interactions go wrong, causing proteins to clump together in ways that damage and ultimately kill brain cells. One promising therapeutic strategy is using short protein fragments called peptides to selectively block these harmful interactions. However, peptides break down quickly in the body and have difficulty entering cells, which limits their usefulness as treatments. Protein-like polymers (PLPs) are an emerging class of materials that mimic peptides while overcoming these limitations. By linking many peptide units together onto a stable backbone, PLPs offer improved durability, multivalency, and better ability to enter cells compared to peptides alone. A major challenge in treating brain diseases is delivering therapeutics specifically affected cell types such as neurons. This project focused on the use of the Tet-1 peptide, a short sequence originally identified for its ability to bind neuronal receptors. We investigated the hypothesis that PLPs derived from the Tet-1 peptide sequence would retain their targeting ability while gaining the advantages of the polymer platform. Tet-1 based monomers were synthesized, assembled into polymers, and fluorescently labeled to track their localization into cells. The polymers were characterized to confirm their structure and consistency. Our results support Tet-1 derived PLPs for cellular targeting and internalization, establishing a promising platform for delivering future therapeutics to brain cells.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 22P

Rodriguez, Chantal Faculty Advisor: Laura Pigozzi

Exploring How Aggressive Immigration Policies and Anti-Immigrant Rhetoric Affect Latino Immigrant Health

Health disparities among Latine immigrants in the United States are shaped not only by individual behaviors but also by political rhetoric, public discourse, and immigration policy. This project examines how anti-immigrant rhetoric and aggressive immigration enforcement function as social determinants of health, producing physical and mental health consequences for Latine immigrant communities. Drawing on scholarship, this work situates contemporary anti-immigrant discourse within the long-standing "Latino Threat Narrative," which racializes Latine immigrants as criminal, threats, and culturally incompatible.

Using the framework developed by Vargas, Sanchez, and Juarez, which links hostile anti-immigrant policy environments to adverse health outcomes, this project explores how rhetoric and policy are embodied through material, psychosocial, and behavioral pathways. These mechanisms include chronic stress, fear of detention and deportation, avoidance of healthcare and social services, unsafe working and housing conditions, and increased rates of anxiety, depression, and somatic symptoms. Notably, the effects extend beyond unauthorized immigrants to authorized individuals and mixed-status families, demonstrating how immigration enforcement produces widespread community-level harm.

The project also highlights how language itself, through criminalizing labels and repeated narratives, contributes to health inequities by shaping lived experience and daily decision-making. By framing immigration policy as health policy, this work underscores the ethical and public health implications of anti-immigrant rhetoric. Ultimately, it argues for greater scholarly attention to immigrant health and for policy approaches grounded in human rights and health equity.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 39P

Rogers, Cami Faculty Advisor: Julia Kalow

What's the junction's function? Quantifying molecular load of dynamic crosslinks in polymer networks.

Covalent adaptable networks (CANs) are an exciting class of polymeric materials capable of self-healing and reprocessing through topological network reconfiguration. They are an emerging alternative to thermosets, which are high-strength materials composed of polymer chains crosslinked by strong, irreversible bonds. Such bonds make thermosets difficult to recycle and thereby are single-time use. CANs aim to provide a solution to the high rates of plastic waste. However, for CANs to be a true alternative to thermosets, they must be able to withstand high levels of mechanical force during use and reprocessing. Despite decades of research, there is a limited understanding of the mechanical failure of CANs on a molecular level. This knowledge gap poses significant issue, as understanding on the molecular level is essential to design robust CANs suitable for the replacement of thermosets. Force-sensitive small molecules, known as mechanophores, undergo predictable chemical transformation upon critical stress. Through the incorporation of intentionally weak, mechanosensing bonds, damage within CANs can be quantified during and after failure of the material's structure, and self-healing can autonomously occur to restore the original properties. Current work in the Kalow lab quantifies the mechanical load of fully dynamic crosslinks within CANs, providing partial insight into the mechanism of mechanical failure. To investigate the load-bearing capabilities of the polymer backbone-crosslinker junction, mechanophore activation is measured via fluorescence within a hybrid static-dynamic crosslink. Greater understanding of this physical property will allow for application of CANs on a bulk scale, making CANs an increasingly accessible and sustainable alternative to thermosets.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 18A

Rong, Edwin Faculty Advisor: David Uttal

Identifying Block Design Task Completion Strategies With Machine Learning

The Block Design Task (BDT) is a spatial intelligence task that requires participants to arrange colorful blocks into a target design, providing a unique opportunity to capture individual differences in spatial thinking. However, past studies on variations in how participants complete the BDT have relied on time-consuming manual video reviews. Machine learning (ML) refers to a category of computational tools to automatically detect patterns in data, which can accelerate analyses of complex behavioral data. This project assesses the viability of sensor-embedded blocks alongside ML methods for more efficient analyses of inter-participant strategic variations in completing the BDT. We collected data from 57 undergraduate students who each completed seven BDT target designs without time constraints. Embedded sensors indicated each block's relative position over time, providing information on each participant's block placement sequences. We developed a custom hierarchical time series clustering algorithm to quantify strategic similarities between participants and identify potential participant groupings. Preliminary results for one target design showed no discernible groupings, but the algorithm's judgments of similarities between participants aligned with qualitative video-based judgments. This suggests that the algorithm can effectively capture inter-participant patterns in BDT strategies regardless of whether underlying groupings exist. Further analyses will be conducted to assess the ML algorithm across the remaining target designs. If our methodology with sensor-embedded blocks and ML-based analyses consistently captures patterns between participants, similar approaches may help expedite and scale future research on how participants approach the BDT.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 64P

Rozwat, Madison

Combination Treatment of DFMO and Navitoclax for Medulloblastoma, a Childhood Brain Cancer

Medulloblastoma is a type of childhood brain cancer that arises in the part of the brain that controls coordination, balance, and other complex motor functions. It is the most common cancerous pediatric brain tumor, and the typical course of treatment includes surgical removal of the tumor, chemotherapy, and radiation. In cases of medulloblastoma relapse, there is a very low survival rate, which is why it is important to find curative treatment options. This research explores the effect of combining two drugs on medulloblastoma cancer cells. Difluoromethylornithine, or DFMO, is a drug that halts cell growth by inducing a state of “senescence” in cells. This is followed by a second drug, Navitoclax, which selectively eliminates these senescent cells. First, it was confirmed that DFMO induces senescence in medulloblastoma cells. To explore the combination effect of these two drugs, medulloblastoma cells were first treated with DFMO for 96 hours, followed by a range of concentrations of Navitoclax for 48 hours. The proportion of cancer cells that survived the treatment was measured, where a lower level of “viability” meant the treatment was more effective at eliminating the cancer cells. The sequential DFMO–Navitoclax treatment reduced the viability of medulloblastoma cancer cells to a greater extent than either drug alone, indicating that this combination may be an effective therapeutic approach.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 51P

Ryan, Jack Faculty Advisor: Andrew Geraci

Precision In Situ Measurement of Rotor Wobble in the ARIADNE Axion Experiment

Dark matter makes up most of the matter in the universe, yet its identity remains a mystery. The ARIADNE experiment at Northwestern will search for a promising dark matter candidate called the axion. To do this, the experiment utilizes a spinning tungsten rotor located extremely close, within 75 micrometers, of a sensitive quantum-based sensor. If the axion exists, it would produce a measurable signal we can detect using a magnetic measurement technique. A challenge of the experiment is the stability of the rotor. Small displacements or “wobble” of the rotor can lead to collisions with the detector, damaging the experiment. A reliable measurement method to track displacements of the rotor on the order of several micrometers is necessary. This system must be able to operate under extreme experimental conditions, including temperatures reaching -452 ºF and high vacuum pressures. We developed a method utilizing a fiber interferometer, a laser sent down an optical fiber, to track the wobble of the rotor with high precision. As the distance between the fiber and rotor changes, the signal we measure shifts in a predictable manner. By calibrating this relationship between the signal and distance, we can convert the signal into precise measurements of the rotor’s position within the experiment. Initial tests show that this method can track the wobble of the rotor with high precision, and we have characterized the wobble of the rotor measuring displacements of 20 micrometers. This technique will be utilized to ensure safe operation of the rotor within ARIADNE.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 60A

Saavedra, Rosa Faculty Advisor: Andrea Graham

Language Matters; How Adults with Food Insecurity Perceive the Term "Binge Eating"

Nearly one fifth of Americans experience food insecurity, meaning they do not have consistent access to nutritious food. Food insecurity often leads to a cycle of food scarcity followed by overeating when food is available; this cycle can contribute to binge eating, a serious but often overlooked health concern. However, most binge eating research has focused on a narrow population (skinny, white, affluent, girls). Consequently, language used to describe binge eating may not resonate with individuals experiencing food insecurity. Therefore, we explored whether the term "binge eating" resonates with this population, and if alternative words like "emotional eating" or "overeating" better reflect their experiences. We conducted a thematic analysis from interview data with 26 adults with food insecurity who received a digital binge eating intervention. Participants described mixed reactions to the term "binge eating." Some felt it accurately described their experiences, while others found it too broad or stigmatizing. Several participants preferred the terms "emotional eating" or "uncontrolled eating" because they felt these better captured their experiences without implying a diagnosis. Some participants identified with "binge eating," but their personal definitions did not consistently match the clinical definition, indicating a misunderstanding. These findings suggest that current language around binge eating may not fully resonate with individuals experiencing food insecurity. When people do not see themselves reflected in terminology, they may be less likely to seek support or may view interventions as irrelevant. This highlights the need for more inclusive, accessible language in eating disorder research and intervention outreach.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 70P

Sachdeva, Ruhani Faculty Advisor: Matthew Glucksberg

Toward Autonomous Breathing: Predicting CO2 in Children with CCHS Using Wearable Sensors

Breathing is something that most people rarely think about, seamlessly adapting to changes in sleep, exercise, or everyday movement. But for children with Congenital Central Hypoventilation Syndrome (CCHS), this automatic control is impaired, requiring reliance on assisted ventilation to breathe.

Advances such as phrenic nerve stimulation have enabled patients to live longer and more active lives by restoring diaphragm function. However, these systems still depend on manual, clinician-driven adjustments, limiting their ability to respond in real time to changing metabolic demands.

This project investigates whether a patient's ventilatory needs can be anticipated before physiological signals reflect these changes. Transcutaneous CO2 (tcCO2), a key clinical measure of ventilation, is inherently delayed due to physiological and diffusion constraints, restricting its use in real-time control.

To address this, we developed a predictive framework using noninvasive wearable signals. We built upon an existing sensing platform and designed an experimental protocol to capture interactions between ventilation and activity. Using synchronized expired CO2 (etCO2) and accelerometry data, we trained a machine learning model (CNN to biLSTM) to learn these physiological patterns.

The model was able the identify the delay between signals and predict future CO2 levels with mean absolute errors of 0.55-0.58 mmHg.

These results demonstrate the feasibility of predicting clinically meaningful CO2 dynamics from wearable data, providing a foundation for closed-loop, anticipatory control of phrenic nerve stimulation. Future work will focus on building a large, diverse physiological dataset across individuals, enabling personalized and adaptive respiratory support across age, activity level, and patient variability.

Presentation Format: Oral
Time: 12:45-2:15PM
Room: Lake Room

Salehi, Nicole Faculty Advisor: Zheng Zhang

Impact of Myeloid-Specific MyD88 Deficiency on Inflammatory Cell Responses Following Lung Ischemia-Reperfusion Injury in Latent MCMV Infection

Cytomegalovirus (CMV) establishes lifelong latency in many individuals and poses significant risks to immunocompromised patients, especially organ transplant recipients. Lung ischemia-reperfusion injury (IRI) is a common consequence of transplantation and induces significant inflammation that can influence viral reactivation. Myeloid differentiation primary response 88 (MyD88) is a key adaptor protein in innate immune signaling pathways and plays a key role in regulating inflammatory responses. Therefore, this study investigates the role of myeloid-specific MyD88 signaling in modulating innate immune responses during latent MCMV infection and lung IRI. To investigate this, a tamoxifen-inducible, myeloid-specific MyD88 knockout mouse model underwent lung IRI. Immune cell populations in lung tissue, bronchoalveolar lavage (BAL), and spleen were analyzed by flow cytometry. DNA was isolated from lung and salivary gland tissue and used in quantitative PCR (qPCR) to assess MCMV viral burden via immediate-early gene (IE-1) copy number. Preliminary flow cytometry data suggest differences in inflammatory cell populations, including neutrophils, CD11b+ cells, and interstitial macrophages between MyD88-deficient and control groups. qPCR analyses of MCMV IE-1 copy number across lung and salivary gland tissues are ongoing. Further analyses will characterize these immune and virologic trends and determine the extent to which MyD88 signaling influences the balance between inflammatory injury and viral control following lung IRI. These findings may provide insight into therapeutic targeting of inflammatory pathways in transplant settings.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 50P

Sanchez, Gabriel Faculty Advisor: Nicolette Bruner

Construction of the Deportable Body: The Ontology of "Reasonable Suspicion" and Noem V. Vasquez Perdomo

This thesis traces the genealogy of the reasonable suspicion legal standard and examines its contemporary application in *Noem v. Vasquez Perdomo* (2025). In *Perdomo*, Supreme Court Justice Brett Kavanaugh permitted race, language, occupation, and location to function as indicators of undocumented status. This thesis argues that Kavanaugh's interpretation of reasonable suspicion is a racialized project, disproportionately targeting Brown-skinned Latine individuals. Drawing on three foundational reasonable suspicion majority opinions alongside analysis of social media documentation of detainments, this thesis demonstrates how the ontology of reasonable suspicion has been reformulated to target Brown-skinned Latine individuals. I introduce the deportable body as a social construct to emphasize how Immigration Enforcement Officers automatically label Brown bodies as "illegal." By situating *Perdomo* within broader histories of racialization, this project ultimately reframes reasonable suspicion not as a neutral legal standard, but as a tool that produces non-belonging and institutionalizes racialized disposability in the United States.

Presentation Format: Oral
Time: 12:45-2:15PM
Room: Rock Room

Seel, Cate Faculty Advisor: David Rapp

Historical Knowledge: Development of Normative Data for Historical Misconceptions Study

People's beliefs about history reflect and shape future-oriented public policy. Some of these beliefs may be misinformed, particularly due to differential information uptake for historical topics. Construal level theory suggests that information about temporally distant, historical events may be recalled more abstractly as compared to information about proximal, contemporary events. Increased psychological distance encourages greater reliance on general, distinguishing features rather than on details. To study how historical information is reproduced, we developed a set of 64 conceptually-matched questions related to 32 unfamiliar distal (e.g., "What city did the Erie Canal make the primary seaport in the U.S. in the late 1800s?") and 32 unfamiliar proximal (e.g., "What port has been most impacted by the Trump administration's import tariffs?") U.S. historical events to be used in misinformation experiments. We normed these materials by asking 160 participants to answer each of the 64 open-ended questions, indicate their confidence in their responses, and rate their familiarity with the topics. Participants' response accuracy ranged from 1.32% to 87.50%. We identified and analyzed the 24 distal/proximal pairs with the lowest accuracy (30.76% average) for use in future studies. Correlational analyses indicated that response confidence was weakly correlated with accuracy ($r = 0.32$), familiarity with the question topic was moderately correlated with accuracy ($r = 0.59$), and familiarity was strongly correlated with confidence ($r = 0.82$). These normed materials will be used in planned experiments to examine the influence of psychological distance on readers' uptake of and reliance on inaccurate historical ideas.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 71P

Shah, Rishi Faculty Advisor: Amy Paller

Efficacy of Tomivosertib (MNK1/2 Inhibitor) in Mitigating Pain in a Mouse Model of Recessive Dystrophic Epidermolysis Bullosa

Recessive Dystrophic Epidermolysis Bullosa (RDEB) is a rare genetic disease that causes extremely fragile skin, leading to frequent blistering, wounds, and scarring. In addition to visible skin damage, many patients experience chronic nerve-related pain, especially during routine care like changing bandages. Current treatments, such as morphine and gabapentin, often provide limited relief and can have significant side effects. In this study, we investigated whether a specific cellular pathway, called the MAP Interacting Protein Kinase (MNK) signaling pathway, contributes to this chronic pain. We used a mouse model of RDEB that closely mimics the human disease, including both skin symptoms and pain-related behaviors. We found that several genes involved in the MNK pathway were more active in pain-sensing nerves of RDEB mice compared to healthy controls, suggesting this pathway may be overactive in the disease. To test whether blocking this pathway could reduce pain, we treated RDEB mice with a drug called tomivosertib, which inhibits MNK signaling. Treated mice showed reduced activity of pain-related genes and exhibited fewer pain behaviors, such as excessive grooming and facial grimacing. These findings suggest that overactivation of the MNK pathway may play a key role in chronic pain in RDEB. Targeting this pathway with drugs like tomivosertib could offer a new and more effective approach for managing pain in patients with this condition.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 36A

Shamah, Haley Faculty Advisor: Michelle Driscoll

All Cracked Up: How Solvents Stress Out Hydrogels

Poly(ethylene glycol)-based hydrogels are widely used in biomedical and soft robotic systems such as drug delivery and tissue engineering due to their ability to absorb solvents while maintaining their structural integrity and then revert back to their size. Here, we study drying, the dynamic process which occurs when solvent evaporates from a fully swollen gel. The drying process induces internal stress gradients that can lead to cracking, peeling, or fracturing, which limits practical applications for these materials.

Here, we investigate drying-induced fracturing and peeling of PEG-based hydrogels, focusing on how solvent volatility, crosslinking density, and gel thickness influences these behaviors. We find that cracking and peeling behavior does not vary significantly across changes in gel thickness or crosslinker density, but it does depend on the volatility of the solvent. More volatile solvents lead to faster drying, as well as a greater extent of material fracturing.

By quantifying the behaviors of drying-induced fractures, our research provides insights into these mechanics and offers strategies to design more durable PEG-based hydrogels. These insights also advance our understanding of solvent-gel interactions in various drying conditions and can be used to help inform the design of systems that rely on structural integrity and minimal instabilities.

Presentation Format: Oral
Time: 10:45AM-12:15PM
Room: Lake Room

Shariff, Sabeen Faculty Advisor: Rayvon Fouche

Storytelling & Video Editing for Protest Tradition: The NBA Bubble Podcast

In the summer of 2020, the NBA created a biosecure "bubble" at Disney World to finish its season amidst a global pandemic. Though often glazed over as a health, athletic, and financial success, this project examines the bubble's reckoning with the pandemic and the Black Lives Matter movement to reevaluate the practice of protest in sports. Drawing on news coverage, athlete statements, social media, and academic scholarship on race and sports media, this podcast investigates three questions: whether the bubble's geographic and economic isolation amplified or constrained political action; whether the two-day work stoppage constituted genuine protest or institutional management of dissent; and whether the negotiated outcome, expanded voting access, represented a meaningful resolution or a deflection from the core demand of ending police violence. The podcast further examines the tension between athletes' economic privilege and their inextricable Blackness, asking whether proximity to wealth severs or deepens connections within communities most harmed by racial violence. Our contributions included reviewing, selecting, and sequencing footage and audio; annotating raw footage; editing video clips using Premiere Pro and Descript; eliminating redundancies; creating promotional short-form content; and balancing humor and tragedy to shape an impactful narrative arc. Effort was dedicated to making the content accessible for all listeners to amplify the impact of sports' role in politics and calls for justice.

This podcast asks who controlled the bubble, and to what end, challenging listeners to reflect on their own social values, expectations of athletes, and the opportunities for progress spurred by protest in sports.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 32A

Shen, Claire Faculty Advisor: Christine Wang

Engineering Galectin-mimicking Nanoparticles for Targeting Lung Cancer

Lung cancer is the leading cause of cancer-related mortality. Despite recent advances in immunotherapy, the clinical response rate for treatments remains below 50%. This warrants the development of targeted therapeutics and biomarker discovery for safe and effective delivery of these agents. Tumor-associated carbohydrate antigens (TACAs) are glycans overexpressed in cancer that contribute to a suppressive tumor microenvironment and represent promising targets. However, their potential remains underexplored due to heterogenous glycosylation. To address this gap, I developed a nanoparticle panel for systematic investigation of glycan-mediated tumor-immune interactions. In this work, I use nanoparticles displaying novel peptides derived from galectins, a class of carbohydrate-binding proteins that bind to β-galactose-containing glycans with varying affinity and specificity. Lectin staining confirmed the presence of β-galactose on human lung cancer cells (A549) which increased following sialidase treatment. Peptides derived from carbohydrate recognition domains (CRDs) of galectins were then conjugated to poly(lactic-co-glycolic acid) (PLGA) nanoparticles which enables multivalent presentation and increased binding avidity. Fluorescence microscopy in A549 cells showed greater Galectin-3 nanoparticle (Gal3s-NP) binding per cell compared to the control (mean 2.17 vs 1.68, $p=0.002$). Flow cytometry further demonstrated increased Gal3s-NP binding in A549 relative to human T cells (1.92 vs 1.29-fold change over control). Thus, I found that galectin-mimicking nanoparticles enable increased specificity for cancer epithelial cells. I expect that our nanoparticle platform will shed insights into dissecting complex tumor-immune interactions with potential to improve targeted delivery of existing and emerging anticancer agents. Future work will expand to additional galectins and ex vivo tissues.

Presentation Format: Oral
Time: 12:45-2:15PM
Room: Lake Room

Shin, Cat Faculty Advisor: Daniel Shanahan

Examining the Relationship Between Musical Sophistication and Technologically Mediating Listening Experiences

With the introduction of all-access music streaming platforms, there has been a noticeable shift in user behavior as listeners move away from radio-broadcast systems toward a more individualized experience. Previous research has investigated how users interact with these platforms and found a correlation with musical sophistication, measured using the Goldsmiths Musical Sophistication Index (Gold-MSI; Mullensiefen, et al. 2014), suggesting that listening behaviors can predict musical sophistication (Ferwerda & Graus, 2018). This study examines the relationship between listener types and musical sophistication using two psychometric instruments: the Streamed Listening Index of Musical Experience (SLIME) and the Gold-MSI. Data collection is still ongoing (anticipated n=300).

These findings have implications for how we theorize the relationship between immersion in music and technologically mediated listening. A deeper understanding of the relationship between musical sophistication and a critical awareness of recommendation systems can help us to better understand the nature of intentional vs. passive consumption in music, which could contribute to our understanding of relationships between training and enculturation, implicit and explicit knowledge, and broader elements of musical absorption (cf. Sandstrom and Russo, 2013). Future work will examine whether this relationship holds across listener subgroups and how specific dimensions of sophistication - such as musical training, engagement, and perceptual abilities - differentially predict algorithmic awareness. Ultimately, this research aims to refine psychometric tools appropriate for studying music listening in algorithmically shaped environments.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 40P

Shin, Wonho

Self similarity of Fractals

What does it mean to measure the complexity of a shape when it is too irregular to be described by notions like length or area? Fractals provide a compelling example: they are geometric sets in which similar patterns repeat at different scales, so that zooming in reveals smaller copies of the overall structure. Classic examples include the Cantor set and the Sierpinski gasket, which are constructed through iterative processes yet produce intricate forms.

A key tool for quantifying this complexity is the *Hausdorff dimension*, which extends the idea of dimension beyond integers and captures how "dense" a set is at arbitrarily small scales. Unlike standard dimensions (such as 1 for a line or 2 for a surface), fractals often have non-integer values.

This project investigates how Hausdorff dimension behaves under transformations. While some transformations preserve dimension, there is no simple general criterion for when it remains unchanged. To address this, I study classical fractals and apply transformations including scaling, nonlinear distortions, and projections, comparing their dimensions before and after transformation.

The results reveal a sharp contrast. Transformations that preserve geometric structure, such as uniform scaling, also preserve Hausdorff dimension. In contrast, transformations that collapse structure, such as projecting a two-dimensional fractal onto a line, can reduce it. These findings show that fractal complexity is stable under some transformations but sensitive to others.

This work highlights the dependence of fractal geometry on representation and points to the open problem of characterizing which transformations preserve Hausdorff dimension.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 74P

Shuford, Emma Faculty Advisor: Marcia Grabowecky

Food Difficulties Associated with Autism Spectrum Disorder and Travel Experiences

Autism Spectrum Disorder (ASD) is often associated with various food difficulties, including pickiness and sensory sensitivities, which often result in a reliance on familiar, preferred foods. These food sensitivities have been studied frequently in children, but there is little known about how food difficulties manifest for adults with autism. Even less is known about how food difficulties associated with ASD affect adults' relationship to traveling and experience of a new place. The present study examined the relationship between travel and ASD-associated food difficulties in the form of an anonymous self-report questionnaire conducted via Qualtrics. The self-report questionnaire included the Swedish Eating Assessment for Autism Spectrum Disorders (SWEAA) to assess issues related to eating and mealtimes in adults with autism, in addition to behavioral and attitudinal questions aimed to more specifically investigate people's travel experiences as related to food and eating, including but not limited to anxiety or stress around food selection when traveling. Food is an integral part of the travel experience, and traveling often disrupts familiar routines and limits access to preferred foods. Participants reported moderate levels of the following: food sensitivities, general preferences for familiar foods and mealtime routines, and food-related difficulties during travel. Studying these food difficulties can increase understanding about the stressors adults with autism may face when traveling, inform more inclusive travel planning, and lead to travel-targeted interventions and general public awareness surrounding complex experiences with food and travel. Future research should explore how to improve travel for adults with autism who experience food difficulties.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 45A

Simon, Ella Faculty Advisor: Joseph Bass

Antisense Oligonucleotide-Mediated Knockdown of ZFP423 in Adipose Tissue Enhances Thermogenesis and Improves Metabolic Health

Circadian rhythms are present among nearly all organisms on Earth and serve to predict the metabolic demands of the day. Disruption of circadian rhythms due to shiftwork, night-time eating, and interrupted sleep increases risk for diabetes and obesity. Recent studies demonstrated that circadian disruption through time-restricted feeding (TRF) during the inactive period disrupts thermogenesis, leading to greater risk of diabetes and obesity. Genetically knocking out zinc finger protein 423 (ZFP423), a gene that is normally responsible for maintaining white adipose tissue, promotes the formation of "beige" adipocytes and enhances thermogenesis. We recently found that knocking out ZFP423 prevents the negative effects of inactive period TRF. We hypothesized that a drug designed to knock down ZFP423 could make an effective therapy for diabetes and obesity, so we designed ZFP423-targeting Antisense Oligonucleotides (ASOs), an RNA-targeting therapeutic, modified to target adipose tissue. Following four weeks of weekly ASO injections, wild-type mice were given a glucose tolerance test and then placed into Sable metabolic phenotyping chambers to measure food intake, oxygen consumption, daily activity, respiratory exchange ratio, and energy expenditure. Lastly, we collected tissue samples from mice to perform qPCR on cDNA from these samples. We found that mice with ZFP423 knocked down exhibited improved glucose tolerance and insulin sensitivity, and reduced adiposity. The mice also showed increased oxygen consumption and energy expenditure without changes in activity. Together, this study indicates that ASO knockdown of ZFP423 enhances thermogenesis. These findings may help identify new therapies for combating obesity and metabolic disease.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 09A

Simwanza, Simz Faculty Advisor: Shelby Hatch

Heavy Metal Contamination in Chicago

Heavy metals are high-density elements that pose a significant toxicity risk to humans at elevated concentrations. Although naturally present in soil at low levels, human activity has contributed to heavy metal accumulation in urban areas, often disproportionately targeting marginalized communities. Since the level of heavy metal contamination in Chicago requires further exploration to improve evaluations of public health, this project aimed to quantify heavy metal contamination across Chicago's 77 community areas and assess its relationship with socioeconomic and racial demographics. We employed community-based methods to comprehensively survey areas of high human engagement with soil, such as playgrounds, and obtained soil samples from 60 community areas. Concentrations of heavy metals were quantified with Inductively Coupled Plasma-Optical Emission Spectroscopy. Using R and ArcGIS software, these results were geospatially integrated with demographic data from the U.S. Census Bureau and heavy metal regulations from the Environmental Protection Agency (EPA). Holistically, our findings suggested that lower-income and majority non-white Chicago neighborhoods have higher concentrations of heavy metal contamination. Notably, community areas, such as Garfield Park and Brighton Park, consistently exhibited contamination levels that violated EPA thresholds for the majority of metals that were analyzed. Some heavy metals, including arsenic and chromium, exceeded EPA limits across all sampled locations. By elucidating heavy metal contamination patterns, our research highlights communities that are at risk and better informs future environmental policy decisions. Importantly, we emphasized collaborating with communities rather than acting as outside experts, which allows our work to empower communities and support local residents.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 77P

Sisbarro, Derek Faculty Advisor: Shi-Yuan Cheng

Therapeutic Targeting of Chromosome Damage in Drug Resistant Brain Cancer

Glioblastoma (GBM) is a deadly brain cancer commonly treated with the chemotherapy drug temozolomide (TMZ), which targets guanine-rich regions of DNA. After TMZ treatment, cancer cells often develop major abnormalities in their nucleus and chromosomes. Normally, chromosome ends, telomeres, are protected by a group of proteins called the shelterin complex, which binds to guanine-rich DNA structures. When this protection is disrupted, it can trigger a telomere crisis, leading to nuclear envelope rupture and a severe form of DNA damage called chromothripsis (CT). Here, we show that TMZ disrupts shelterin binding and increases both nuclear envelope rupture and chromothripsis. We tested the idea that TMZ treatment makes cancer cells more dependent on nuclear transport systems, which move molecules in and out of the nucleus. This dependency may make TMZ-resistant cells more vulnerable to drugs that block nuclear export, such as KPT-330 (Selinexor), which targets the export protein XPO1. To study this, we used DNA sequencing, immunofluorescence, and cell proliferation experiments in both TMZ-sensitive and resistant GBM cells. We also analyzed samples from patients treated with KPT-330. We found that TMZ-resistant cells have shorter telomeres and clear signs of chromothripsis, including nuclear damage and protein mislocalization. These cells were significantly more sensitive to KPT-330 than TMZ-sensitive cells. Importantly, patient data showed reduced chromothripsis markers after treatment, and higher initial levels of these markers were linked to longer survival. Overall, our results suggest that TMZ-induced DNA damage increases reliance on nuclear transport, creating a therapeutic opportunity for targeting resistant GBM with KPT-330.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 82P

Smirnova, Anastasia Faculty Advisor: M Olvera de la Cruz

Tunable Permeability of Peptide-Brush-Polymers

Peptide therapeutics have been in use since the early 20th century, with the advent of insulin. They attach very precisely to important receptors in a human cell that trigger a variety of crucial intracellular effects- this is leveraged to treat metabolic disorders, cardiovascular disease, etc. However, they tend to dissolve in the bloodstream or fail to enter the human cell. Peptide-brush polymers (PBPs) improve on this- they have a tree or brush-like structure that imparts improved stability, with better stability the more hydrophobic their backbone is. However, current literature doesn't describe how this affects permeability- their ability to cross into the human cell.

In this project, we wished to investigate this effect through molecular dynamics simulations, a popular computational method. To do so, several model PBPs of varying backbone hydrophobicity and a cell membrane were constructed using the GROMACS software package. Then, the motion through the cell membrane was simulated and the energetic and structural changes characterized. The PBP's repulsion or attraction at relevant coordinates was quantified through weighted histogram analysis method and visualized using the software package Visual Molecular Dynamics. It was found that the greater the hydrophobicity difference between the backbone and side chains, the more difficult it was for the PBP to adapt from the membrane to the aqueous environment. We also wish to further investigate the effect of side chain density. In the future, this may be used to design more permeable and adaptive PBPs.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 01P

Snyder, Alexandria Faculty Advisor: Marco Gallio

Substrate Temperature Preference During Egg-Laying in Drosophila

Ectotherms like fruit flies cannot regulate their internal body temperature and instead rely on behavioral strategies to avoid temperatures outside of their optimal range. Unlike adult flies, *Drosophila* larvae are limited in their ability to escape unfavorable conditions, and thus, female flies have to select a suitable substrate on which to lay eggs to increase the survival of the progeny. However, we know little about the specific choices a female fly makes when choosing the egg-laying substrate. The Gallio Lab studies the molecules and neurons underlying temperature-guided behaviors in *Drosophila*. In this project, we explored how substrate temperature influences where female fruit flies lay their eggs. We developed a new 2-choice temperature assay in which gravid female flies are given a choice between substrates kept at different temperatures. Using this assay, we found a strong preference for egg laying at 25C over lower or higher temperatures. Surprisingly, the main thermosensory organ of the fly, the antennae, is not required for this behavior. To identify the relevant thermosensory neurons, we have tested the role of candidate cell types using genetic tools to silence their activity. Lastly, we found that *Drosophila* species adapted to different thermal environments prefer to lay eggs at temperatures matching their native environments. This work suggests the presence of novel thermosensory neurons affecting decision-making during egg-laying behavior in *Drosophila*.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 90P

Sohini, Rohan Faculty Advisor: Yogesh Goyal

Modeling the Variability in Breast Cancer Resistance Outcomes

Within a population of genetically identical cancer cells, rare cells can survive drug treatment, leading to relapse in cancer patients. Over the past century, researchers have developed probabilistic models to understand how cancer cells grow. Initial models assumed that all resistant cells were the same, so they modeled them as having the same growth probabilities. Later work introduced additional states like persisters which are slow-proliferating resistant cells, in contrast to fast-proliferating cells.

However, with over 20,000 protein-coding genes in DNA, with each acting as a dial that shapes a cell's identity, there are potentially *thousands* of meaningful states a cell could occupy. Moreover, despite common assumptions, a cell's ability to switch between states, and follow different growth laws, is realistically not governed by a constant probability.

Thus, I implemented a theoretical framework that does not require a fixed number of states. Instead, I model a cell's probability to divide or die by drawing the probabilities from a distribution. By treating the probabilities as random variables, I am lifting the assumption that cells follow discrete states. Using my model, I inferred these distributions from publicly available data on breast cancer cells treated with two drugs, a targeted therapy and a chemotherapy.

I found that in small doses, the targeted therapy can actually promote cancer cell growth. For both drugs, cells become more volatile at medium doses, making less predictable decisions to divide or die. Lastly, I recovered that the chemotherapy results in greater cell death, which is a known result.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 02P

Stanciu, Nicolas Faculty Advisor: Dominique Licops

Inclusive Writing in the French-speaking World: A Study of Governmental and University Policies in Canada, France, and Belgium

French inclusive writing (*ecriture inclusive*) comprises various linguistic strategies-including the feminization of job titles, gender-neutral terminology, and the use of the mid-dot (*professeure*)-intended to challenge the French-language grammatical convention of the masculine as a universal default. While these practices have gained visibility across the French-speaking world, their official adoption remains inconsistent. This study examines the formal incorporation of inclusive writing practices within governmental and academic institutions in Canada, France, and Belgium, aiming to map the intersection between public policy and institutional implementation.

A comparative analysis was conducted using official government documents, including legislative decrees, ministerial circulars, and institutional writing guides, alongside university communication policies and adopted writing guides from each country.

It was found that while all three governments broadly support traditional strategies such as feminization and the use of epicene terms, significant restrictions are placed on the mid-dot and neopronouns. A restrictive stance is maintained in France, where inclusive writing has been prohibited in official texts since a 2017 ministerial circular. In Canada, comprehensive guidelines are provided through an interministerial framework, though compliance is voluntary. In Belgium, a supportive 2021 decree was initially adopted, but restrictions regarding the mid-dot have recently been implemented within educational settings.

Furthermore, governmental policy is found to correlate with university behavior: French universities exhibit the highest level of restraint, Canadian institutions generally adhere to regional linguistic authority, and Belgian universities demonstrate the most progressive adoption of inclusive forms. These findings highlight the role of governmental policies in shaping institutional inclusive writing practices.

Presentation Format: Oral
Time: 10:45AM-12:15PM
Room: Armadillo Room

Stone, Dylan

Reassessing v-Blockers in HFpEF: Prescribing Patterns And Outcomes In The Cardiometabolic Center Alliance Registry

Beta-blockers are medications that slow the heart rate and reduce blood pressure, and they are well-established treatments for one form of heart failure where the heart pumps too weakly (HFrEF). However, their benefit is far less certain in heart failure with preserved ejection fraction (HFpEF), a distinct and increasingly common condition in which the heart pumps normally but has difficulty relaxing to fill with blood. Despite evolving evidence questioning the routine use of beta-blockers in HFpEF, many patients continue to receive them. To examine this practice, we analyzed data from 620 adults with HFpEF enrolled across 11 clinical sites in the Cardiometabolic Center Alliance (CMCA) registry between March 2019 and May 2025. Beta-blocker use was classified into dosage tiers, and patient characteristics and one-year outcomes - including hospitalizations for heart failure and episodes of fainting - were compared across groups. We found that nearly three-quarters of patients (72.8%) were prescribed beta-blockers, most at moderate doses. Patients with coronary artery disease or atrial fibrillation were more likely to be on them, yet dosing did not consistently reflect recent cardiac events such as heart attacks. Importantly, one-year rates of heart failure hospitalization and fainting were similar between treated and untreated patients. These findings suggest that beta-blockers may be overused in HFpEF patients, and that clinicians have meaningful opportunities to reassess and potentially reduce or discontinue therapy in carefully selected patients - a step that could minimize side effects and shift focus toward more effective, evidence-based lifestyle treatments.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 43A

Sun, Jasmine Faculty Advisor: Abigail Adams

Exposing Hidden Risks: The Impact of Laws Targeting Nonfatal Strangulation on Intimate Partner Violence

Intimate partner violence (IPV) is widespread yet substantially underreported to law enforcement. However, among cases of IPV that escalate to become lethal, the vast majority involve prior law enforcement contact for domestic violence in the years preceding homicide. These moments of contact present critical opportunities for law enforcement to act before lethal escalation occurs, but effective intervention depends on the justice system to respond appropriately.

Nonfatal strangulation is a particularly dangerous form of IPV and one of the strongest predictors of future homicide, but it often leaves little visible physical evidence, limiting law enforcement's ability to identify and respond effectively. In response, many states have adopted statutes specifically targeting nonfatal strangulation to strengthen legal recognition and penalties for this form of violence.

My thesis examines the effects of these laws on reporting and identification, law enforcement responses, and escalation to homicide. Exploiting the staggered adoption of statutes targeting strangulation across states in the U.S., I employ a difference-in-differences framework using incident-level data combined from the National Incident-Based Reporting System (NIBRS) and Uniform Crime Reports Supplementary Homicide Reports (SHR). I find that strangulation statutes increase both identification of IPV strangulation cases and arrest rates in severe IPV incidents. Additionally, these laws reduce the rates at which females are killed by male partners, suggesting that increased intervention can mitigate escalation. Heterogeneity analysis by victim race highlights how policy impacts differ across varying social and institutional contexts.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 63A

Sun, Jessica Faculty Advisor: Haoqi Zhang

PATH: A Structured Reflection Tool for Understanding and Addressing the Interplay among Emotions, Beliefs, Metacognition, and Actions in CS1

During programming struggles, many introductory Computer Science students resort to ineffective behaviors such as brute-forcing and seeking direct answers. However, researchers lack understanding of how students' learning process breaks down. Our work proposes the Multilevel-Regulation framework, in which we argue that beyond content difficulty, these ineffective behaviors arise due to students' self beliefs shaping their emotional responses during struggle, and that limits access to effective metacognitive strategies, leading to ineffective actions and outcomes. We embodied this framework in *PATH* a structured reflection tool that helps students understand this interplay. In our experiment, we compared *PATH* (structured reflection) with free-form journaling to examine how students' understanding of their struggles and proposals for future strategies differed after using the two tools. Through qualitative coding, we found that *PATH* was more effective in helping students 1) understand how struggle was shaped by their interplay of beliefs, emotions, and metacognitive skills, and 2) develop a specific suite of strategies across those dimensions. We also identified three distinct pathways of student struggle: 1) Time Anxious Pathway, where low self-efficacy leads to panic and rushed, unplanned actions such as seeking direct answers; 2) Motivational Withdrawal Pathway, where low self-efficacy leads to demotivation and brute forcing; and 3) Performance-Protective Fixation Pathway, where fear of failure leads to guilt and fixation on ineffective approaches. This analysis deepens understanding of common struggle patterns among introductory CS students and calls on educators and researchers to design cross-dimensional interventions that help students overcome their particular pattern while learning to program.

Presentation Format: Oral
Time: 12:45-2:15PM
Room: Arch Room

Taboada Abrego, Ericka Faculty Advisor: Pamela Souza

Understanding how hearing loss and cognitive change impact older adult communication

As individuals age, communication can become increasingly difficult, particularly for those experiencing cognitive decline, such as dementia. This project aims to examine whether hearing aids, communication strategies, or a combination of both are most effective in improving communication between older adult partners.

To investigate this, participants are older adults with hearing loss and either dementia or mild cognitive impairment. Participants and their communication partners (e.g., their spouse) complete a conversation task, using paired images designed to prompt conversation. Conversations are video-recorded and analyzed for efficiency (e.g., time to identify image differences) and communication patterns (e.g., how partners resolve confusion or missed words). Participants are randomly assigned to either learn communication strategies before using over-the-counter hearing aids or begin with hearing aids before receiving communication training. This approach allows for comparison of how the order of interventions may influence communication outcomes.

Although data collection is still ongoing, early observations suggest that both hearing support and communication strategies facilitate clearer interactions. However, their effectiveness may depend on the order in which they are introduced.

These findings have important implications for improving the quality of life in aging populations. Identifying effective, accessible interventions could support not only individuals with dementia but also their partners and caregivers, making everyday communication more manageable and meaningful.

Work on this project is supported by funding from the National Institute on Aging.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 47P

Tagoe, Jaeda Faculty Advisor: Almaz Mesghina

Trust or Bust? Investigating Cultural Variations in Physician-Patient Interactions

This study examines how empathy is valued in healthcare across cultures. Most research has been conducted in Western cultures that endorse individualistic, independent values, whereas non-Western societies hold collectivist, interdependent values. To bridge the gap, this study explored whether patients' perceptions of patient-physician interactions differ in Accra, Ghana, a more collectivist society, as compared to the United States, a more individualistic society. We hypothesize that a lower perceived physician empathy should result in lower patient satisfaction, especially in Ghana (Hypothesis 1). Because Ghanaians tend to be more collectivist, who might exclude physicians from their in-group, Ghanaian patients should display a lower perceived physician empathy when compared to the United States (Hypothesis 2a). However, if Ghanaians include their physician in their "in-group," Ghanaian patients should display a higher perceived physician empathy when compared to the United States (Hypothesis 2a). Lastly, Participants within the American sample who obtain a greater Collectivism score should predict greater satisfaction with higher perceived physician empathy (Hypothesis 3).

Data from this study were first collected from patients visiting a hospital in Accra, Ghana (N = 26). Then, identical data were collected from the United States (N = 44). This showed that both Ghanaian and American patients value empathy, with Ghana displaying a stronger correlation, which positively predicted higher satisfaction. Furthermore, American patients who scored higher in collectivism reported greater patient satisfaction. We hope that these findings encourage medical educators to consider patients' cultural perceptions and preferences, and the implications this may have for their satisfaction with their healthcare.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 69P

Tchernikov, Anastasia Faculty Advisor: Reesha Patel

Corticotropin-Releasing Factor in the Medial Prefrontal Cortex Modulates Fear Generalization in a Sex-Specific Manner

Post-traumatic stress disorder (PTSD) is a severe psychiatric condition characterized by symptoms such as fear generalization, where threat responses extend beyond the initial traumatic stimulus to similar but safe stimuli. The medial prefrontal cortex (mPFC) is a critical brain region involved in fear learning, regulation, and expression, making it significant for understanding fear disorders. Corticotropin-releasing factor (CRF), a neuropeptide involved in stress responses, is expressed in a subset of inhibitory interneurons within the mPFC and is often elevated in the cerebrospinal fluid of individuals with PTSD, suggesting persistent hyperactivity of central CRF neurons. In the mPFC, CRF primarily acts on corticotropin-releasing factor receptor type 1 (CRF1), which is located on excitatory projection neurons that influence downstream fear circuits. Despite this, the precise role of mPFC CRF-CRF1 signaling in fear behaviors, particularly fear generalization, remains unclear. We hypothesized that fear expression would increase mPFC CRF release, and that manipulating mPFC CRF-CRF1 signaling would alter fear behaviors. Using an auditory Pavlovian fear conditioning paradigm in mice, we measured CRF release with fiber photometry, manipulated CRF neurons using optogenetics, and pharmacologically altered CRF1 signaling. Fear conditioning increased both freezing behavior and CRF release during acquisition and generalization. In male mice, inhibiting mPFC CRF neurons reduced freezing during generalization, while activation increased it; these effects were not observed in females. Pharmacological CRF1 inhibition also showed a trend toward reduced freezing in males. These findings reveal a sex-specific role for mPFC CRF-CRF1 signaling in fear generalization, advancing our understanding of the neural mechanisms underlying PTSD.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 54P

Tendilla, Brandon Faculty Advisor: Nivedita Arora

Functionalizing Yarns with Dip-Coating for Smart Knitted Haptic and Tactile Interfaces

The human skin is lined with several mechanoreceptors that are constantly converting things like pressure, vibration and textures into tactile signals that our brain can interpret. Replicating a small portion of this sensitivity into everyday materials would make sensing possible without being intrusive. Clothes are almost all knitted and universally needed by everyone, which makes it the perfect choice to begin embedding sensing into what we use. However, the process is limited by the material as there are no commercially available conductive yarns that are machine knittable, flexible, and insulated in a way that can be useful for research and prototyping. This project addresses this problem by developing a low-cost dip coating process that produces custom insulated conductive yarns compatible with knitting machines. We studied the effects that varying mechanical, chemical, and environmental parameters had on coating uniformity, thickness, and electrical isolation. By characterizing this behavior, we have developed a fabrication process that meets all requirements for being integrated into everyday knitted items. The initial results of these custom yarns have been tested in a proper knitting machine which proved the abrasion resistance was enough to maintain proper electrical insulation after being used to knit. These knitted items were used to create capacitive sensing pressure arrays that mimic mechanoreceptors and triboelectric nanogenerators for low power embedded systems. These results demonstrate that by solving the yarn fabrication problem, a new class of embedded smart knitted textiles applications can be created, from haptic feedback interfaces to pose tracking for AI databases.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 71A

Tribush, Shayna Faculty Advisor: Lilah Shapiro

Policy and Perception: How Legislation Shapes Gen Z Attitudes Toward Sex Work in the Netherlands and the U.S.

Debates around sex work exemplify morality politics, as conflict is driven by ideological disagreements rather than technical disputes. Legislation addressing sex work often aims to influence social norms, relying on the assumption that law influences perception so that legalizing sex work will normalize it in society. This study examines that assumption by comparing attitudes toward sex work among Gen Z women in the Netherlands and the United States, two countries with distinct policy frameworks. While existing research links policy and public attitudes, it remains unclear whether law actively shapes perception or primarily reflects preexisting cultural values.

I conducted 22 semi-structured interviews with college-educated, secular Gen Z women (10 Dutch, 12 American), focusing on this group because age, gender, and religiosity are among the strongest predictors of attitudes toward sex work. I analyzed interviews using a grounded theory approach. Three primary themes emerged: the moral framing of sex work, the gendering of the industry, and the navigation of political correctness. Across all three, American participants more frequently drew on empowerment paradigms and feminist framing, while Dutch participants more often invoked schemas of necessity and exploitation. Both groups displayed internal contradictions, voicing support for sex workers while expressing moral judgement for clients. These findings suggest that attitudes are shaped less by formal legal frameworks than by the cultural contexts in which participants most commonly encounter sex work.

Presentation Format: Oral
Time: 10:45AM-12:15PM
Room: Rock Room

Truong, Anna Faculty Advisor: Mesmin Destin

"I'm Grateful that I Had a First-Gen Experience": A Qualitative Study of the Strengths of First-Generation Professionals

As first-generation college students graduate, they face unique challenges when they enter the workforce. These first-generation college students become first-generation professionals, and research has shown that they face barriers in hiring and advancement due to deficit narratives attached to their first-generation status. In contrast to these narratives, this study explores the strengths of first-generation professionals, gained from their backgrounds, and how they use them to succeed in their careers. Employing a qualitative approach, this study incorporates semi-structured interviews with 17 first-generation professionals (10 female, 7 male; 41% Hispanic, Latine, Spanish, 35% White, 18% Black, 12% Asian) with work experience (median = 18 years) in the nonprofit, philanthropy, and higher education sectors. Results showed that when determining their career path, first-generation professionals greatly benefitted from internship experiences, career exposure, and mentors to guide them. Participants attribute their professional success to various strengths of being first-generation such as being resourceful, adaptable, and perceptive. Although they face challenges extending from college regarding finances, career preparation, and networking, they rely heavily on supportive supervisors and colleagues, as well as their aspirations to give back to their communities, as motivation in their careers. Notably, many participants express feeling a disconnect between their experiences and their parents' experiences, compounding the isolation they already feel in the workplace, as they navigate the professional world. Thus, this research highlights the key role universities and workplaces can play in recognizing the strengths of first-generation professionals, and creating environments that foster belonging and opportunities for mentorship and social support.

Presentation Format: Oral
Time: 10:45AM-12:15PM
Room: Rock Room

Tu, Vivian Faculty Advisor: Thomas Meade

Size-Dependent Delivery of Gold Nanoparticle Single-Chain Variable Fragment Conjugate for Early Diagnosis of Alzheimer's Disease

Alzheimer’s disease (AD) is the leading cause of dementia, and its prevalence is expected to rise substantially in the coming years. Current symptom-based clinical diagnostics lack sensitivity for early-stage detection, contributing to significant underdiagnosis of early-stage AD, when therapeutic interventions are most effective. Although magnetic resonance imaging (MRI) is routinely used to assess brain volume, existing clinical contrast agents lack the sensitivity and molecular specificity required for early detection. Thus, there is a clinical need for biomarker-based strategies to diagnose AD at its earliest stages. In collaboration with the Klein lab, the Meade lab has developed a highly sensitive, multimodal molecular imaging probe targeting amyloid-β oligomers (AβOs), a key neurotoxic species in AD. This probe consists of gold nanoparticles (AuNPs) conjugated to NUsc1, an AβO-specific antibody fragment. We have successfully synthesized and characterized the size and concentration of these probes using UV-Vis spectroscopy, dynamic light scattering (DLS), inductively coupled plasma optical emission spectroscopy (ICP-OES), and transmission electron microscopy (TEM) imaging. Confocal reflectance microscopy images of *ex vivo* brain sections immunostained with NUsc1-AuNPs show robust, specific signals with a clear concentration-dependent response. Ongoing studies aim to evaluate probe binding and detection of AβOs in primary neuron cultures and a murine model (5xFAD) of AD. To assess the sensitivity and binding efficiency of the probe *in vivo*, delivery and distribution of AuNPs will be monitored using both IVIS and micro-CT, and quantitative biodistribution of gold in brain tissue will be measured via ICP-MS.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 07P

Udwadia, Nicholas Faculty Advisor: Jonathan Guryan

The Effects of Name, Image, and Likeness (NIL) Policy on College Football Winning Percentages

In a landmark hearing in 2021, the Supreme Court unanimously ruled against the NCAA's restrictions on compensation and benefits for college athletes. This ushered collegiate sports into a new age of expanded student athlete power, as players are now able to monetize their name, image, and likeness (NIL). The advent of NIL policies created a new vector by which college athletic programs compete to attract the most talented national recruits. Considering the infancy of NIL policy, much research needs to be done in order to analyze the economic and competitive implications. This economics thesis paper aims to discover the effects of NIL policy on college football winning percentages. Given that universities have differing levels of financial resources and alumni-giving, economic theory indicates that greater collegiate NIL funding should lead to the recruitment of better talent and the accumulation of more wins. This paper applies production function maximization and Tullock Contest competitive dynamics to model how universities compete with one another to maximize their utility function given a range of key input variables. After collecting estimated university level NIL expenditure data along with other institutional financial metrics, regression analysis and difference-in-difference inspired computational methods were utilized. The results indicate that NIL expenditure has a negligible effect on winning percentages. This raises significant implications for college athletic programs because it indicates that NIL policy is a vehicle of social welfare redistribution, as athletes are receiving greater producer surplus, while universities do not gain any additional wins on the aggregate.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 70A

Vandayar, Kavin

Understanding the Interaction Between Self-perceived Hearing Metrics, Demographic Characteristics, and Hearing Aid Retention

Purpose: Over-the-counter (OTC) hearing aids (HA), a more affordable and recently available alternative, are designed to address a longstanding treatment gap for hearing loss in adults. Prior research has demonstrated lower retention rates for OTC HA. The purpose of this study was to examine the factors driving retention for OTC and audiologist-fit HA - specifically, the influence of self-perceived hearing metrics and demographic characteristics on the decision to retain devices following a six-week trial.

Method: Data was drawn from a multisite randomized controlled trial (N = 584) in which participants with perceived mild-to-moderate hearing difficulty were fit using one of three fitting conditions: audiologist best-practices (AB) or one of two self-fit methods, consumer decides (CD), or efficient fitting (EF). Participants had the option to return hearing devices after a six-week post-fit period. Predictor variables included self-reported handicap measured via the Hearing Handicap Inventory for the Elderly (HHIE), self-perceived benefit assessed using the Abbreviated Profile of Hearing Aid Benefit (APHAB), and participant demographic characteristics including age, education level, and income.

Results: Data analysis is ongoing. Lower self-perceived device benefit and higher self-reported hearing handicap are anticipated to influence likelihood to retain devices regardless of fitting condition. Demographic characteristics are also expected to contribute to retention likelihood. As reported in the literature, retention is anticipated to be lower for OTC HA.

Conclusions: Understanding interactions between self-perceived hearing metrics, demographic characteristics, HA fit method, and HA retention can help shape the design of hearing healthcare policy and practice.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 40A

Vasile, Karina Faculty Advisor: Colin Hisey

Investigating Spatial Restriction as a Cue to Improve In Vitro Syncytium Formation

The placenta plays a critical role in fetal development by enabling the exchange of nutrients and gases between the mother and fetus. Its outermost layer, called the syncytium, forms a continuous, multinucleated barrier that carefully regulates what can pass between maternal and fetal blood. Because the placenta is the most species-specific mammalian organ, animal models often do not accurately reflect human placental function. At the same time, recreating a fully fused syncytium in the lab has remained a major challenge, limiting our ability to study how this barrier controls drug transfer and impacts fetal and maternal health. To address this gap, this project aims to develop a lab-based model of the maternal-fetal interface using a "mini-syncytium-on-chip". We focus on how spatial restriction, confining cells within defined shapes, affects the formation of the syncytium. This approach mimics the natural environment of the placenta, where tissue structures stretch and thin during pregnancy. Using micropatterning techniques, we create small, biologically relevant shapes on glass surfaces and seed cells onto them to observe how they fuse and organize. Initial optimization of pattern geometry and seeding density has shown that a shape diameter of 30 m with 8104 cells works best to promote the formation of syncytium-like structures. This model provides a scalable platform for studying placental cell behavior in a controlled setting, with potential applications in drug safety testing and the development of therapies for pregnancy-related complications.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 09P

Vincent, Andrew Faculty Advisor: Melville Ulmer

Determining the trajectory and origin of the X-ray binary source Hercules X-1

Hercules X-1 (Her X-1) is a binary star system that is a prominent X-ray source. Its position 3 kiloparsecs above the galactic plane suggests it was ejected from its origin, either the galactic plane or a globular cluster of stars, due to the supernova of its neutron star. Recent data from the Gaia satellite, which found precise astrometric values for stars, has new values of the position and angular velocity of Her X-1. These can be coupled with radial velocity measurements to calculate the orbit of Her X-1, which potentially allows for its origin to be determined and its supernova remnant (the remaining material from its neutron star's supernova) to be found. Using this data, the orbit of Her X-1 up until one billion years ago has been calculated through orbit integration with the Gala python package. Twelve different intersection points of the orbit with the galactic plane were found, with the first being approximately 65.7 million years ago. In addition, a globular cluster origin for Her X-1 before this first intersection has been ruled out. Given this minimum age, a supernova remnant for the system cannot be identified due to limits in detection. Future work will determine if, and if so, which velocities of Her X-1 at its intersection points are consistent with prior models of ejection velocities and if a globular cluster origin is possible. This research will add to our knowledge of X-ray binaries and help determine the accuracy of mass distribution models of the galaxy.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 58A

Vinokour, Emma Faculty Advisor: Danielle Tullman-Ercek

Optimizing MS2 Virus-Like Particle Assembly Around Nucleic Acid Cargo for Drug Delivery

Virus-like particles (VLPs) derived from the MS2 bacteriophage are promising nanocarriers for the delivery of therapeutic molecular "cargo" to diseased cells. The *in vitro* cargo-loading process requires disassembly of the VLP protein capsid and removal of its native RNA, followed by spontaneous capsid re-assembly around exogenous cargo, typically nucleic acids. However, there is no standardized framework for packaging and quantifying the encapsulated cargo across the literature. Here we compared multiple quantification techniques for MS2 VLPs and found that traditional absorbance-based measures of MS2 concentration can be cargo-dependent and therefore unreliable for assessing encapsulation yield, particularly across heterogeneous cargoes. We determined that area-under-the-curve absorbance from chromatographic separation provides a consistent proxy for encapsulation yield when supported by mass-based reporting of the cargo-to-protein ratio (as compared to a molar-based ratio). We also identified parameters for a more reproducible re-assembly protocol, including an optimal capsid disassembly incubation time. We then explored how cargo type and structural features influence differences in cargo-packing efficiency during *in vitro* VLP reassembly. Specifically, we examined VLP assembly with different nucleic acid cargoes and observed distinct yield and stability profiles, indicating that both chemical composition of (i.e. DNA vs. RNA) and structural determinants, such as a stem-loop motif, modulate packaging efficiency and capsid stability. These findings suggest that maximized VLP assembly and cargo loading require both accurate yield quantification and an understanding of the biophysical principles governing nucleic acid-capsid interactions. This work holds great potential to inform the sustainable and scalable design of emerging VLP-based drug delivery platforms.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 52A

Virsik, Kate Faculty Advisor: Luisa Marcelino

Examining Functional Genomic Variation in Symbiodiniaceae associated with Acropora cervicornis Coral

Coral reefs are essential ecosystems that support marine biodiversity, protect coastlines, and sustain fisheries. Corals harbor symbiotic photosynthetic dinoflagellate algae in their tissues that provide more than 90% of the coral's energy and carbon. Rising ocean temperatures can breakdown this relationship by causing corals to expel their algal symbionts due to increased toxic byproducts, often leading to starvation and death. However, some algal symbionts are heat tolerant, providing food to their hosts even at high temperatures. Identifying thermotolerant symbionts and measuring potential shifts in their community composition is therefore crucial for predicting coral holobiont (coral and associated symbionts) responses to environmental stress. I examined algae symbiont community composition in the critically endangered Caribbean coral Acropora cervicornis. RNA sequencing data from heat-sensitive and heat-resistant coral holobionts were analyzed to test whether symbiont community composition and functional profiles differed between the two groups. Sequencing reads were aligned to a draft genome of Symbiodinium fitti, the dominant symbiont in A. cervicornis. After filtering for high-confidence alignments (MAPQ >= 30), approximately 20,000 symbiont genes were retained per holobiont and annotated by function. Comparisons of gene expression profiles between symbionts associated with heat-sensitive and heat-resistant corals reveal potential functional variation in symbiont strains. Understanding these differences can inform conservation strategies, such as prioritizing heat-tolerant coral-symbiont pairings for outplantation to repopulate depleted reefs and improve coral survivorship in a warming ocean.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 27P

Vivona, Isa Faculty Advisor: Stephanie Eisenbarth

Investigating differences in cytoskeletal organization between dendritic cell subtypes during their migration.

Dendritic cells (DCs) are immune cells that migrate from tissues in the lung or gut to the lymph nodes, where they activate other immune cells. They can be classified as type 1 DCs (DC1s) and type 2 DCs (DC2s). To reach the lymph nodes, DCs must travel through the extracellular matrix (ECM), a complex 3-dimensional web of proteins. The migratory capacity of DCs through the ECM is essential to the immune response. During migration, DCs must protect their nuclei from ECM compression by reorganizing the cytoskeleton, including actin, microtubules, and intermediate filaments. The protein Dedicator of Cytokinesis 8 (DOCK8) is involved in a pathway that activates actin in response to compression. Previous work in mice has shown that DOCK8 is critical for DC migration in DC2s, but not DC1s. To further investigate how DC subtypes use their cytoskeleton differently during migration, I used under-agarose migration assays to simulate 3-dimensional DC migration and immunofluorescence to visualize cytoskeletal structures. I compared the localizations and colocalizations of cytoskeletal structures in DC1s and DC2s under untreated conditions, DOCK8 deficiency, and inhibition of additional DOCK8 pathway proteins. My findings suggest that DC subtypes exhibit distinct cytoskeletal rearrangements around their nuclei, with varying dependence on the DOCK8 pathway. Mutations in the DOCK8 pathway have severe implications for human disease, including several immunodeficiency disorders. Studying the role of these factors, specifically in dendritic cell subtypes, can help identify the components necessary for effective immune responses.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 51A

Vujevich, Sofia Faculty Advisor: SonBinh Nguyen

Lauroyl-MBEH-Loaded Nanoparticles as a Targeted Anti-Melanoma Treatment

To treat melanoma and metastases of this cancer, I propose to target the melanogenic pathway, where the natural amino acid L-tyrosine is converted into melanin, using drugs like monobenzyl-ether of hydroquinone (MBEH), an L-tyrosine mimic drug. Such targeting is highly selective, avoids common chemotherapy side effects, and enhances antitumor immune activity. Nanoparticles (NPs) loaded with lauroyl-MBEH, a novel MBEH derivative, can offer additional selectivity and cytotoxicity through their small size, abilities to target cancer cells, and capability of delivering a high concentration of drug to such cells. To determine the effectiveness of my proposed treatment, free MBEH is compared against the drug-loaded NPs. While this research is ongoing, I hypothesize that free MBEH is cytostatic while drug-loaded NPs are cytotoxic. To confirm my hypothesis, biochemical assays, like MTT and LDH, are used to quantify cell viability and cytotoxicity, respectively. In the MTT assays I've done so far, I observed that the percent cell viability decreases as the concentration of free MBEH increases. When compared to cell images of each treatment, the cells are not dying, but rather their growth is inhibited, indicating a cytostatic effect. In the LDH assay, percent LDH released and free MBEH concentration show a direct relationship, however LDH released tends to plateau after 50μM. To further support the hypothesis, mechanistic techniques, like Western blot analysis, will be used in upcoming experiments to quantify expression of three programmed cell death (apoptosis) proteins—caspase 3, BAX, and Bcl-2—and a proliferation protein, Ki-67, to further support my hypothesis.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 80P

Walsh, Marissa Faculty Advisor: Daniel Arango

t6A Inhibition Decreases Cell Proliferation of Cisplatin Resistant Ovarian Cancer Cells and May Alter Protein Translation Dynamics

t6A is a modified nucleotide in tRNAs that has been found to play a significant role in ensuring proper translation of mRNAs into proteins within cells. t6A has been found to promote translation of mRNAs starting at the canonical AUG start codon site. Past research has also found that t6A expression is amplified in ovarian cancer cells compared to normal, non-cancerous cells. We hypothesize that inhibition of t6A in ovarian cancer cells increases non-canonical translation of mRNAs. Thus, we predict that t6A inhibition can increase the synthesis of non-self proteins known as neoantigens, triggering immune system responses that kill ovarian cancer cells with increased efficiency. This project focuses on how t6A inhibition impacts protein translation within cisplatin sensitive and resistant ovarian cancer cell lines. We analyzed cell proliferation via a colony formation experiment, and we analyzed protein translation via a flow cytometry experiment. We found that cisplatin sensitive cells had little change in cell proliferation with t6A inhibition. Alternatively, we found that cisplatin resistant cells had a significant decrease in cell proliferation with t6A inhibition. We also found that protein translation in the cisplatin resistant cells had a significant increase with t6A inhibition. We believe that cisplatin resistant ovarian cancer cells treated with t6A inhibition have reduced proliferation due to altered protein translation, indicative of increased production of neoantigens, as predicted in our hypothesis. Further investigation will reveal the specific impacts of t6A inhibition, allowing for a clearer understanding of immunotherapeutic applications of t6A inhibition.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 13P

Wang, Rebecca

Understanding the Clinical Experiences of Human Trafficking and Domestic Violence Survivors in Illinois

Human trafficking (HT) and domestic violence (DV) are critical public health crises nationwide and in Illinois. Healthcare professionals are uniquely positioned to identify victims actively in need of life-saving support. Prior research has found that up to 90% of trafficking survivors accessed healthcare during their trafficking situation, yet fewer than half reported being asked about their personal circumstances. A comprehensive literature search regarding the intersection of healthcare and HT or DV survivors yielded limited studies, and were conducted from the perspective of healthcare professionals. There remains a gap in research regarding survivors' healthcare experiences in Illinois. We aim to address this using a cross-sectional study of adult HT & DV survivors across four Illinois shelters regarding their experiences in healthcare. Surveys were developed from screening tools and national clinical guidelines for HT & DV identification.

Data collection is still on-going with an anticipated total of 300 responses. We hypothesize participants will report healthcare encounters during periods of active abuse or exploitation, with relatively low rates of standardized screening and intervention. We anticipate variability in survivor comfort and perceived provider recognition, highlighting missed opportunities for identification and support. We anticipate future findings to provide early evidence to inform trauma-informed screening practices, health care provider education, and policy efforts aimed at improving healthcare-based identification and support for survivors of human trafficking and DV in Illinois.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 41A

Wang, Winnie Faculty Advisor: Wendi Gardner

Compensatory Mechanisms of Avoidant Attachment

Individual differences in attachment style present unique challenges for their partners. What keeps partners of the insecurely attached committed to their relationships? Anxious attachment is characterized by uncomfortability with distance from their partner, while avoidant attachment is uncomfortability being too close to a partner. Although prior research has revealed ways that anxiously attached individuals may support their partner's satisfaction, the potential compensatory mechanisms that keep partners committed to an avoidant partner remain under-explored.

In Study 1, I explored whether the autonomy needs of an avoidant partner may benefit their romantic partners with space and time to pursue their own goals as well as to maintain broader social support networks outside the relationship. 200 U.S. adults in committed romantic relationships responded to behaviors aligned with avoidant and/or anxious attachment, as well as measures of attachment and relationship satisfaction,. Results revealed that, contrary to expectation, even the autonomy supporting avoidant-attachment-related behaviors were perceived generally more negatively, although people's own attachment style influenced their perceptions of other's behaviors.

Study 2 surveyed 300 U.S. adults to examine whether these differences can be explained by a difference in frequency of attachment-related behavior, as well as whether awareness of a partner's attachment style influences the perception of their behaviors. I also explored what kinds of values people perceive certain relationship behaviors express. While results are pending, I hypothesize that while avoidant-attachment-related behaviors are more damaging, they may occur less frequently.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 90A

Wang, April Faculty Advisor: Nikos Hardavellas

Hardware- and Noise-Aware Quantum Compilation of Fermionic Hamiltonians

Quantum computing is a relatively new computing paradigm that promises increased efficiency over classical computers for a variety of applications, including the simulation of quantum systems. Many quantum systems that scientists would like to simulate are *fermionic systems*, including most molecular and material systems. To conduct this simulation, the fermionic Hamiltonian, a mathematical description of the system, needs to be compiled into machine instructions. Effective compilation must be *hardware-aware*, meaning that it considers hardware constraints including qubit connectivity, coherence times, and gate fidelity. It should also be *noise-aware* in order to anticipate and minimize errors.

In this project, we build a compilation pipeline that makes such considerations. Our approach improves upon the recently proposed Treespilation pipeline, which vertically incorporates different stages of fermionic Hamiltonian compilation and directly synthesizes hardware-aware circuits using a ternary tree representation. We extend this work by including more detailed hardware awareness and accounting for noise earlier in the compilation process.

We show that total Pauli weight can be used as a computationally cheaper metric for optimizing the ternary tree representation. We also develop optimization strategies to minimize the estimated number of costly qubit interactions and maximize their fidelity without requiring full circuit synthesis. Finally, we incorporate noise-awareness by favoring more reliable qubit couplings, reducing the average error rate of two-qubit gates by around 25% across multiple benchmarks. These results demonstrate the utility of ternary tree structures for fermionic simulation, as well as the benefits of incorporating hardware awareness early in the compilation pipeline.

Presentation Format: Oral
Time: 12:45-2:15PM
Room: Lake Room

Weingust, Ethan Faculty Advisor: Linsey Seitz

From Pollution to Products: Upgrading CO2 Emissions Using Electricity

Humanity faces a monumental challenge in mitigating global warming, a crisis heavily driven by the systems we rely on for everyday materials and chemicals. The commodity chemical industry, for example, is deeply entrenched in using natural gas as its primary feedstock for everything from plastics to pharmaceuticals. Transitioning away from this fossil fuel-based status quo is a daunting challenge, but recent developments in green, electricity-driven chemical manufacturing offer a highly promising alternative.

Instead of treating carbon dioxide (CO2) purely as a pollutant, our work reimagines it as a valuable, renewable resource. In this work, we demonstrate a sustainable method to synthesize a crucial precursor to acrylic acid, a widely-used chemical used in adhesives, paints, and diapers, using electricity and CO2 as inputs. Through a process called electrocatalysis, we can harness renewable electricity to drive chemical reactions that traditionally require fossil fuels. The benefit here is twofold: we can simultaneously reduce emissions and transition away from fossil fuel-based power in major chemical processes.

While converting CO2 into new materials is notoriously difficult due to its high stability, our project explores how targeted electrochemical carboxylation reactions can effectively break CO2 down and rebuild it into useful molecules. By bridging the gap between harmful emissions and valuable products, this research outlines a tangible pathway to drastically reduce the chemical industry's carbon footprint.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 19P

Wen, Marisa Faculty Advisor: Elnur Babayev

Epigenetic Loss of Fibrillin-1 Drives Ovarian Fibrosis During Reproductive Aging

Ovarian aging results in a rapid decline in oocyte quality and quantity, contributing to infertility, miscarriage, and broader health consequences such as impaired cardiovascular, musculoskeletal, and immune function. Ovarian fibrosis is a hallmark of reproductive aging, yet both its upstream drivers and the epigenetic mechanisms underlying this process remain poorly understood. Using unbiased Spatial Assay for Transposase-Accessible Chromatin with sequencing (ATAC-seq), we identified Fbn1, which encodes the extracellular matrix protein fibrillin-1 (FBN1), as epigenetically downregulated in aged murine ovaries. FBN1 also plays a key role in sequestering latent transforming growth factor-beta (TGF-β) complexes in the extracellular matrix, suggesting that its loss may dysregulate profibrotic signaling. To investigate this, we analyzed ovaries from young (8-10 weeks) and aged (10-11 months) CD1 mice. RNA in situ hybridization demonstrated significantly reduced Fbn1 transcript expression in aged ovaries, particularly within stromal and theca-interstitial compartments. Immunohistochemistry further confirmed reduced stromal FBN1 localization with aging. Western blot analysis showed decreased total FBN1 protein levels and increased active TGF-β1 in aged tissue. Together, these findings suggest that epigenetic loss of FBN1 may impair extracellular sequestration of TGF-β, leading to increased profibrotic signaling and ovarian fibrosis. This work identifies extracellular matrix remodeling as a potential upstream driver of ovarian aging and highlights epigenetically downregulated FBN1 as a promising therapeutic target for preserving reproductive function. Our ongoing experiments explore elastic fiber organization, comprised of FBN1-based microfibrils and elastin, to characterize ovarian fibrosis.

Presentation Format: Oral
Time: 10:45AM-12:15PM
Room: Lake Room

Wertz, Abby Faculty Advisor: Steven Schwulst

Microglia Depletion Does Not Impact CD8+ T Cell Infiltration Into The Young Mouse Brain 8 Hours Post-Traumatic Brain Injury

Traumatic Brain Injury (TBI) accounts for approximately one-third of injury-related deaths in the United States. In aged mice, microglia, the brain's resident immune cells, facilitate recruitment of CD8+ T cells to the brain following TBI. CD8+ T cells destroy damaged cells, and their infiltration contributes to worse motor and cognitive outcomes. Microglia depletion in aged mice reduces CD8+ T cell infiltration. Since CD8+ T cell infiltration is not typically observed in young mice, we hypothesized that microglia depletion would not affect CD8+ T cell recruitment post-injury. Young male and female mice were placed on either a microglia depletion diet or a control diet. Mice underwent TBI via controlled cortical impact using a piston that struck the brain, or sham injury, in which an incision was made, but TBI was not induced. Mice were euthanized 8 hours post-injury, and brain and blood samples were collected. Flow cytometry confirmed microglia depletion and quantified CD8+ T cells. Microglia depletion did not result in significant differences in CD8+ T cell populations 8 hours post-TBI compared to control diet groups (p > 0.05). Additionally, no significant differences were observed between the TBI and sham groups at this time point (p > 0.05). These findings support our hypothesis that microglia depletion does not influence CD8+ T cell infiltration in young mice at 8 hours post-injury. These results further suggest that CD8+ T cell infiltration after TBI is an age-dependent mechanism. These findings may reflect similar immune responses in humans and inform the development of targeted therapeutics.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 82A

Williams, Margarita Faculty Advisor: Ivy Wilson

Remixing and Reliving: The interethnic power of intergenerational sampling in Bronx and London Hip-Hop

As a musicophile, the use of third-spaces as a safe space for creative expression and cultural connection led me to question hip-hop as a genre of ceaseless reinvention. As the middle ground between rap and R&B, hip-hop uses its musical space to coin sounds of earlier generations through the sampling (or borrowing) portions of older tracks and repurposing them for new songs, particularly prior to 1993 and the enactment of federal sampling restrictions.

In analyzing the foundations of hip-hop through a case study of A Tribe Called Quest during 1988-1993, my research argues that hip-hop sampling choices are not politically or racially motivated, and instead fundamentally represent the interconnectedness of the hip-hop community and the various ethnic communities of the Bronx, particularly the interactions between Hispanic and African diasporic populations.

Through a previous interest in the unique racial diversity of the Bronx, this research conducts a comparative analysis between NYC hip-hop and London hip-hop, as London is equally as diverse but with a large population of Caribbean and West Indian peoples versus NYC's heavy Puerto Rican population. This comparative analysis examines the weight of ethnic diversity in both the samples used and communities involved through ethnomusicological and historical research, completed through sample analysis, multimedia archives, and historical recountings.

In this, I aspire to emphasize community world-making as a foundation for hip-hop creation, and thereby inspiring its reclamation of older genres through sampling and ethnic diversity through sample selection and setting a precedent for interethnic collectives and safe spaces.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 43P

Wilson, Amelia Faculty Advisor: Karina Van Bogart

Loneliness and cognitive functioning in everyday life among mid-to-later life adults

Loneliness, the subjective feeling of being disconnected and alone, is a significant global public health concern linked to poorer health and well-being. For example, growing research shows that lonelier people, on average, have worse cognitive health in older age and face an elevated risk for dementia. However, it is less clear how feelings of loneliness are associated with cognitive functioning in everyday life. In the current research, we addressed two main research questions: 1) Do individuals with more unstable momentary loneliness show worse cognitive functioning across the 20-day study period?, and 2) In moments when loneliness is higher, is cognitive function worse in that moment and ~3 hours later? Participants (N = 157, ages 45–74) completed Ecological Momentary Assessments (EMAs) five times per day for 20 days, yielding approximately 15,700 momentary observations. Cognitive functioning was assessed via mobile tasks measuring processing speed (Digit Symbol Matching), episodic memory (Visual Paired Associates), and working memory (N-Back). Multilevel models controlling for showed no evidence for associations in our main research questions. In addition to discussing these findings, I will share other work that is in progress that seeks to dive deeper into the timescales and potential thresholds of associations between loneliness and cognitive function in daily life.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 58P

Wojno, Shreya

Missing from the Model: The Gender Imbalance in Undergraduate Economics at Northwestern University

Prior research highlights a "leaky pipeline" in economics fields, with women underrepresented at senior levels in finance and academia. This imbalance is mirrored early on; at the undergraduate level, economics consistently has one of the largest gender divides. Considering the impact undergraduate experiences have on postgraduate decisions, this is a natural intervention point. Yet research has overlooked what deters women during early economics experiences. This study addresses this gap by examining how perceptions of gender dynamics shape female students' experiences in economics at Northwestern University.

I conducted 16 semi-structured interviews with Northwestern undergraduate economics students, both male and female ranging from sophomores to seniors. The interviews were analyzed using a grounded theory approach. Primary findings have uncovered three key themes: (1) Northwestern's economics department perpetuates a culture that structurally disadvantages female students through certain community norms and identity expectations, (2) the economics extracurricular environment prioritizes select students (ie, males with prior finance background), driving social competition and character assimilation for students outside these identities, (3) there is a prevalent social hierarchy of "acceptable" careers; students are encouraged to pursue industries such as finance and consulting, with those who don't seen as less successful. These themes reveal the importance of extracurricular and social considerations in addition to curricular ones, emphasizing the impact of the comprehensive economics experience on female matriculation. This project aims to contribute to scholarship on the origins of the economics leaky pipeline, uncovering potential opportunities and solutions for building a more inclusive community.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 34P

Wolk, Zach Faculty Advisor: Ann Bradlow

Foreign Accent Detection With and Without Language Familiarity: Evidence from Korean

It is well established that first-language (L1) listeners can distinguish between L1 and second-language (L2) speech in their own language, even from brief or content-masked samples. Furthermore, in "Identifying a Foreign Accent in an Unfamiliar Language," Major (2007) found that listeners unfamiliar with a language, namely Portuguese, could detect L2 speech in that language. However, this conclusion rests on the English–Portuguese pairing, leaving open whether that ability extends to more typologically distant languages. The present study asks whether listeners unfamiliar with Korean can distinguish between L1 and L2 speech in Korean. In a binary forced-choice task, 22 listeners classified Korean speech samples as native or nonnative, corresponding to L1 and L2, respectively. Listeners were stratified by familiarity with Korean and English: L1-Korean/L2-English (KE), L1-English unfamiliar with Korean (EX), and L1-Other/L2-English unfamiliar with Korean (XE). The 30 talkers comprised 15 L1-Korean/L2-English talkers and 15 L1-English/L2-Korean talkers. Listeners across all three groups distinguished L1 from L2 talkers, with the magnitude of separation varying systematically with Korean familiarity, confirming that this ability generalizes beyond Portuguese. Whereas Major (2007) found that accentedness ratings were comparable regardless of familiarity with Portuguese, the ability to distinguish L1 from L2 Korean speech varied substantially across listener groups: scores were highest (greatest separation) for KE and lowest for XE. Further, all groups showed a consistent bias toward the L1 response, with weakest bias in KE and strongest bias in XE. These results support the existence of both language-general and language-specific cues to L2 speech.

Presentation Format: Oral
Time: 10:45AM-12:15PM
Room: Armadillo Room

Xu, Monica Faculty Advisor: Niall Mangan

Salmonella in the Human Gut: Understanding Its Survival Strategy via an Encapsulated Metabolic Pathway

Salmonella is a type of bacteria that causes severe diseases in the human digestive tract. It is one of the four major causes of diarrhoeal diseases globally. Studying their survival yields crucial insights in the medical and engineering industries. In *Salmonella enterica*, the 1,2-propanediol utilization (Pdu) pathway supports energy production under an inflamed environment. The pathway is encapsulated by a protein shell commonly referred to as microcompartment (MCP). It allows the pathogen to sequester a toxic intermediate and provides a competitive advantage over commensal microbiota. This project investigates the performance of the Pdu pathway through both theoretical and experimental approaches.

A mathematical model was established based on the flux balance of encapsulated reactions to estimate pathway performance. At steady state, we found an optimal permeability using experimentally informed parameters. We also interpreted the permeability tradeoff by asymptotic approximations and simulated results with different substrate levels. These works would guide mass production bioengineering. However, the steady state assumption is limited. To formally analyze *Salmonella*'s Pdu pathway in the human host, we extend the model to capture dynamic timescales, where the MCP number appears as an important variable. Thus, we tested the effect of environmental conditions on MCP formation via fluorescent microscopy. MCP number increases with substrate concentration, and notably, we identified a threshold concentration that triggers significant compartment assembly. Ultimately, we aim to use the integration of mathematical analysis and microbiology experiments to promote further hypothesis development and translate into strategies for disease control and designing efficient engineered metabolic systems.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 57A

Xu, Angela Faculty Advisor: John Rogers

Fingertip-Sized Wireless Button Sensors for Health Monitoring in Neonatal and Pediatric Patients

The button sensors project aims to develop fingertip-sized devices for wirelessly monitoring vital signs of preterm patients in the Neonatal Intensive Care Unit (NICU). Three discrete units-ECG/EEG (chest/head), PPG (head), and mechano-acoustic (peripheral)-work in tandem to provide comprehensive measurements of key vitals: heart rate, respiration, blood oxygen level, EEG, heart rate variability, sleep, motion, and temperature. Each unit includes electronic components encapsulated by medical-grade silicone or epoxy, which protects sensitive neonatal skin from abrasion and ensures longevity of the device. One significant technical achievement of the project was successfully miniaturizing photoplethysmography (PPG) for the head unit, enabling measurement of peripheral oxygen saturation with increased space optimization and power-efficiency. The sensors are supported by two power supply options-wireless power transfer technology or a miniature magnet-connected battery-which can be easily interchanged to reduce data dropouts. Data streams continuously and wirelessly to a smartphone app and uploads to the cloud, which allows clinicians to monitor changes in real time.

By implementing wireless communication, this design overcomes the limitations of conventional wired systems with rigid interfaces, protecting neonates from further skin injuries and vascular complications. Ultimately, by offering a fully wireless, ultra-miniaturized device and an intuitive real-time monitoring platform, this integrated system represents a significant advancement in enabling parent kangaroo care, streamlining clinical workflows, allowing unobstructed movement, and delivering accurate, minimally intrusive monitoring in even the most sensitive patients.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 15P

Xu, Amy Faculty Advisor: Adriana Weisleder

Measuring What Matters: Evaluating Language Exposure Tool in Bilingual Toddler Research

In the United States, a quarter of children hear a language other than English at home, and this number is growing. Extensive research shows that dual language exposure affects children's language development. However, less is known about how dual language exposure changes throughout toddlerhood as children's own language production grows, and what its relationship is to language dominance.

I will explore these gaps through a longitudinal study of Spanish-English bilingual toddlers, BiTODLS (Bilingual Toddlers' Development of Language Skills), conducted in the Child Language Lab under the direction of Principal Investigator Dr. Adriana Weisleder to assess the relationship between language exposure and language dominance in toddlers aged 2 to 2.5 years. Language exposure was measured using the Language Exposure Assessment Tool (LEAT), a parent interview that assesses children's exposure to English and Spanish from different speakers and environments during a typical week. Children's vocabulary was measured using the MacArthur-Bates Communicative Development Inventories (CDI), which is a parent-reported measure of a child's English and Spanish vocabulary. These measures provided an estimate of children's language exposure and language dominance based on vocabulary.

In my work as a Research Assistant, I helped analyze, code, and interpret LEAT and CDI data. Preliminary results show wide heterogeneity in changes in children's language exposure over time, as well as inconsistencies between language exposure and language dominance. To address these observations, I will include an exploratory analysis of potential factors driving these differences, examine the limitations of LEAT, and propose ways to strengthen these measures.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 38P

Yang, Eric Faculty Advisor: Hai Zhou

Deductive Proof vs. Model Checking for Safety Verification of Autonomous Physical Systems

Autonomous systems are widely used in practice, from drones to robotic arms in factories, because they improve productivity and efficiency across many settings. Yet even when their controllers come with theoretical guarantees, real-world environments remain noisy and uncertain.

Formal verification uses mathematical logic to prove that a formal model satisfies a specification, or to exhibit a counterexample when it does not. This project studies how formal verification can be used to analyze the safety of a 3D drone model controlled by a hardware-oriented controller. The goal is to understand which verification methods are most effective for systems that combine continuous dynamics, discrete control, and safety constraints.

We model the drone mathematically, design a simple controller that keeps it within a safe region, and analyze the resulting closed-loop system using three approaches: explicit-state model checking in TLA+/TLC, symbolic model checking in nuSMV, and deductive proof in Lean 4. We also study the model across three tiers of complexity, with each tier introducing richer physics, such as wind, or more involved controller behavior. This allows us to compare the methods in terms of guarantee strength, specification, proof effort, and scalability as the system becomes more complex.

Although the project is still ongoing, it already provides a common framework for comparing major verification techniques on the same control problem. The broader contribution is a clearer view of the tradeoffs between these approaches for safety-critical autonomous systems.

Presentation Format: Oral
Time: 12:45-2:15PM
Room: Lake Room

Yin, Angel Faculty Advisor: Nathan Gianneschi

Targeted Synthesis of Protein-like Polymers as JAK2 Inhibitor in Intracellular Pathway of Alopecia

Alopecia areata, a condition that causes hair loss, along with other inflammatory skin diseases, occur when the immune system becomes overactive and attacks the body's tissues. One pathway involved in this process includes the protein JAK2. When JAK2 is overactive, it turns on another protein STAT1, which causes inflammation to continue. While there are FDA-approved JAK2 inhibitors, many of these drugs work by blocking the protein's energy-binding site, known as the ATP site. Many other healthy proteins share similar ATP sites and these drugs can affect unintended targets and cause side effects. This study tested a different approach using protein-like polymers (PLPs), synthetic molecules built to imitate SOCS1, a natural protein that normally helps shut down the overactive JAK/STAT1 signaling pathway. Rather than blocking the ATP site, PLPs bind to a different region of JAK2 that allows selective targeting. PLPs of different sizes were created, and their structure was confirmed using techniques of mass spectrometry, high-performance liquid chromatography, and proton nuclear magnetic resonance spectroscopy. The PLPs were then tested in human skin cells, where they were able to enter cells on their own, remain stable inside cells, and reduce harmful STAT1 activation. The PLPs showed no signs of cell death at tested doses, and it was discovered that a 10-peptide-containing polymer was more effective than a larger 20-peptide-containing polymer. The study suggests that PLPs can become a safer, more precise way to treat inflammatory skin diseases by stopping overactive immune signaling without the side effects of many drugs.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 21P

Yoo, Hye-won Faculty Advisor: Wendi Gardner

Exploring Dyadic Perspectives of Friendship Jealousy on Relationship Maintenance and Individual Well-Being

Friendships are instrumental to well-being in adulthood, yet there is a distinct lack of research exploring friendship conflict past adolescence. One common source of friendship conflict - friendship jealousy, the emotion that arises when one's role in a friendship is threatened by a third party - has been particularly understudied in adult populations. Therefore, this project aims to explore the impact of friendship jealousy on members of a friendship dyad. This study used a mixed-methods approach. Study 1A used a Large Language Model (GPT-5.4) to classify comments (N = 12,024) from 20 high-engagement TikTok videos about friendship jealousy. Study 1B conducted interviews with emerging adults (N = 39) about their experiences as both the experiencer and receiver of friendship jealousy. Results indicated that experiencers endorsed friendship jealousy as acceptable, while receivers rejected it. Experiencers reported higher levels of shame, while receivers reported higher levels of exhaustion. Qualitatively, experiencers attached self-worth to holding a unique role in the friendship, but social stigma around jealousy led to its suppression, contributing to higher levels of shame. Receivers filtered their behavior to avoid hurting their friend, contributing to higher levels of exhaustion. Two pathways for resolution emerged: perspective-taking and receptivity to reassurance within the dyad, and reliance on external social networks. These findings suggest a clear experiencer-receiver divide in the acceptability, emotional impact, and expression of friendship jealousy, which may be reconciled through reassurance, perspective-taking, and broader social support. Not only does this extend friendship jealousy research, but it also proposes possible interventions.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 89A

Yun, Kris Faculty Advisor: Yan Chen

Is SpeechLLM Safety Sufficient?

AI voice assistants are frequently used as personal assistants or help center personnel. Unlike traditional text-based LLMs, AI voice assistants use speechLLMs that process spoken audio. Because of their novel processing of audio rather than text, speechLLMs introduce new security vulnerabilities that are not well understood. In this poster, we ask the question: can someone manipulate these systems to behave harmfully by embedding hidden or misleading instructions in audio? We find that the answer is yes, and that audio-based AI systems like speechLLMs face a wider range of potential manipulation than text-based ones.

To study this, we built a testing framework that evaluates how well these AI systems resist harmful attacks delivered through sound. We tested both publicly available and commercial models using a consistent set of conditions to ensure fair comparison across systems.

We also created AABench, a new dataset containing over 4,000 example prompts designed to test whether these AI systems can be tricked into ignoring their safety guidelines. These prompts were drawn from six different sources covering a range of manipulation strategies.

Our results reveal how no single protection method works reliably across all types of manipulation attempts; a defense that works well against one strategy may fail against another. This finding suggests that current audio-based AI systems like speechLLMs are more fragile than they might appear, and that stronger, more general protections are needed before these systems can be safely deployed at scale.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 62A

Zaidi, Arianna Faculty Advisor: Richard Carthew

Characterizing the Secretory Mechanism of a Novel Peptide Through an RNAi Screen in Drosophila

Advances in sequencing and proteomic technologies have revealed a class of understudied peptides translated from short open reading frames in eukaryotes. One of these, the novel peptide Dafcin, is expressed in the ovaries of the fruit fly *Drosophila* and has been shown to traffick a fused GFP cargo from the ovarian follicle cells to the oocyte. However, Dafcin-GFP's secretion mechanism is poorly understood as it lacks the N-terminal signal peptide characteristic of canonical secretory proteins. Dafcin-GFP is predicted to adopt an amphipathic a-helical configuration, a common feature of molecules that interact with lipid membranes, which may facilitate interactions with lipid bilayers and enhance transport out of follicle cells. While untagged GFP markers remain in follicle cells, Dafcin-GFP is transported to the oocyte. Additionally, in stage 10B follicle cells, Dafcin-GFP associates with the Golgi complex, indicating a possible role for the Golgi in mediating Dafcin-GFP secretion, although definitive evidence is lacking. To investigate the hypothesis that Dafcin-GFP utilizes a noncanonical secretion pathway, I have employed RNAi knockdown of secretion genes in the follicle cells. To assess the secretion properties of Dafcin-GFP, I have identified a canonically secreted GFP-marked protein that can be compared to Dafcin-GFP: wgSS-sfGFP, a GFP fusion protein that contains the N-terminal signal sequence of the protein Wingless. By systematically analyzing the secretion patterns of Dafcin-GFP, this study aims to clarify its transport mechanisms and to deduce broader implications for noncanonical secretion pathways.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 81A

Zhang, Oliver Faculty Advisor: William Klein

A Supersensitive ELISA for Detecting Amyloid-v Oligomers in Alzheimer's Disease and Its Potential Diagnostic Utility

Alzheimer's disease (AD), the most common form of dementia worldwide, is characterized by the abnormal accumulation of soluble oligomers of v-amyloid peptide (AvO) and neurofibrillary tangles formed by hyperphosphorylated Tau. There is significant evidence supporting that AD pathogenesis begins years before symptoms appear, with neurotoxic AvOs playing a central role in disease triggering by contributing to synaptic dysfunction and neuronal death. Even though AvOs are hypothesized as a key biomarker for early AD diagnosis, diagnostic challenges remain, as non-invasive methods for early and definitive confirmation are scarce, along with antibodies that specifically target and discriminate among soluble toxic species. Amongst the antibodies selected against ABOs, the single chain fragment variable (scFv) NUsc1 demonstrates reliable selectivity to AD-relevant neurotoxic AvOs and serves as a prime candidate for their detection. We conducted a supersensitive ELISA with phage-bound NUsc1, where we used the phage-displayed version of the scFv fragment to detect AvO concentrations in AD mice brain homogenates, human brain homogenates, and human CSF samples. pbNUsc1 identifies AvOs in mice and human brain samples, clearly distinguishing MCI and AD patients from age-matched controls. Furthermore, pbNUsc1 can detect AvOs in CSF from McGill rats and human patients, providing a straightforward, less invasive readout beyond the brain. These findings support the potential utility of this supersensitive assay as an early diagnostic tool for AD. By specifically targeting soluble, neurotoxic AvOs using the pbNUsc1 antibody, this assay may provide a clinically relevant method for diagnosing and detecting AD.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 36P

Zhang, Jimmy Faculty Advisor: Han Liu

Optimizing dimensionality reduction in gene expression using autoresearch

High dimensionality presents a common challenge in biological data analysis, characterized by a large number of features in a dataset. Traditional statistical methods often struggle in this setting, motivating dimension reduction techniques such as Principal Component Analysis (PCA) that aim to preserve the original variance structure and relationships in a low-dimensional representation of the data. While PCA applies a simple linear transformation, more complex methods use flexible nonlinear transformations that can produce a better representation. Autoresearch is a cutting-edge machine learning approach that harnesses the coding capabilities of large language models (LLMs) to automate the iterative process of experimentation and evaluation traditionally performed by a researcher. I apply this framework to identify a suitable low-dimensional representation for gene expression data from the human brain, optimizing the classification accuracy of gene probes on chromosomes. Within this project centered around agentic coding, my work focuses on designing structured prompts and constraints that guide the LLM through iterative code modification and model evaluation. I highlight key differences when using autoresearch by comparing against a baseline established through manual experimentation. My analysis emphasizes the ability and efficiency of autoresearch to achieve classification accuracy comparable to the baseline. The resulting low-dimensional representation of the data provides insight into relationships between features that correspond to different brain regions. This work also serves as a proof of concept for autoresearch as an effective and scalable approach to model selection, offering potential applications in science and other contexts where such iterative experimentation is necessary.

Presentation Format: PM Poster
Time: 1:30-3:00PM
Room: Louis Room
Number: 23P

Zhong, Angela Faculty Advisor: Yang Qu

Not All Setbacks Are Equal: How the Brain Responds to Failure Shapes Recovery and Motivation in Adolescents

Why do some teenagers bounce back after a setback while others spiral? Adolescence is a developmental period marked by increased sensitivity to feedback and elevated risk for depression and anxiety, yet the neurocognitive mechanisms shaping this vulnerability remain only partly understood. One promising candidate is how the adolescent brain processes negative feedback, an experience that is ubiquitous and affectively salient during this stage of development. The present study examines whether individual differences in neural sensitivity to negative feedback predict behavioral recovery and associated psychological outcomes in a diverse sample of middle school adolescents (N = 76, mean age = 13.31 years). Participants completed a trial-by-trial arithmetic task with immediate performance feedback during an fMRI (functional magnetic resonance imaging), allowing continuous measurement of brain activity throughout the task. Neural processing of failure was operationalized as BOLD (blood-oxygen-level-dependent) response to incorrect feedback trials across regions of interest implicated in learning and affective regulation (e.g., amygdala, ventromedial prefrontal cortex). Behavioral performance on the task and self-reported psychological measures were assessed alongside validated measures of depressive and anxious symptoms. By integrating neural, behavioral, psychological, and clinical measures, this study characterizes how individual responses to negative feedback relate to learning and engagement in adolescents. These findings clarify mechanisms linking feedback processing to affect, motivation and adaptive behavior during early adolescence.

Presentation Format: Oral
Time: 12:45-2:15PM
Room: Arch Room

Zhu, Selina Faculty Advisor: Catherine Woolley

Estrogen Receptor-a Expression in Serotonergic Neurons of the Dorsal Raphe in Male and Female Mice: Implications for Estrogenic Modulation of Serotonergic Circuits

Estrogen signaling in the brain plays a critical role in modulating mood, stress responses, and susceptibility to neuropsychiatric disorders, yet the cellular mechanisms underlying these effects remain incompletely understood. A key step in addressing this gap is identifying where estrogen receptors (ERs) are expressed. To overcome limitations of nonspecific ER antibodies, we used gene-targeted knock-in mice expressing an estrogen receptor-alpha (ERα)-enhanced yellow fluorescent protein (eYFP) fusion, enabling detection with anti-GFP antibodies (which recognize eYFP). We investigated the extent to which ERα is expressed in serotonergic neurons of the dorsal raphe nucleus (DRN), a key hub for serotonin production, in male and female ERα-eYFP mice. Using double immunofluorescent staining in mouse brain sections (40 μm), we labeled ERα-eYFP and tryptophan hydroxylase 2 (TPH2), the rate limiting enzyme in serotonin synthesis. Tile scans of every third section across the DRN were acquired through confocal microscopy. ERα-eYFP single-labeled, TPH2 single-labeled, and double-labeled cells were quantified using QuPath software across three experiments in male and female mice (n=3 per group). Approximately 39% of TPH2-positive neurons co-expressed ERα, while ~65% of ERα-positive cells were serotonergic in both sexes. ERα-expressing serotonergic neurons were concentrated in the middle DRN, forming distinct dorsal and ventral clusters. These results suggest a potential anatomical substrate for modulation of serotonergic signaling by estrogens. By clarifying ERα distribution within the DRN, this study advances understanding of interactions between serotonin and estrogens and provides a foundation for future work on how hormonal regulation of specific neuronal populations contributes to behavior and neuropsychiatric disease.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 22A

Zwiebach, Molly Faculty Advisor: Vijay Mittal

Assessment of Comorbid Depressive Symptoms in Clinical High-Risk for Psychosis Research

Clinical high risk for psychosis (CHR), the syndrome putatively preceding the onset of psychosis, has clinical utility in identifying and categorizing early psychotic-like symptoms. A better understanding of the CHR state can clarify the trajectory of psychosis and support the development of interventions that can treat, and potentially prevent, progression to a psychotic episode. Studying depressive symptoms and disorders in relation to psychosis risk is important given the high incidence of symptom overlap and co-morbidity. The present study examines whether and how depression is assessed in studies with CHR populations. In this meta-analysis, we extracted demographic characteristics and information on depression assessment instruments from identified studies involving CHR samples. Overall, we found that 37.83% of CHR studies, totaling 696 studies, included a measure of depression from 1998 to 2023. DSM-5-based diagnostic criteria were the most frequently used diagnostic instrument, both overall and across decades (used in 36.93% of studies). We further evaluated patterns of depression instrument use in relation to key demographic variables. Dominant findings indicated that studies conducted in Europe ($r = 2.37$) and Oceania ($r = 2.21$) measured mood symptoms at a significantly higher rate than other global regions. Additionally, psychosis transition rates were significantly lower in samples assessed for mood symptoms compared to those that were not (Cohen's $d = 0.31$). Together, these findings highlight important gaps in the representation and assessment of comorbid depressive symptoms in CHR research. They further identify methodological and demographic areas that warrant additional investigation to improve characterization of this population.

Presentation Format: AM Poster
Time: 10-11:30AM
Room: Louis Room
Number: 85A